

Symmetric 4,6-Dialkyl/arylamino-5-nitropyrimidines: Theoretical Explanation of Why Aminolysis of Alkoxy Groups Is Favoured over Chlorine Aminolysis in Nitro-Activated Pyrimidines.

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ABSTRACT

A new synthetic route to obtain symmetric disubstituted alkyl/arylamino-pyrimidines under mild conditions is presented, which can be used to generate new purine libraries for drug discovery. We investigated the unexpected reaction of 6-alkoxy-4-chloro-5-nitropyrimidines with primary amines, which produced disubstituted dialkyl/arylamine pyrimidines instead of the expected 6-alkoxy-4-alkylamine-5-nitropyrimidines. To clarify this reaction, a computational study of the reaction mechanism was carried out and it was found that the Meisenheimer complex may play a key role in this reaction.

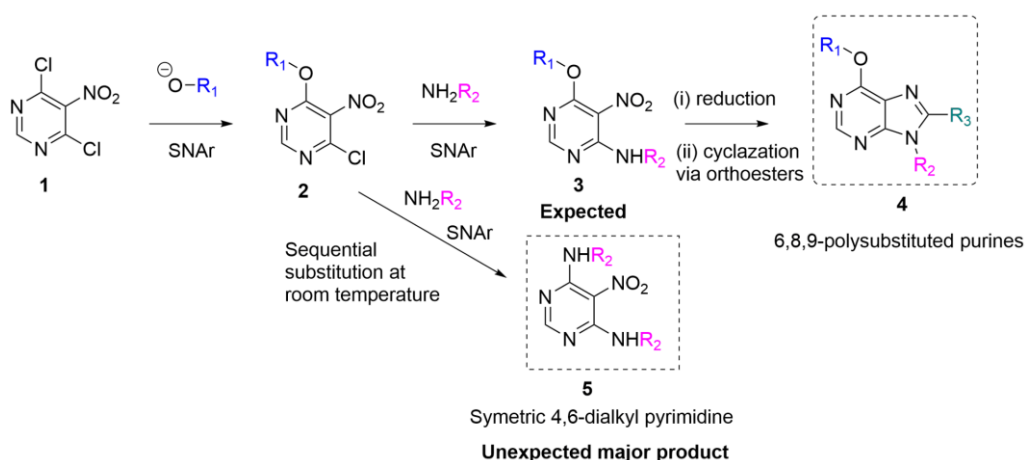
1. Introduction

Purines are among the most widely used structures in medicinal chemistry. This is due to the large number of enzymes that use purine cofactors, known as purine-utilising enzymes (PUEs), which are involved in the catabolism of purines and pyrimidines¹. Additionally, purinergic receptors play a key role in signal transduction following their activation by extracellular purines.

Therefore, purines are considered privileged structures in medicinal chemistry, with more than 15 FDA-approved purine drugs. For example, Vidaza, 5-fluorouracil, tecadenoson, and regrelor, to name a few². Therefore, medicinal chemists have developed several synthetic routes to obtain structurally diverse purine libraries. The use of polysubstituted pyrimidines as core elements prior to the purine ring formation is one of the most widely used approaches. Polyhalogenated pyrimidines are the preferred starting materials because they allow halogen substitution by a variety of nucleophiles. In particular, pyrimidines activated by electron-withdrawing groups such as nitro and nitroso promote highly efficient aromatic nucleophilic substitutions. Alkoxides are interesting moieties because they act as nucleophiles in the aromatic substitution of substituted

pyrimidines^{3,4} and as leaving groups in these reactions^{5–8}. However, the latter has not been extensively studied, as it may be counterintuitive, as they are poor leaving groups in standard S_N2. The reason for the inversion of reactivity when comparing S_N2 and S_NAr is the energetically favourable rearrangement step that occurs via the elimination of the alkoxide group. Therefore, the rate-determining step in the S_NAr reaction is nucleophilic addition⁹.

Our group has presented various 6,8,9-polysubstituted purine libraries prepared by a one-pot synthesis starting from 5-amino-4-chloro-6-alkylamino pyrimidines to obtain 6-alkoxy-8,9-dialkyl/aryl purines. This one-pot synthesis followed two different pathways depending on the size of the *N,N*-dialkyl amide used in the reaction¹⁰. Although this reaction is very useful for generating specific purine libraries, it has limitations in introducing structural variability at positions 6 and 8 of the purine ring. To gain more control over these positions, we designed a synthetic pathway starting with 4,6-dichloro-5-nitropyrimidine **1** (Scheme 1). The first reaction in this pathway is the aromatic substitution of a chlorine atom with an alkoxy group to give 4-alkoxy-6-chloro-5-nitropyrimidine **2**. Because di-substitution of the remaining halogen does not proceed under mild conditions, this reaction is very well controlled. The next step would be the aromatic substitution of chlorine at position **4** of the pyrimidine ring with alkylamines, which are very good nucleophiles, to give the expected 6-alkoxy-4-alkylamine-5-nitro-pyrimidine **3**. After reducing the nitro group, the ring would be closed with a variety of orthoesters to give the 6,8,9-polysubstituted purines **4**. However, under mild conditions, the dialkyl amine disubstituted product **5** was preferentially obtained. This was surprising, as the disubstitution of two chlorine atoms in the reaction of 4,6-dichloro-5-nitropyrimidine **1** with alkylamines requires strong conditions¹¹.



Scheme 1. A proposed synthetic pathway designed to obtain 6,8,9 polysubstituted purines **4**.

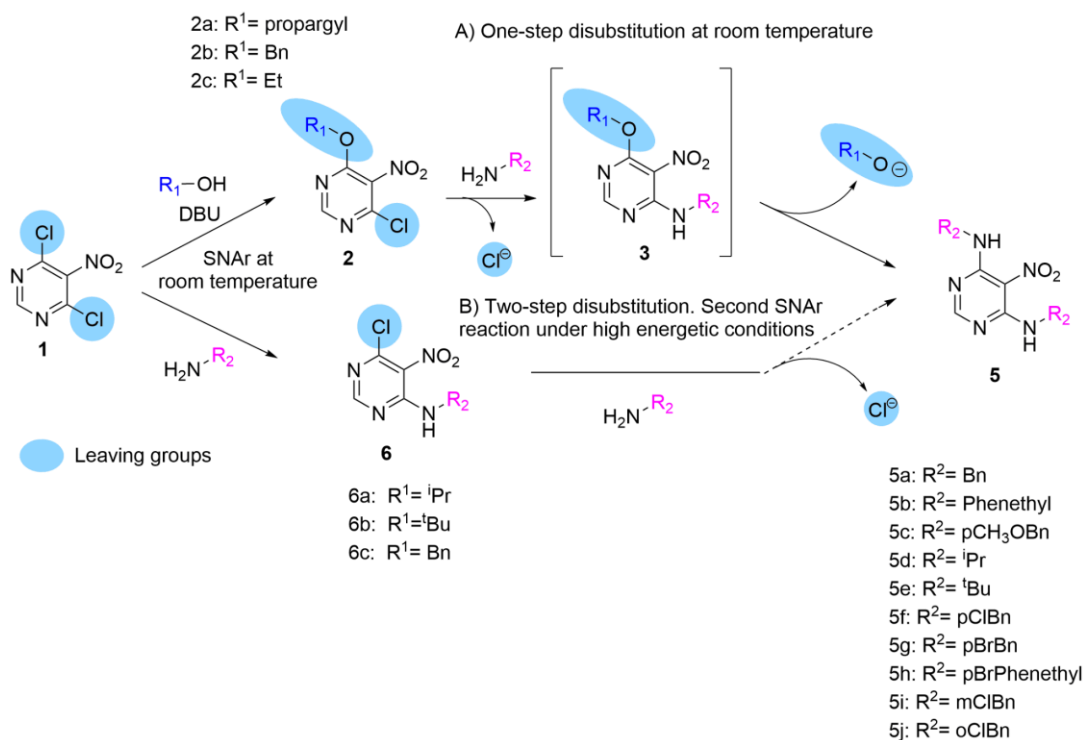
Therefore, the aim of this work was to elucidate the role of alkoxides as leaving groups in the aromatic substitution of activated aromatic rings and to provide a robust synthetic route to obtain symmetric pyrimidines under mild conditions, which can then be transformed into novel polysubstituted purines.

2. Results and Discussion

2.1 One-pot synthesis of symmetrical 4,6-dialkylamino-5-nitropyrimidines

We first synthesised pyrimidines **2** by the dropwise addition of solutions of prop-2-yn-1-ol, benzylic alcohols, and ethanol with DBU to a solution of 4,6-dichloro-5-nitropyrimidine **1** in anhydrous THF at 0°C to give **2a**, **2b**, and **2c**, **respectively**, in moderate yields. Having obtained 6-alkoxy-4-chloro-5-nitropyrimidines, we expected a smooth reaction between pyrimidines **2** and the primary amine at room temperature to give 4-alkylamine-6-alkoxy-5-nitropyrimidines **3**. However, when 1.5 equiv. of *N*-benzylamine was added at room temperature to a solution of pyrimidine **2a** in DCM with TEA, and disubstituted product **5a** was rapidly identified by MS in the crude reaction. By adding two equivalents of amine to the same reaction vessel, the conversion of **2a** to **5a** was completed without traces of monosubstituted product **3a** being detected. This was surprising because the disubstitution of polyhalogenated pyrimidines with nucleophiles always requires strong conditions⁸. Therefore, after purifying and fully characterising the symmetric pyrimidine **5a**, we wanted to see if this reaction would also occur when the other two 6-alkoxy-4-chloro-5-nitropyrimidines **2b** and **2c** were reacted with *N*-benzylamine under the same reaction conditions (Scheme 2). Both reactions afforded symmetrical dibenzylamino-5-nitropyrimidine **5a**.

Therefore, we investigated other primary amines as nucleophiles to determine whether this reaction could be widely used to synthesise symmetric 4,6-disubstituted pyrimidines. Eleven other primary amines were used, and in all cases, the corresponding disubstituted purines **5b-5j** were obtained in good yield (Scheme 2).



Scheme 2. Comparison of the leaving group capabilities of alkoxy groups and chlorine atoms when found in the same molecule.

Our results were consistent with those reported by Marchal et al⁸. They reported the use of activated 2-amino-4,6-methoxy-5-nitrosopyrimidine as a starting material to obtain symmetrical and asymmetrical 2-amino-4,6-dialkylamine-5-nitro-pyrimidines.

To better understand this reaction, three 4-alkylamine-6-chloro-5-nitropyrimidines **6a-c** that were synthesised from pyrimidine **1** following a procedure described elsewhere by us^{10,12} (Scheme 2) were used. This allowed us to compare the leaving group abilities of the chlorine and alkoxy moieties in these systems. As described above, the 6-alkoxy-4-chloro-5-nitropyrimidine **2** gave rise to pyrimidines **5** by sequential substitution first of the chlorine atom to give the compounds **3** and then, simultaneously, in the same reaction vessel, the second substitution of the alkoxy group to give the compounds **5**. However, the nitropyrimidines **6** remained unreactive under these conditions. This could lead to the erroneous conclusion that alkoxy groups are generally better leaving groups than chlorine atoms in S_NAr. In the case of compound **2** reacting with primary amines, the chlorine atom rather than the alkoxy group is the preferred leaving group of the two. When benzylamine reacts with pyrimidines **2**, the first aromatic substitution of the chlorine atom and the second of the alkoxy groups occur under mild conditions, but when the same benzylamine reacts with pyrimidine **1**, the first aromatic substitution of the chlorine atom occurs; however it is

then unable to carry out the second substitution of the second chlorine atom under the same mild conditions. Therefore, when both groups are together in the same ring, as in pyrimidines **2**, the chlorine atom is a better leaving group when reacting with a primary amine, but then comparing pyrimidines **3** and **6** where both have the same alkylamine at position 6 of the ring, the alkoxy group is a better leaving group.

2.2 Theoretical study

While traditionally S_NAr reactions were thought to follow a stepwise mechanism involving Meisenheimer complexes¹³, recent developments in experimental and computational chemistry have identified a rapidly growing class of nucleophilic aromatic substitutions that proceed by concerted (cS_NAr)^{14–18} rather than classical stepwise mechanisms. Moreover, recent studies have presented the hypothesis that most S_NAr reactions should follow a concerted mechanism, with the Meisenheimer complex being a transition state rather than an intermediate.^{19,20}

With this dichotomy in mind, a theoretical study was carried out to investigate various questions about the mechanism of reactions between chloronitropyrimidine derivatives and primary amines. For simplicity, the hydrocarbon chains of the amine reagent and the alkoxy group were truncated at the α -carbon atom. The transition states (**TSij**) and other intermediates such as the Meisenheimer complex (**MCij**) are numbered according to the number of the starting product (i) and the final product (j) of the reaction under investigation.

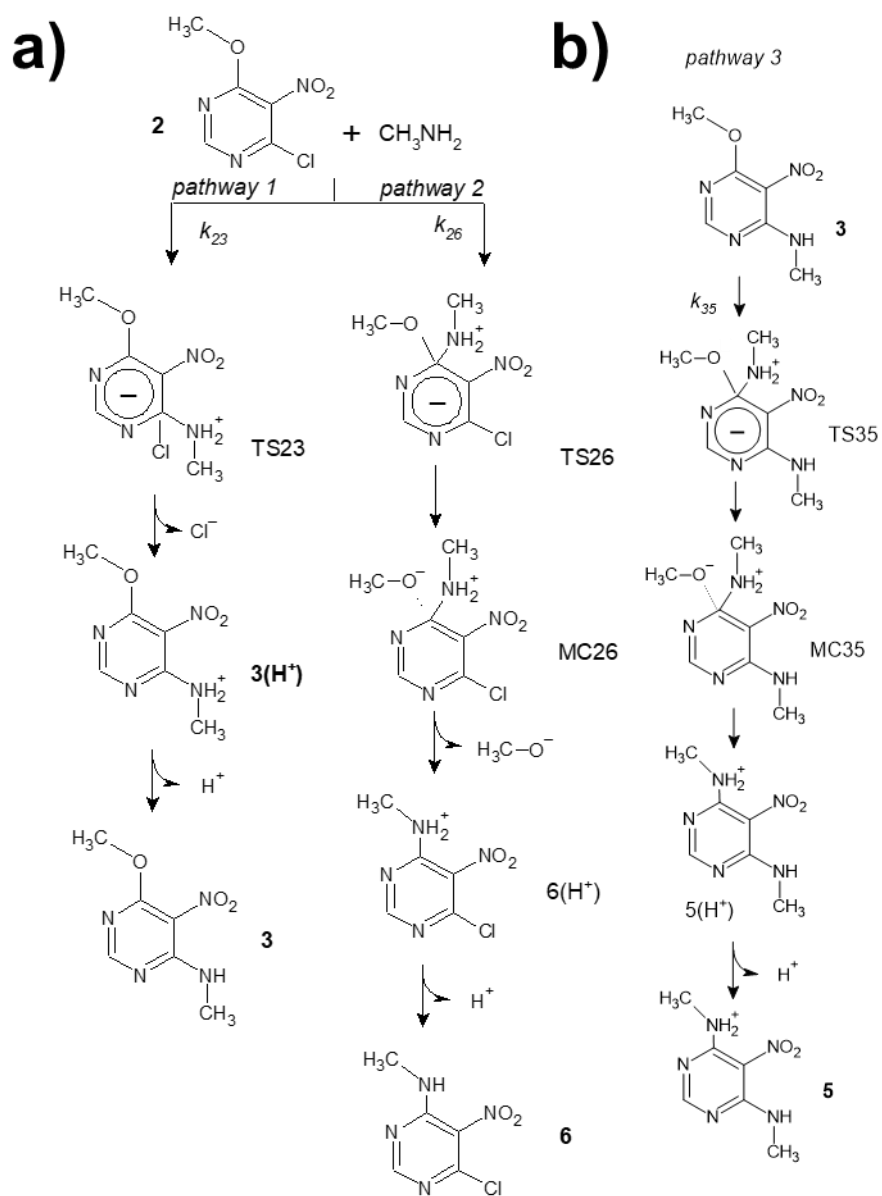
Scheme 3 shows the reaction steps studied in the reaction from **2** to **5**. For the first reaction step - the amine substitution of the product **2** - two different reaction pathways were analysed, where the chlorine atom or the alkoxy moiety is the leaving group (pathways 1 and 2, respectively). The activation barrier of pathway 1 is lower than that of pathway 2 (see Figure 1a and Table 1), indicating that the former is kinetically favoured. The rate constant estimated for **TS23** by TST is 50-fold higher than that for **TS26**. This is in agreement with the literature showing that chlorine atoms are better leaving groups than alkoxy chains^{21,22}. Interestingly, we found a Meisenheimer complex (**MC26**) in pathway 2, prior to the formation of the intermediate **6(H⁺)**, but not in pathway 1. This finding is in agreement with Gallardo-Fuentes et al. who reported that Meisenheimer complexes are not formed when the leaving group is a chlorine atom²³. In the second reaction step, from **3** to **5** via **TS35**, a barrier ~4 kcal mol⁻¹ higher than in the previous step (**TS23**) was found, followed by a Meisenheimer complex (**MC35**) (Figure 1b). The free energy of the ionic intermediates **5(H⁺)** and methoxide is significantly higher than that calculated for **3** and methylamine (here it is necessary to consider the limitations of DFT methods in combination with the CPCM approach to predict solvation energies of ions). Here, the presence of the complex

MC35 in the reaction coordinate could facilitate the transition towards **5(H+)** and be key in the 'apparently' straightforward reaction from **2** to **5**, where **3** was not detected experimentally. The expulsion of the leaving group from the Meisenheimer complex and the subsequent charge rearrangement could be limiting steps to reach **5(H+)**. A detailed study of the role of Meisenheimer complexes in S_NAr reactions can be found in reference²³. In Figure 2, it is interesting to observe how the nitro group in **2** is highly twisted with respect to the benzene ring ($\tau = 56.9^\circ$) and the dihedral angle becomes zero for **5**. The molecular structure of **5** is highly stabilised by two intramolecular hydrogen bonds with O---H distances of 1.87 Å (shorter than the sum of the van der Waals radii) (shown in green in Figure 2)²⁴

Finally, we calculated the transition states of other S_NAr processes where chlorine is the leaving group, such as the reaction from **1** to **6** (**TS16**) and from **6** to **5** (**TS65**). In all cases the energy barriers are comparable to those calculated for the reaction from **2** to **3** (**TS23**) and no Meisenheimer complexes were found in the reaction coordinate. Interestingly, the one-step disubstitution from **1** to **5** is not observed experimentally, although a lower energy barrier is calculated for the second reaction step (from **6** to **5**) than in the case of the reaction from **3** to **5**. This shows the complexity of this type of reaction. Further studies are needed to investigate the role of intermediates such as the Meisenheimer complex in S_NAr reactions and if they are a transition state rather than an intermediate as suggested recently^{19,20}.

Table 1. Free energy reaction barriers ($\Delta G^\ddagger$) calculated at the M06-2X/6-31G* level of theory and rate constants (k_{ij}) estimated according to the thermodynamic formulation of TST (Transition State Theory) for the studied reaction pathways.

Transition State (rate constant)	$\Delta G^\ddagger$ (kcal mol ⁻¹)	k (L mol ⁻¹ s ⁻¹)
TS23 (k_{23})	13.13	1.48×10^3
TS26 (k_{26})	15.44	29.9
TS35 (k_{35})	17.41	1.08
TS16 (k_{16})	11.36	2.92×10^4
TS65 (k_{65})	12.50	4.29×10^3



Scheme 3. (a) Competitive mechanisms for the $\text{S}_{\text{N}}\text{Ar}$ reaction between **2** and methylamine (reaction pathways 1 and 2); (b) Reaction mechanism for the methylamine substitution of the intermediate products **3** (reaction pathway 3).

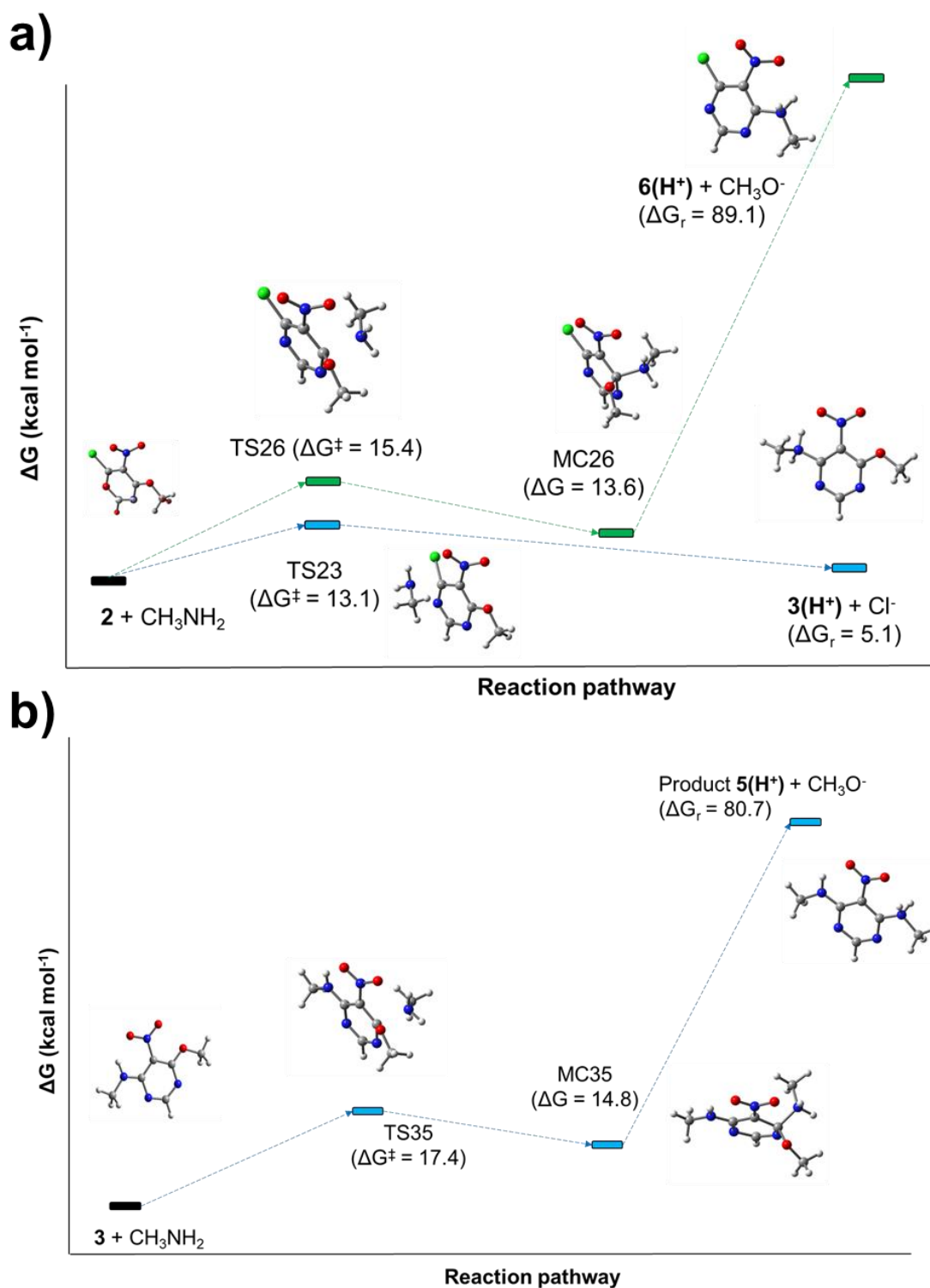


Figure 1. Calculated relative free energies at 298.15 K, with respect to the reagents of the transition state and intermediates for (a) the reaction pathways 1 and 2, and (b) the reaction pathway 3. Calculations were performed at the M06-2X/6-31G* level of theory including solvation effects (dichloromethane).

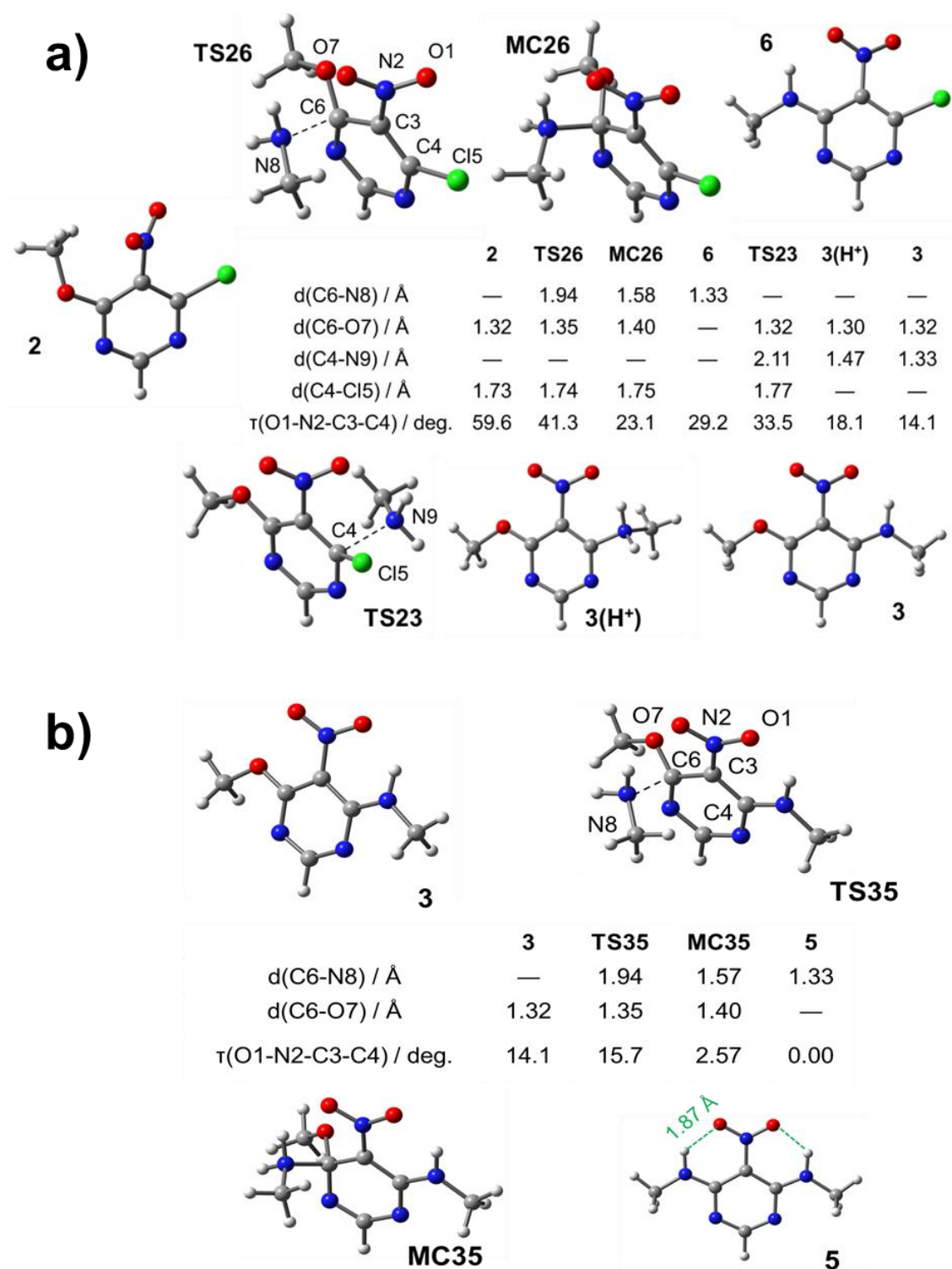


Figure 2. Molecular structure calculated for different species involved in reaction pathways 1, 2 and 3 (Scheme 3). Calculations were performed at the M06-2X/6-31G* level of theory including solvation effects (dichloromethane).

3. Conclusions

During the execution of the original plan to construct polysubstituted purine rings, it was found that when 6-alkoxy-4-chloro-5-nitropyrimidine **2** reacted with primary amines at room temperature, the preferred compound was not the expected 6-alkoxy-4-alkylamine-5-nitropyrimidine **3**, but the disubstituted dialkylamine pyrimidine **5**. This unexpected reaction was validated with three alkoxy groups and 12 primary amines, confirming the universality of this reaction. A computational study of the sequential S_NAR reactions from **2** to **5** showed that the Meisenheimer complex **MC35** could play a key role in the reaction mechanism. In contrast, no Meisenheimer complexes were observed when the leaving group contained chlorine. This would suggest that the first reaction occurs through a step-wise mechanism while the second through a concerted one. However, further studies need to be done as Meisenheimer complexes could be a transition state in a concerted mechanism. This new synthetic route opens the possibility of generating disubstituted pyrimidines under mild conditions, which can then be used to generate new purine libraries of great interest to the drug discovery industry. This novel chemistry can also be used to explore other nucleophiles, such as thiols, to increase the chemical diversity of the pyrimidine and purine libraries.

4. EXPERIMENTAL

4.1 General

Reaction courses and product mixtures were routinely monitored by TLC on silica gel Merck 60-200 mesh silica gel. Melting points were determined on a Stuart Scientific SMP3 apparatus and are incorrect. ¹H-NMR spectra were obtained in CDCl₃, solution on a Varian Direct Drive (400 MHz and 500 MHz). Chemical shifts (δ) are given in ppm upfield from tetramethylsilane. . ¹³C-NMR spectra were obtained in CDCl₃, on a Varian Direct Drive (125 MHz). All products reported showed ¹H-NMR and ¹³C-NMR spectra in agreement with the assigned structures. Mass spectra were obtained by electrospray (ESY) with a LCT Premier XE Micromass Instrument (high resolution mass spectrometry).

4.2. General procedure for the preparation of compounds **2(a-c)**:

To a solution of commercially available 4,6-dichloro-5-nitropyrimidine (0.025 mol, 1 equiv.) in anhydrous THF (20 mL), a solution of different alcohols (1 equiv.) plus DBU (5.8 mL, 1.5 equiv.) in THF (20 mL) is added dropwise. The addition takes place in cold (0° C) for about one hour and then the solvent is removed. After two extractions with methylene chloride, the organic phase was washed with an aq. saturated solution of NaCl and dried on Na₂SO₄. Evaporation under vacuum

gave a solid that was purified by column chromatography eluting with ethyl acetate/petroleum ether solutions.

4.2.1. 4-chloro-5-nitro-6-(prop-2-ynyloxy)pyrimidine (2a): Yellow solid, yield 51%, mp: 76°C. δ H (400.45 MHz, CDCl_3): 8.69 (1H, s, NCHN), 5.16 (2H, d, $J = 2.2$, O- CH_2 -), 2.6 (1H, t, $J = 2,4$, $\text{C}\equiv\text{CH}$). δ C (100.70 MHz, CDCl_3): 160.38, 157.48, 152.20, 132.74, 77.11, 76.05, 56.72.

4.2.2. 4-(benzyloxy)-6-chloro-5-nitropyrimidine (2b): White solid, yield 47%. δ H (400.45 MHz, CDCl_3): 8.64 (1H, s, NCHN), 7.41-7.34 (5H, m, Ph), 5.59 (2H, s, - CH_2 -). δ C (100.70 MHz, CDCl_3): 161.26, 157.55, 151.84, 134.17, 129.09, 128.92, 128.37, 71.01.

4.2.3. 4-chloro-6-ethoxy-5-nitropyrimidine (2c): Yellow solid, yield 16%, mp: 53°C. δ H (400.45 MHz, CDCl_3): 8.61 (1H, s, NCHN), 4.60 (2H, q, $J = 7.2$, - OCH_2 -), 1.43 (3H, t, $J = 7.2$, - OCH_2 - CH_3). δ C (100.70 MHz, CDCl_3): 161.45, 157.64, 151.58, 66.17, 14.20.

4.3. General procedure for the preparation of compounds 5(a-l):

A solution of **2(a-c)** (7 mmol, 1 equiv.) in CH_2Cl_2 (15 ml) containing TEA (initially 1.5 equivalents of each and in later reactions, when the disubstitution was checked, 2 equivalents). In this solution the different amines (10.54 mmol, 1.5 equiv. and later 2 equivalents) dissolved in CH_2Cl_2 (5 mL) were added at room temperature for half an hour. Evaporation under vacuum gave a solid that was purified by column chromatography eluting with ethyl acetate/petroleum ether solutions.

4.3.1. N^4, N^6 -dibenzyl-5-nitropyrimidine-4,6-diamine (5a): Yellow solid, yield 55% (from product **2b**, the yield is 33%, from product **2c**, the yield is 82%), mp: 105°C. δ H (400.45 MHz, CDCl_3): 9.58 (2H, bs, NH x2), 8.17 (1H, s, NCHN), 7.39-7.31 (10H, m, Ph x2), 4.85 (4H, d, $J = 5.6$, - CH_2 -Ph). δ C (100.70 MHz, CDCl_3): 160.00, 158.80, 157.46, 137.46, 128.96, 127.88, 127.81, 113.14, 45.49. ES+HRMS: Calculated $\text{M}+\text{H}=336.1461$ $\text{C}_{18}\text{H}_{18}\text{N}_5\text{O}_2$. Obtained: 336.1460.

4.3.2. 5-nitro- N^4, N^6 -diphenethylpyrimidine-4,6-diamine (5b): Yellow solid, yield 41% (from product **2b**, the yield is 13%), mp: 124°C. δ H (400.45 MHz, CDCl_3): 9.55 (2H, bs, NH x2), 8.28 (1H, s, NCHN), 7.36-7.24 (10H, m, Ph x2), 4.00 (4H, q, $J = 6.8$ y 6.00 NH- CH_2 - CH_2 Ph x2), 3.00 (4H, t, $J = 6.8$ NH- CH_2 - CH_2 Ph x2). δ C (100.70 MHz, CDCl_3): 158.42, 156.77, 137.75, 129.11, 128.92, 127.25, 116.86, 44.23, 35.23. ES+HRMS: Calculated $\text{M}+\text{H}=364.1774$ $\text{C}_{20}\text{H}_{22}\text{N}_5\text{O}_2$. Obtained: 364.1783.

4.3.3. N⁴,N⁶-bis (4-methoxybenzyl)-5-nitropyrimidine-4,6-diamine (5c): Yellow solid, yield 72%, mp: 158°C. δ H (400.45 MHz, CDCl₃): 9.57 (2H, bs, NH x2), 8.22 (1H, s, NCHN), 7.30 (8H, d, J=8.4 Hz, Ph x2), 6.90 (4H, d, J=8.4 Hz, Ph x2), 4.80 (4H, d, J= 5.2, -CH₂-Ph-OCH₃), 3.83 (6H, s, -CH₂-Ph-OCH₃ x2). δ C (100.70 MHz, CDCl₃): 159.45, 157.00, 129.32, 114.41, 112.90, 55.47, 45.26. ES+HRMS: Calculated M+H=396.1672 C₂₀H₂₂N₅O₄. Obtained: 396.1669.

4.3.4. N⁴,N⁶-diisopropyl-5-nitropyrimidine-4,6-diamine (5d): Yellow solid, yield 92%, (from product **2b**, the yield is 60%, from product **2c**, the yield is 72%) mp: 128°C. δ H (400.45 MHz, CDCl₃): 9.27 (2H, bs, NH x2), 8.11 (1H, s, NCHN), 4.51 (2H, m, -CH(CH₃)₂ x2), 1.30 (12H, d, J= 5.2, -CH(CH₃)₂ x2). δ C (100.70 MHz, CDCl₃): 159.42, 156.52, 112.41, 43.73, 22.79. ES+HRMS: Calculated M+H=240.1461 C₁₀H₁₈N₅O₂. Obtained: 240.1467.

4.3.5. N⁴,N⁶-di-tert-butyl-5-nitropyrimidine-4,6-diamine (5e): Yellow solid, yield 85%, mp: 109°C. δ H (400.45 MHz, CDCl₃): 9.55 (2H, bs, NH x2), 8.02 (1H, s, NCHN), 1.52 (18H, s, -(CH₃)₃ x2). δ C (100.70 MHz, CDCl₃): 157.87, 157.52, 113.16, 53.56, 29.23. ES+HRMS: Calculated M+H=240.1461 C₁₂H₂₂N₅O₂. Obtained: 240.1467.

4.3.6. N⁴,N⁶-bis(4-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5f): Yellow solid, yield 95%, mp: 160°C. δ H (400.45 MHz, CDCl₃): 9.57 (2H, bs, NH x2), 8.15 (1H, s, NCHN), 7.32 (4H, d, J= 8.4 Hz, Ph x2), 7.27 (4H, d, J= 8.4 Hz, Ph x2), 4.81 (4H, d, J= 6.00, -CH₂-Ph-Cl). δ C (100.70 MHz, CDCl₃): 159.69, 157.32, 135.94, 133.78, 129.22, 129.11, 113.14, 44.84. ES+HRMS: Calculated M+H=404.0681 C₁₈H₁₆N₅O₂Cl₂. Obtained: 404.0716.

4.3.7. N⁴,N⁶-bis(4-bromobenzyl)-5-nitropyrimidine-4,6-diamine (5g): Yellow solid, yield 68%, mp: 175°C. δ H (400.45 MHz, CDCl₃): 9.55 (2H, bs, NH x2), 8.15 (1H, s, NCHN), 7.47 (4H, d, J= 8.4 Hz, Ph x2), 7.21 (4H, d, J= 8.4 Hz, Ph x2), 4.78 (4H, d, J= 6.00, -CH₂-Ph-Br). δ C (100.70 MHz, CDCl₃): 159.99, 157.48, 136.59, 132.04, 129.53, 121.78, 44.77. ES+HRMS: Calculated M+H=491.9671 C₁₈H₁₆N₅O₂Br₂. Obtained: 491.9667.

4.3.8. N⁴,N⁶-bis(4-bromophenethyl)-5-nitropyrimidine-4,6-diamine (5h): Yellow solid, yield 30%. δ H (400.45 MHz, CDCl₃): 9.36 (2H, bs, NH x2), 8.14 (1H, s, NCHN), 7.44 (4H, d, J= 6.8 Hz, Ph x2), 7.12 (4H, d, J= 6.8 Hz, Ph x2), 3.87 (4H, q, J= 5.6, 6.0 -CH₂-CH₂-Ph-Br x2), 2.93 (4H, t, J= 5.6, -CH₂-CH₂-Ph-Br x2). δ C (100.70 MHz, CDCl₃): 159.30, 157.29, 137.59, 132.26, 130.88, 121.10, 43.32, 36.17. ES+HRMS: Calculated M+H=519.9984 C₂₀H₂₀N₅O₂Br₂. Obtained: 519.9981.

4.3.9. N⁴,N⁶-bis(3-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5i): White solid, yield 93%, mp: 138°C. δ H (400.45 MHz, CDCl₃): 9.57 (2H, bs, NH x2), 8.15 (1H, s, NCHN), 7.32-7.21 (8H, m, , Ph x2), 4.82 (4H, d, J= 6.00, -CH₂-Ph-Cl). δ C (100.70 MHz, CDCl₃): 159.97, 157.50, 139.62,

134.81, 130.20, 128.06, 127.92, 125.89, 113.29, 44.81. ES+HRMS: Calculated $M+H=404.0681$ $C_{18}H_{16}N_5O_2Cl_2$. Obtained: 404.0719.

4.3.10. N^4,N^6 -bis(2-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5j): Yellow solid, yield quantitative, mp: 196°C. δH (400.45 MHz, $CDCl_3$): 9.68 (2H, bs, NH x2), 8.16 (1H, s, $NCHN$), 7.43-7.40 (4H, m, Ph), 7.29-7.22 (4H, m, Ph), 4.94 (4H, d, $J=6.00$, $-CH_2-Ph-Cl$). δC (100.70 MHz, $CDCl_3$): 159.76, 157.47, 134.99, 133.99, 130.12, 129.86, 129.31, 127.15, 113.32, 43.39. ES+HRMS: Calculated $M+H=404.0681$ $C_{18}H_{16}N_5O_2Cl_2$. Obtained: 404.0684.

4.3.11. N^4,N^6 -diethyl 5-nitropyrimidine-4,6-diamine (5k): Yellow solid, yield 89%, mp: 85°C. δH (400.45 MHz, $CDCl_3$): 9.29 (2H, bs, NH x2), 8.09 (1H, s, $NCHN$), 3.63 (4H, m, $-CH_2-$ x2), 1.28 (6H, t, $J=7.3$, $-CH_3$ x2). δC (100.70 MHz, $CDCl_3$): 159.59, 157.16, 112.59, 36.38, 14.43. ES+HRMS: Calculated $M+H=212.1147$ $C_8H_{14}N_5O_2$. Obtained: 212.1148.

4.3.12. N^4,N^6 -diisobutyl 5-nitropyrimidine-4,6-diamine (5l): Yellow solid, yield 83%, mp: 78°C. δH (400.45 MHz, $CDCl_3$): 9.43 (2H, bs, NH x2), 8.05 (1H, s, $NCHN$), 3.42 (4H, dd, $J=1.2$ and 5.6 , $-CH_2-$ x2), 1.94 (2H, m, CH x2), 0.97 (12H, d, $J=3.9$, $-(CH_3)_2$ x2). δC (100.70 MHz, $CDCl_3$): 159.75, 157.67, 112.87, 48.99, 28.28, 20.29. ES+HRMS: Calculated $M+H=268.1774$ $C_{12}H_{22}N_5O_2$. Obtained: 268.1774.

5a and **5d** products have been synthesised from the three alcohols **2a**, **2b** and **2c** with yields of 55%, 33% and 82% for **5a** and 92%, 60%, and 72% for **5d**.

4.4. Compounds 6 (a-c):

These compounds were synthesised following the procedure described elsewhere by us^{10, 12}.

4.5. Computational details

Gaussian16 Rev. C.01 has been employed for all the calculations²⁵. The reagent and transition state (TS) geometries were optimised at the M06-2X level of theory²⁶, along with the 6-31G* basis set. M06-2X is especially recommended for thermodynamic and kinetic calculations²⁷. An initial conformational analysis was carried out for the reagents to obtain the lowest energy conformation. The nature of the stationary points was assessed by means of the normal vibration frequencies calculated from the analytical second derivatives of the energy. First-order saddle points, which are related to transition states, must show an imaginary value for the hessian eigenvector primarily describing the product formation step, whereas the real minima of the potential energy hypersurface, which are related to stable species, have to show positive real

values for all of the vibrational frequencies. In addition, intrinsic reaction coordinate calculations at the M06-2X/6-31G** levels were carried out to assess each reaction pathway. Solvent effects (dichloromethane) were included by using the continuum solvation model CPCM²⁰, which describes the quantum mechanical charge density of a solute molecule interacting with a continuum description of the solvent.

Rate constants (k_i) were calculated through the thermodynamic formulation of the Transition State Theory (TST):

$$k_i = \left(\frac{k_B T}{h} \right) \exp \exp \left(- \frac{\Delta G^\ddagger}{RT} \right) \quad (5)$$

where k_B and h are the Boltzmann and Planck constants, respectively. Gibbs free energy barriers ($\Delta G^\ddagger$) were computed at 298.15 K.

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6. ASSOCIATED CONTENT

*SI Supporting Information NMR and HRMS spectra for all new compounds

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Notes: The authors declare no competing financial interest.

8. ACKNOWLEDGMENTS

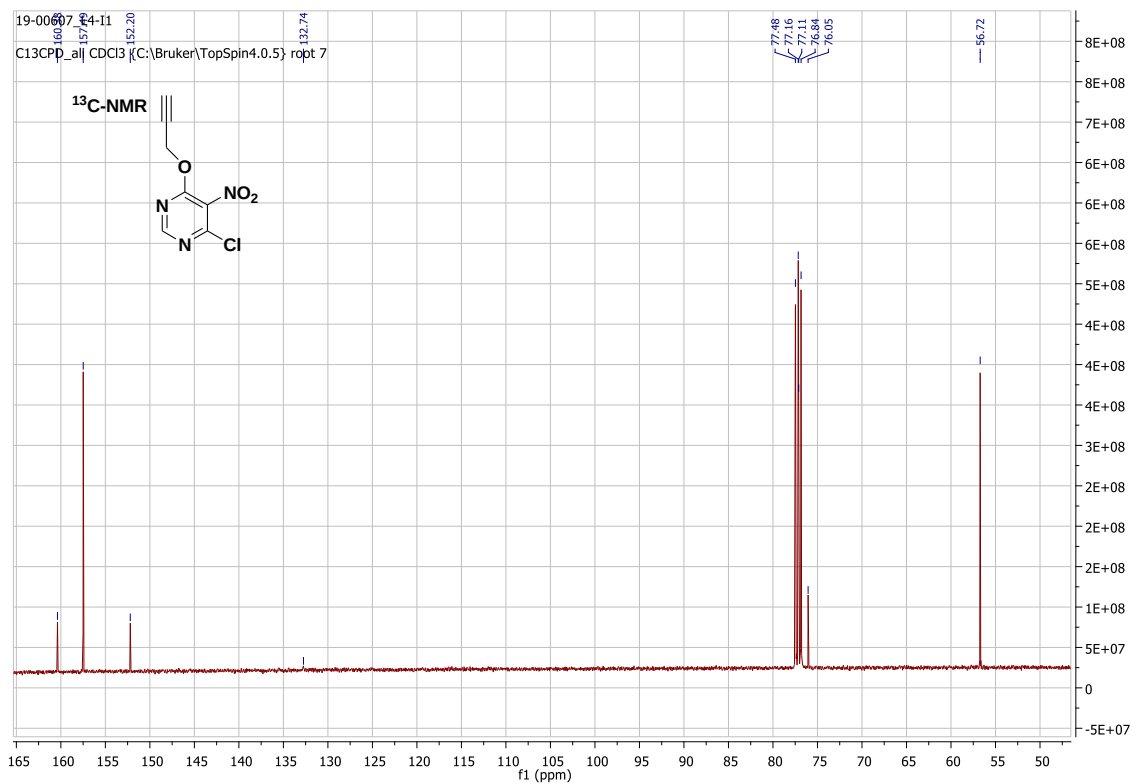
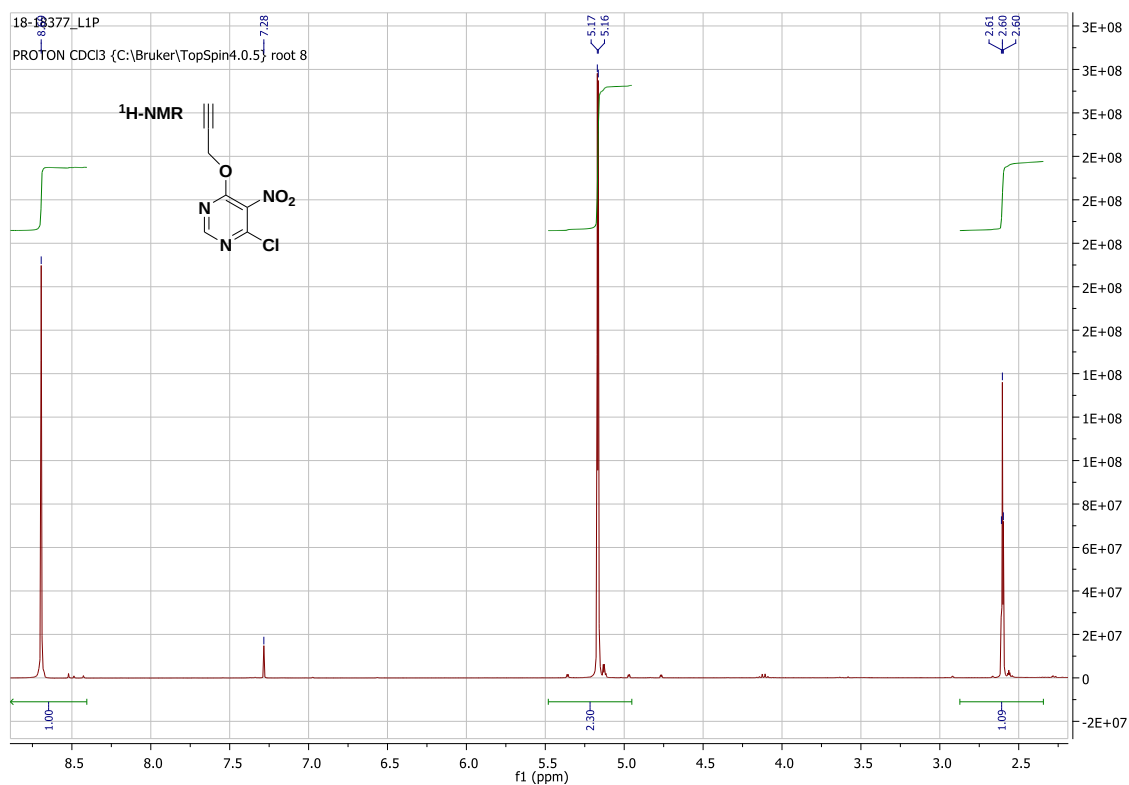
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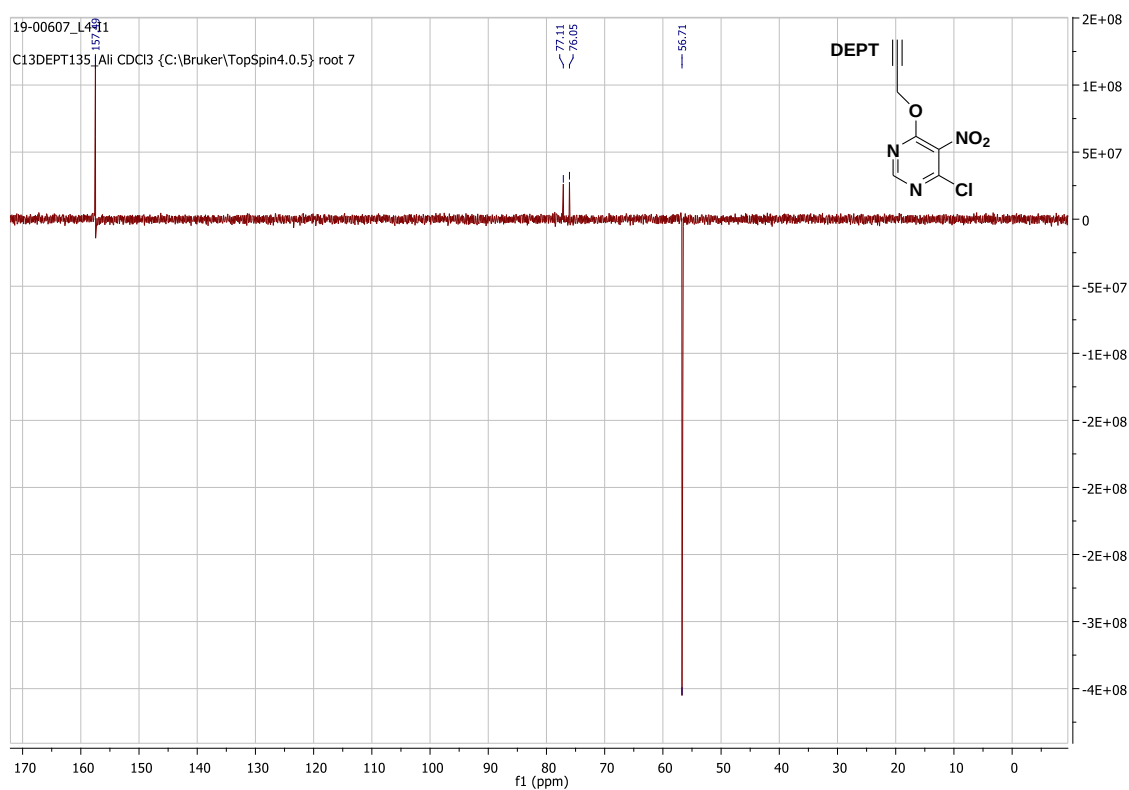
Supporting Information

Symmetric 4,6-Dialkylamine-5-Nitro/Aminopyrimidines: Theoretical Explanation of Why Aminolysis of Alkoxy Groups Is Favoured over That of Chlorine in Nitro-Activated Pyrimidines.

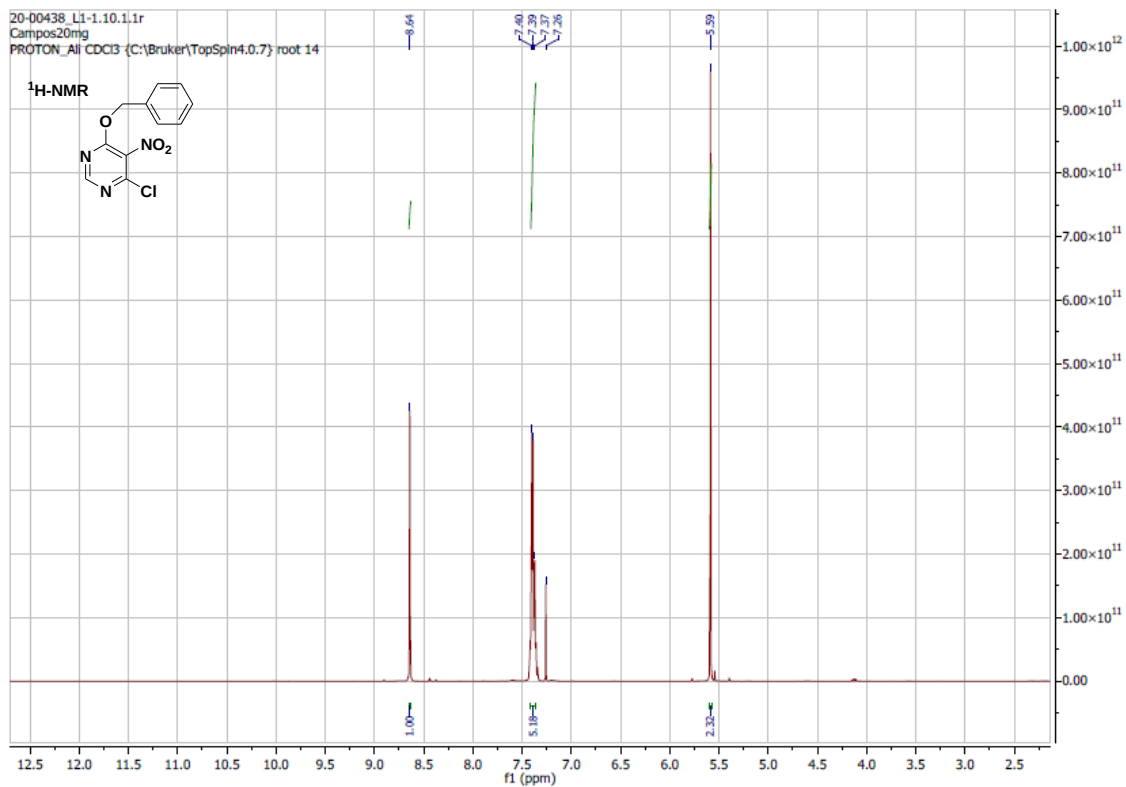
Authors: Laura Córdoba Gómez, Alvaro Lorente-Macias, María José Pineda de las Infantas y Villatoro, Andrés Garzón-Ruiz* and Juan J. Diaz-Mochon*

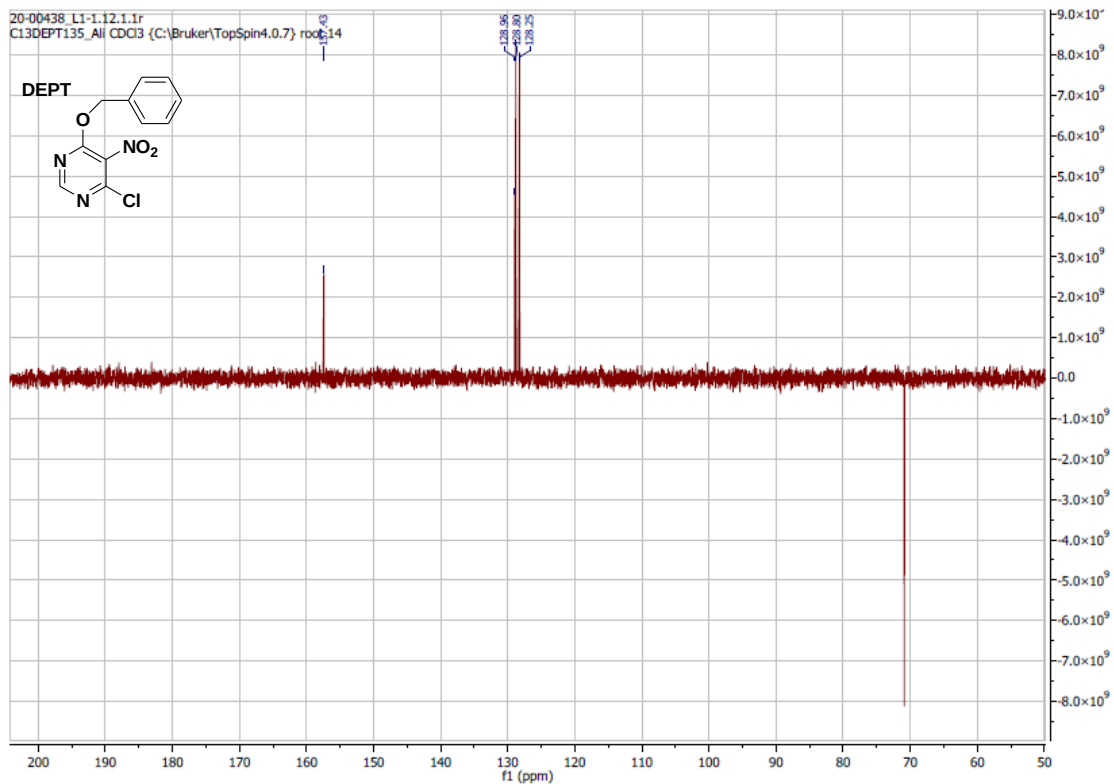
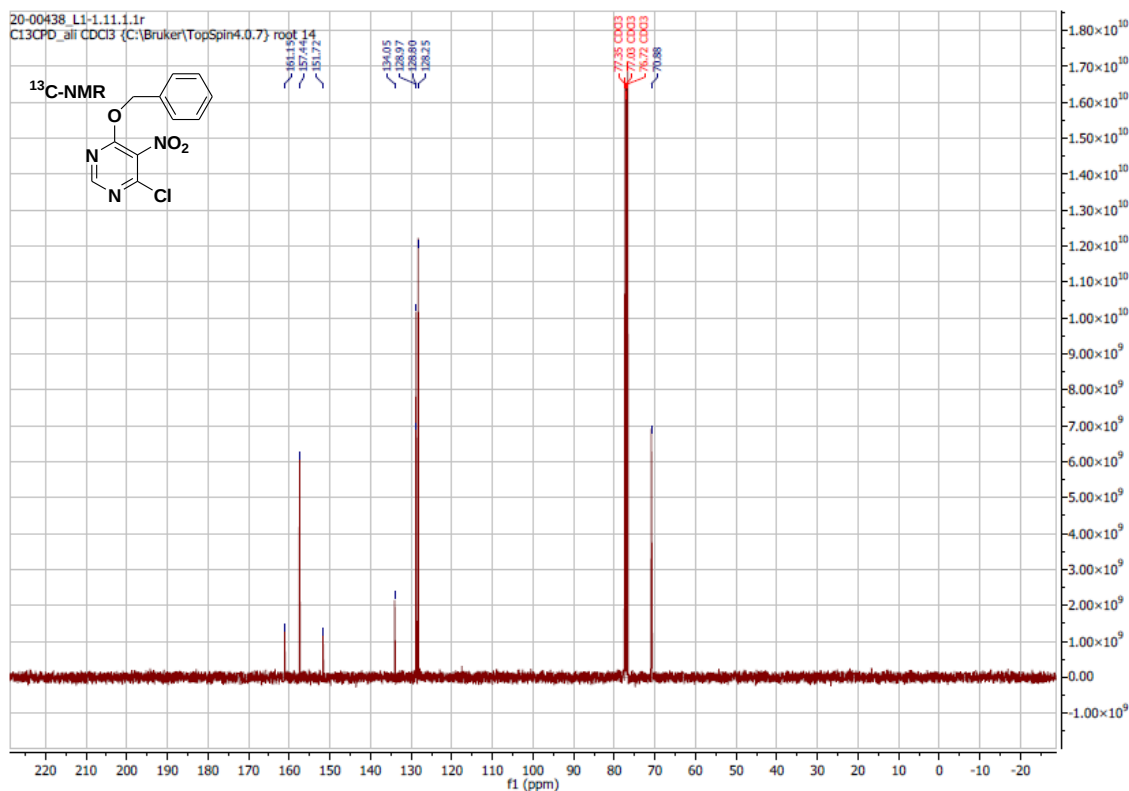
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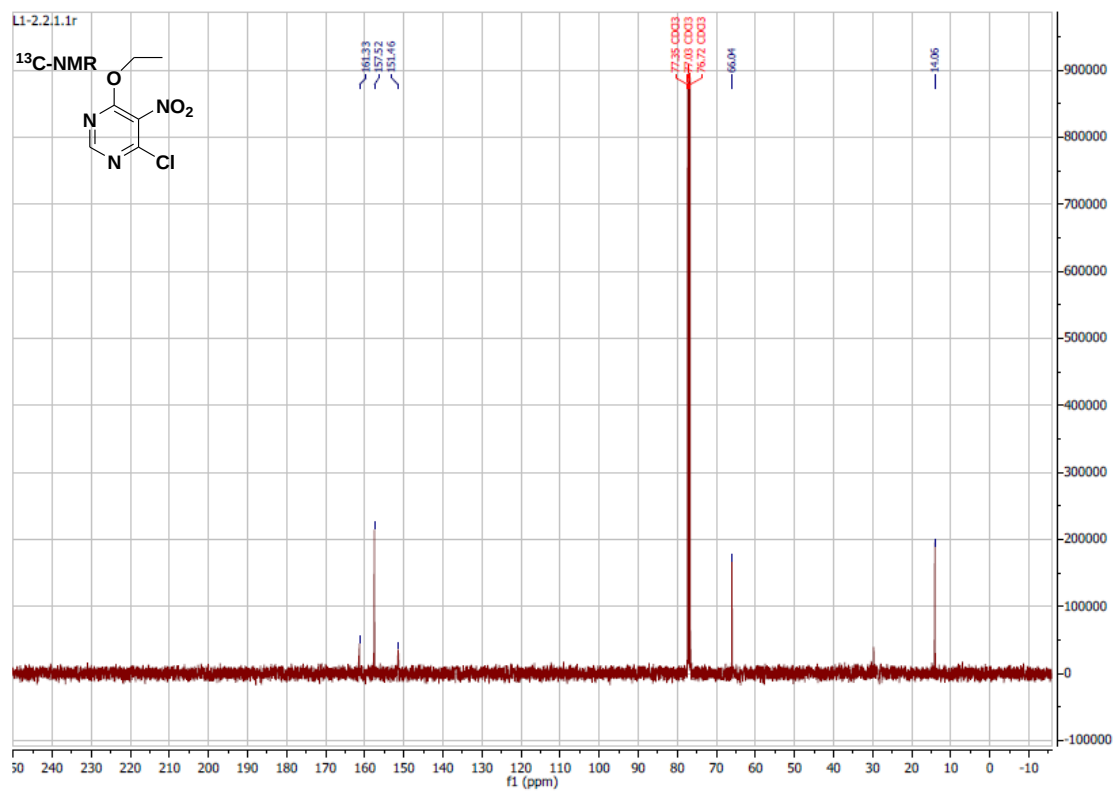
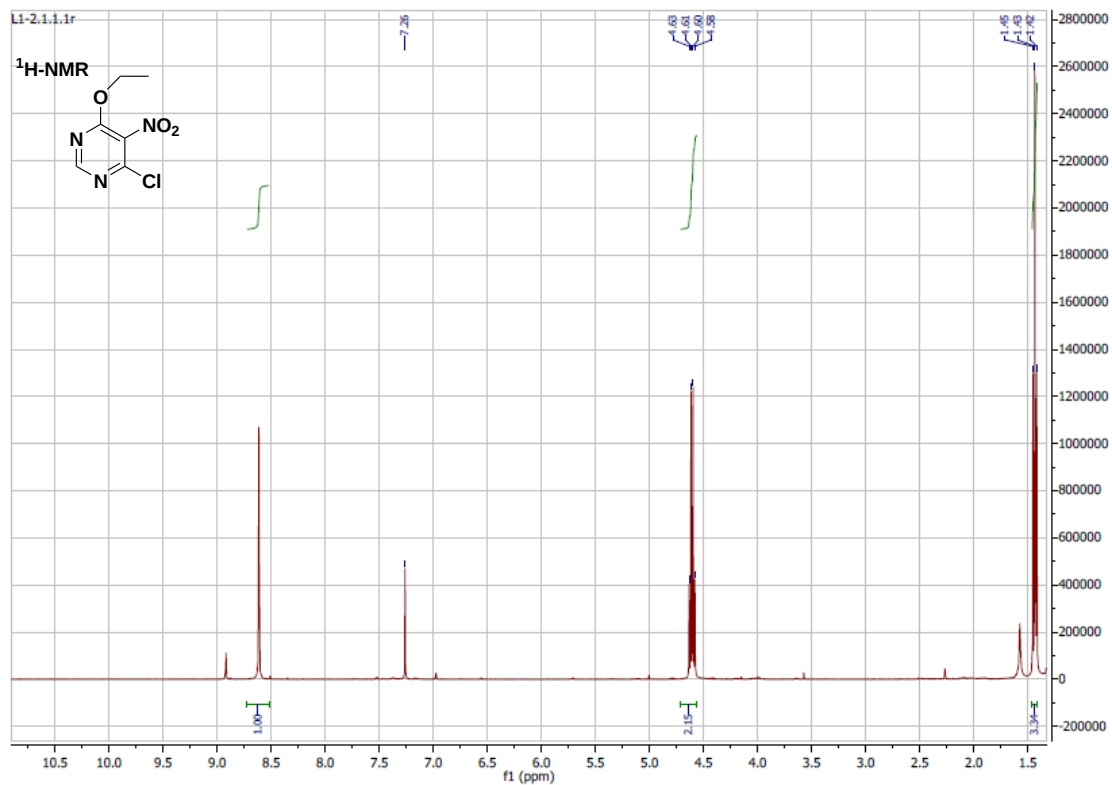


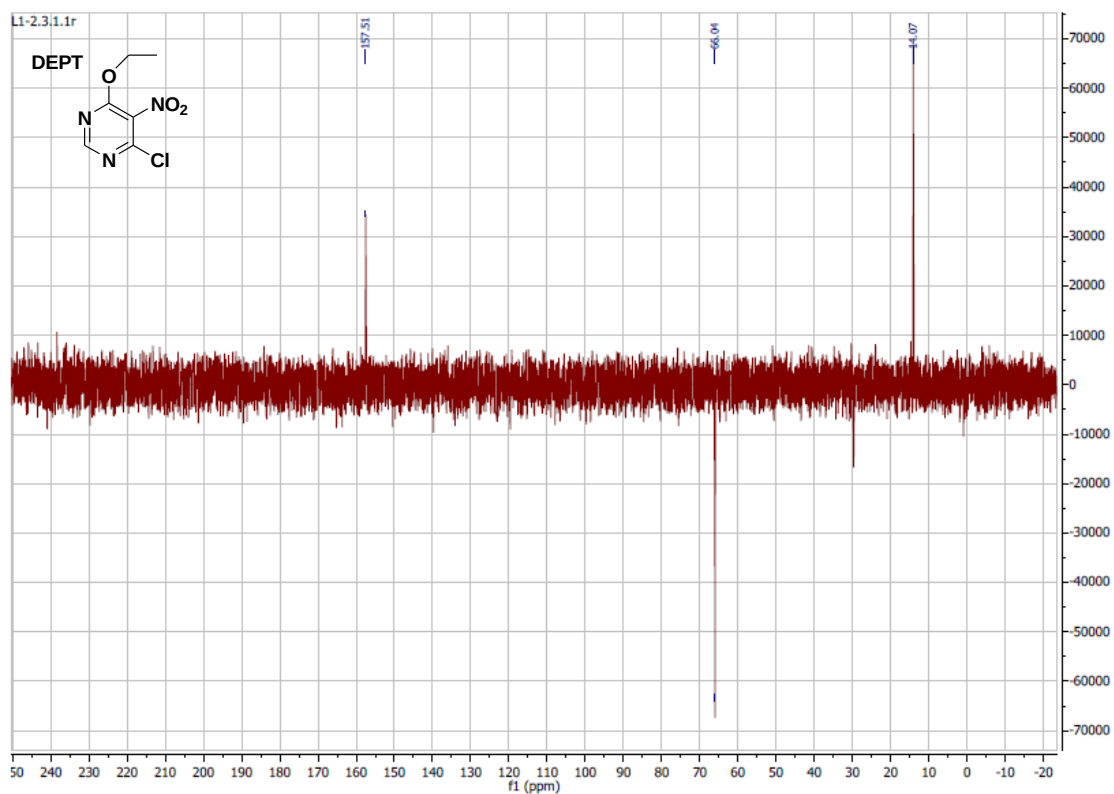
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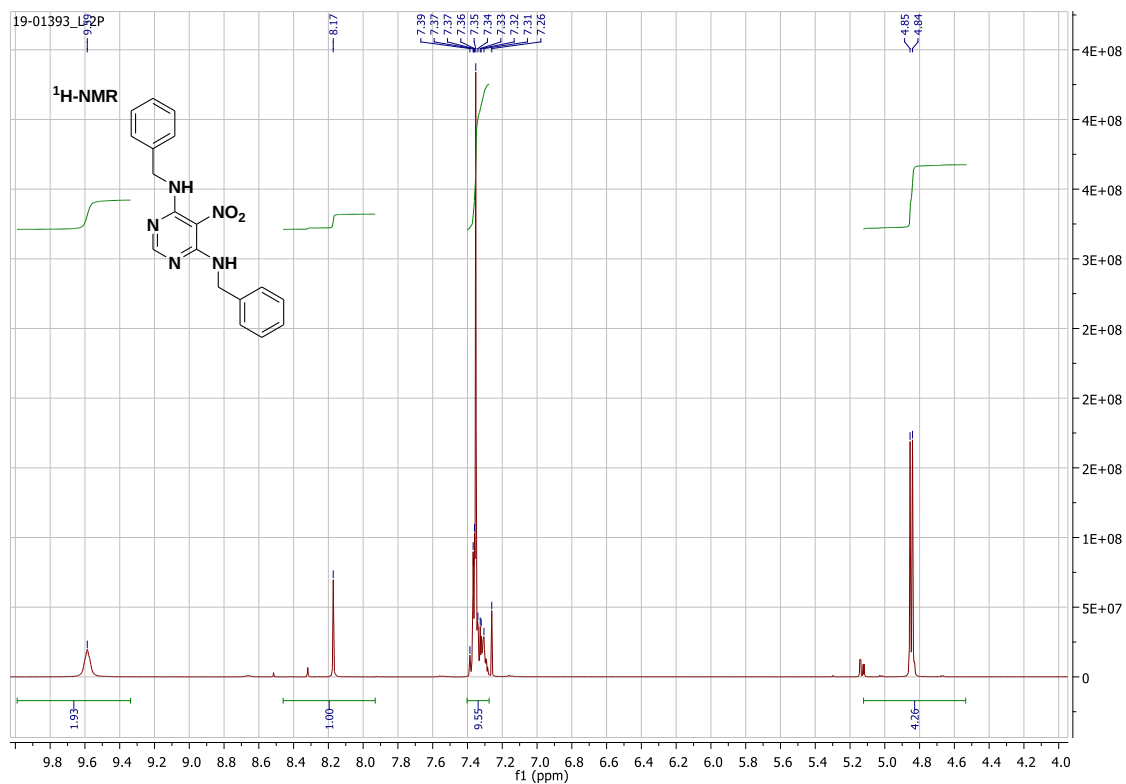


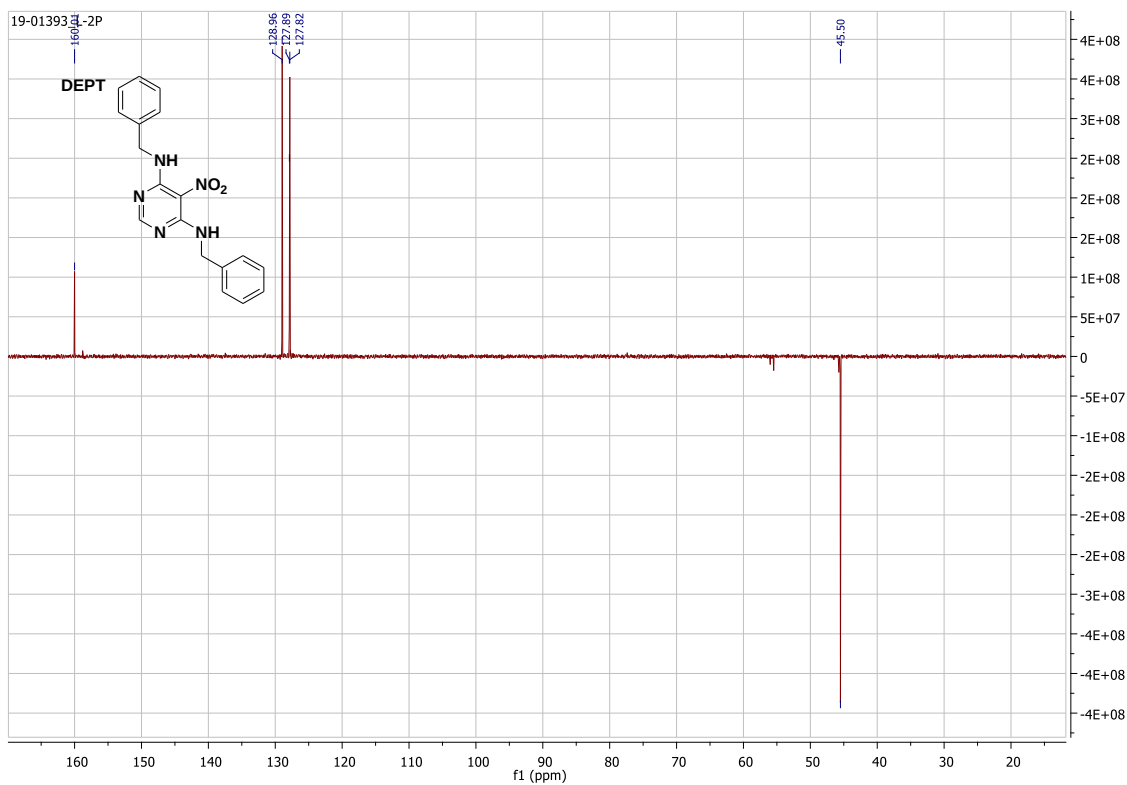
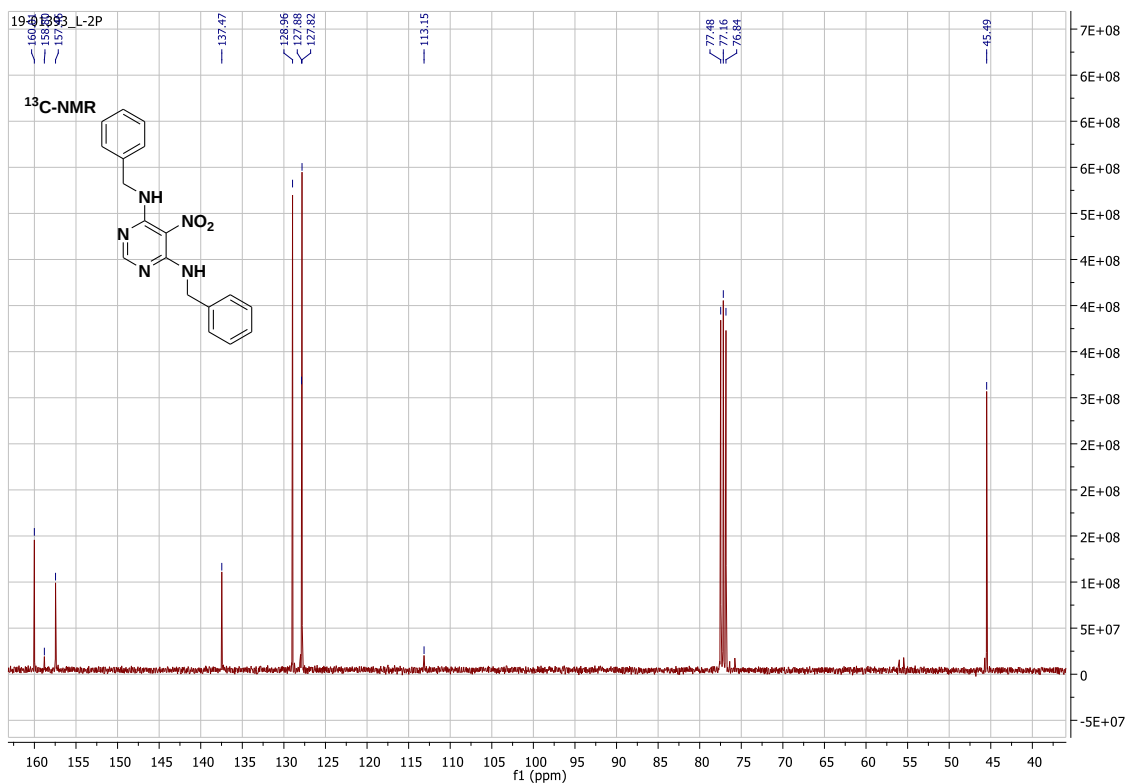
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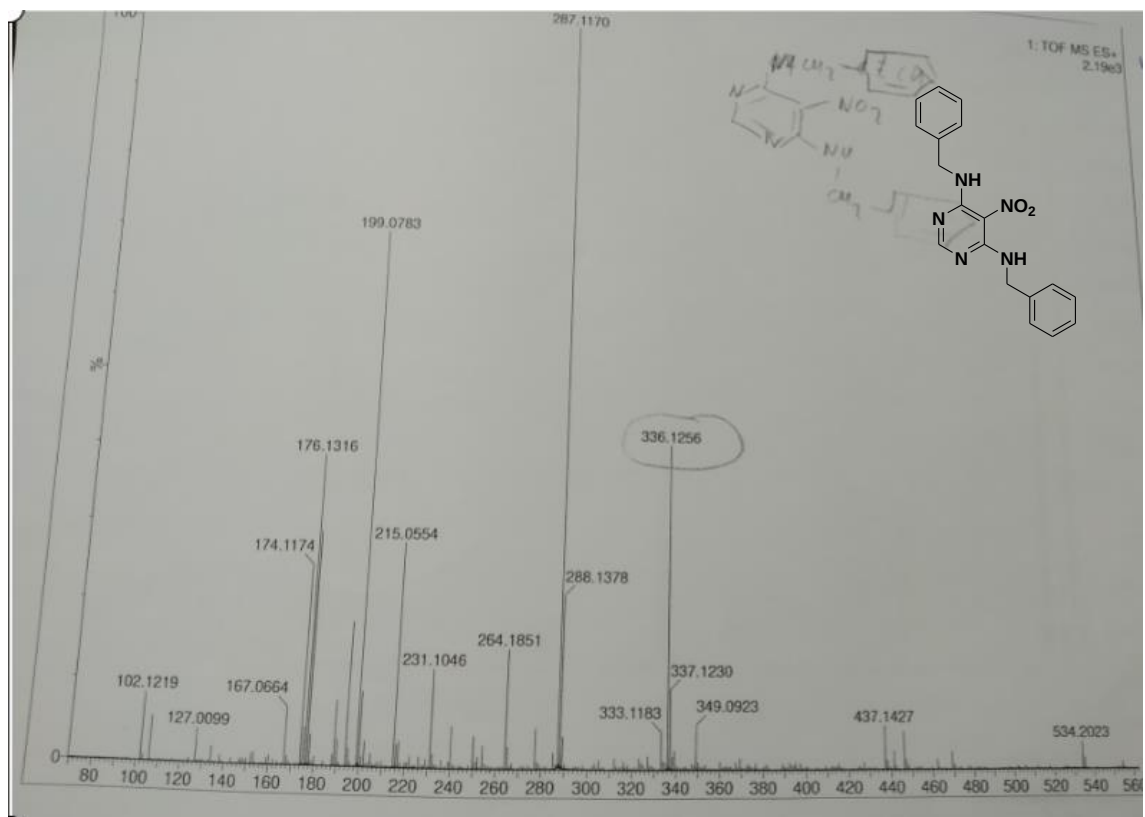
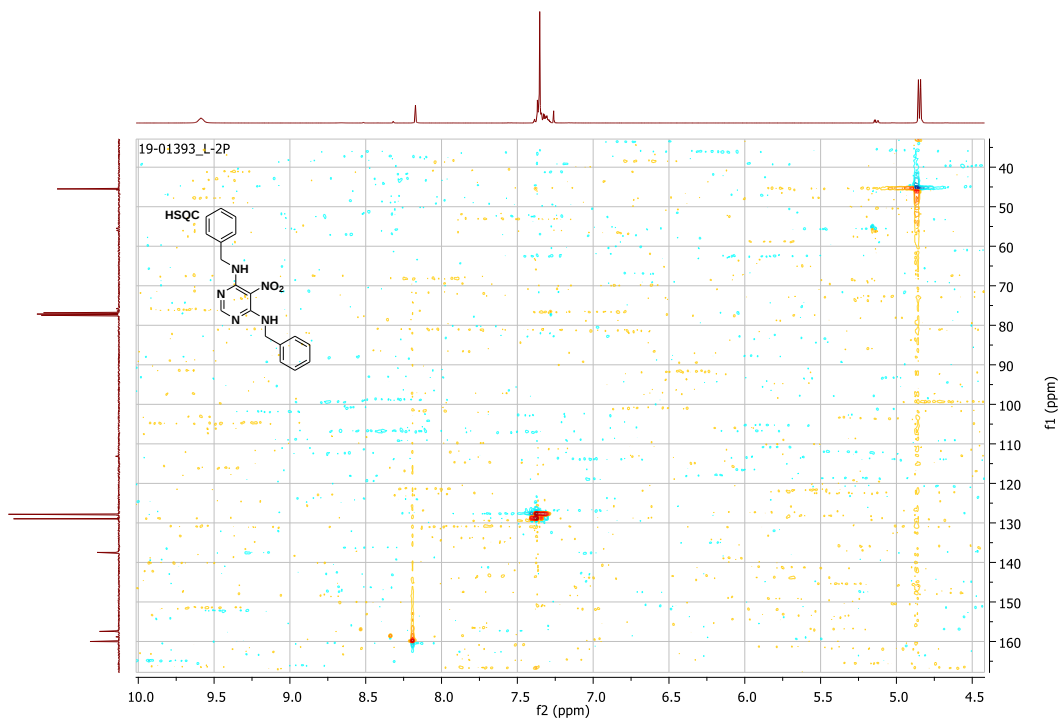




- **N⁴,N⁶-dibenzyl-5-nitropyrimidine-4,6-diamine (5a):**







Elemental Composition Report

Page 1

Single Mass Analysis

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Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

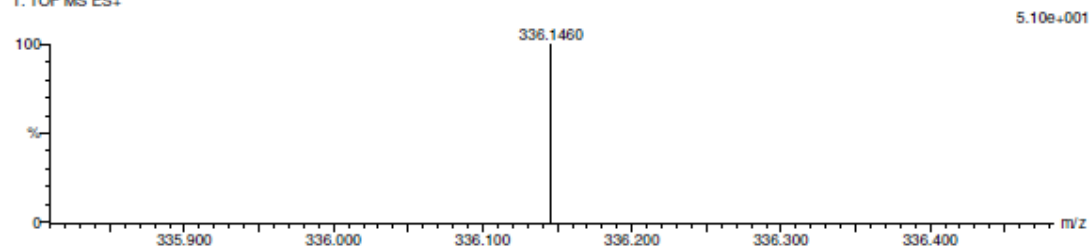
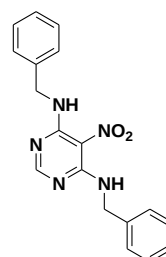
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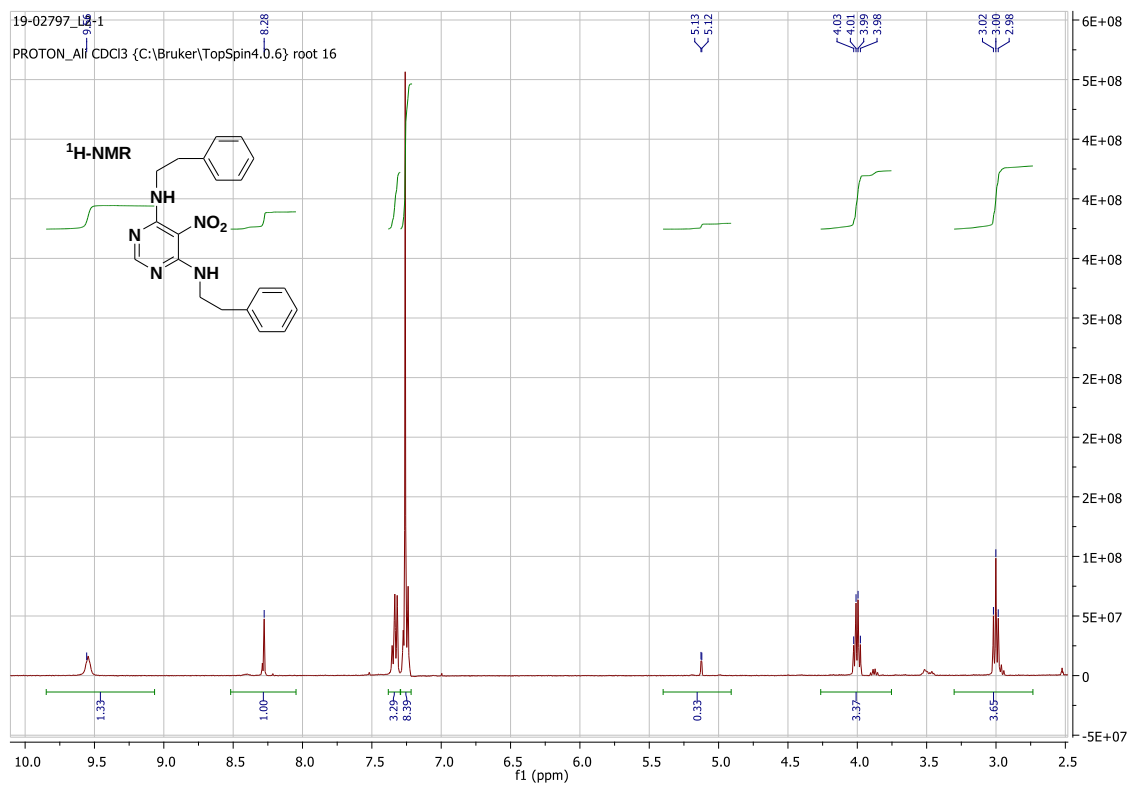
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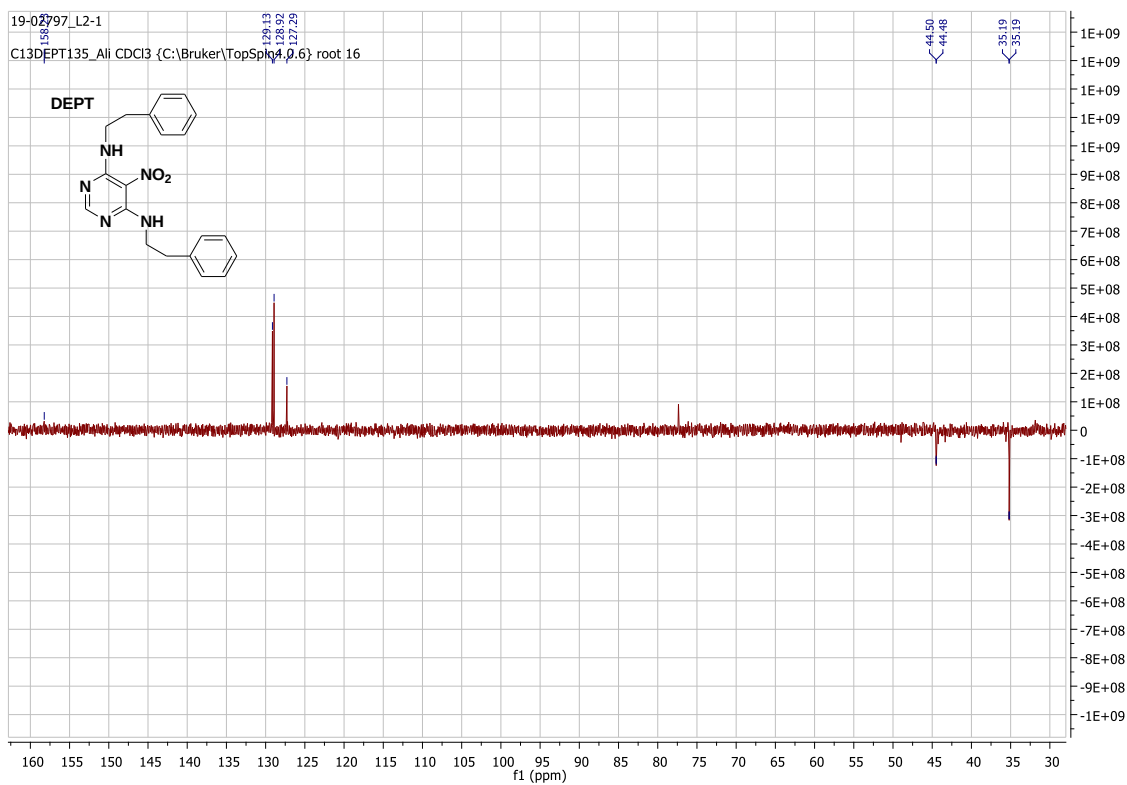
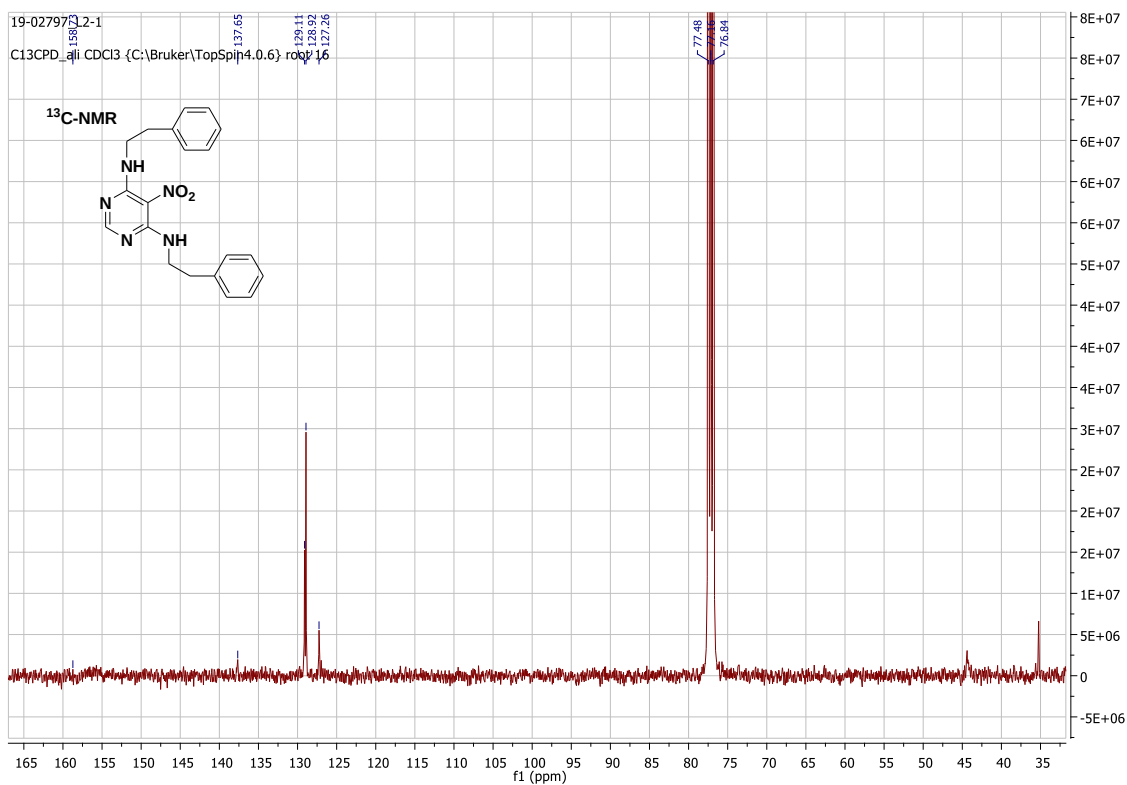
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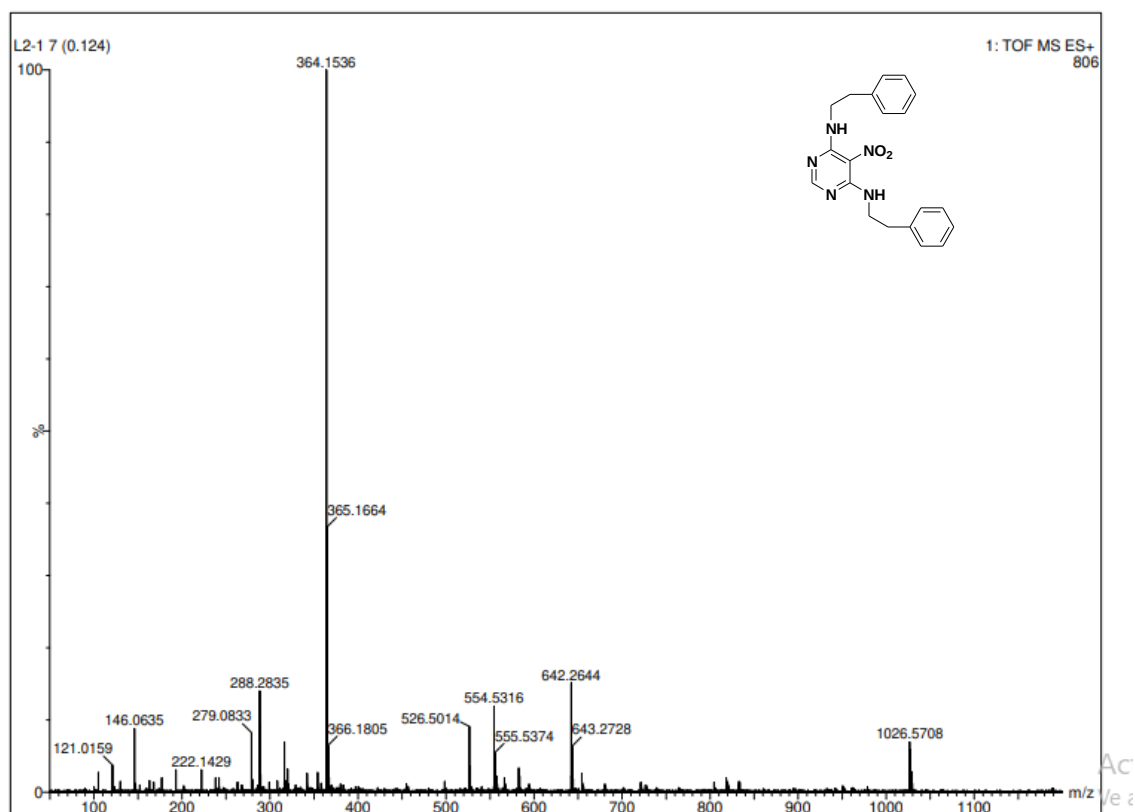


Minimum:				-1.5				
Maximum:		5.0	10.0	50.0				
Mass	Calc. Mass	mDa	PPM	DBE	i-FIT	i-FIT (Norm)	Formula	
336.1460	336.1461	-0.1	-0.3	12.5	16.7	0.7	C18 H18 N5 O2	
	336.1436	2.4	7.1	9.5	16.6	0.6	C16 H19 N5 O2 Na	

- 5-nitro-N⁴,N⁶- diphenethylpyrimidine-4,6-diamine (5b):







Elemental Composition Report

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

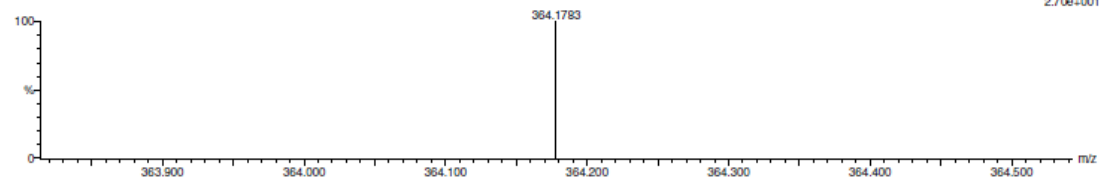
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Elements Used:

C: 0-20 H: 0-1000 N: 0-8 O: 0-5 Na: 0-1

L2-1-2 7 (0.141) AM (Top, 1, H1, 5000.0, 0.00, 1.00)

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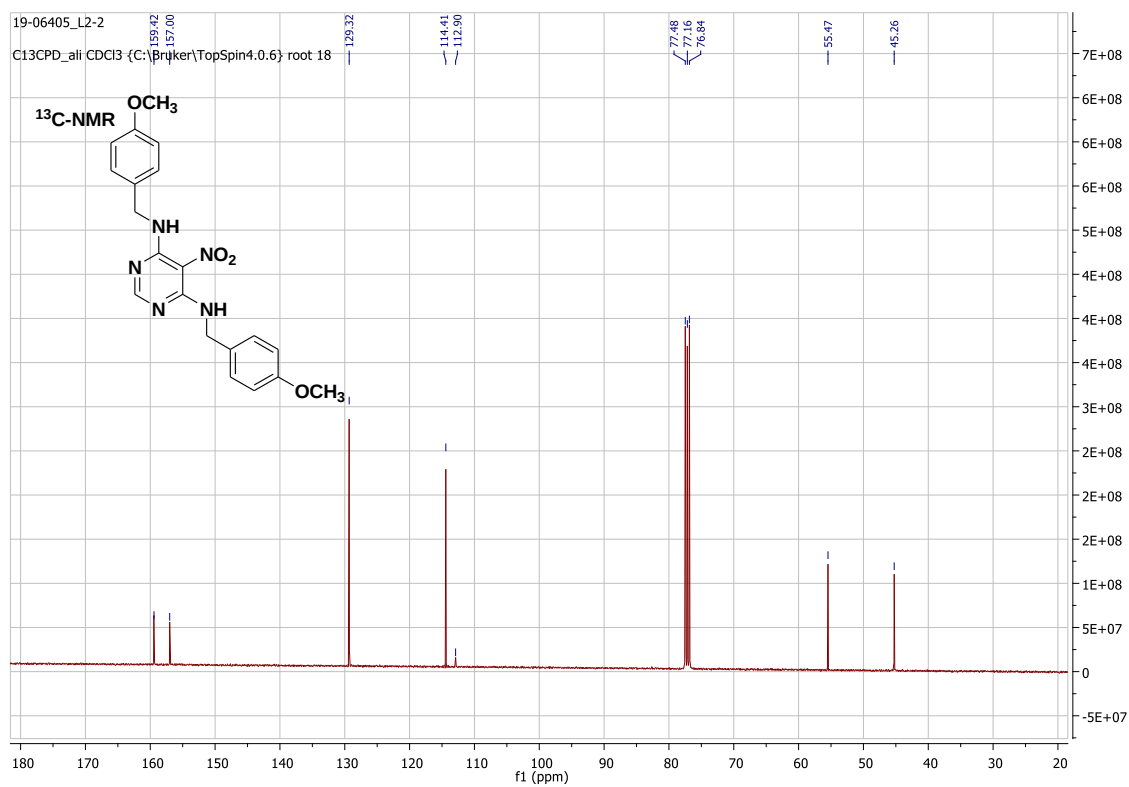
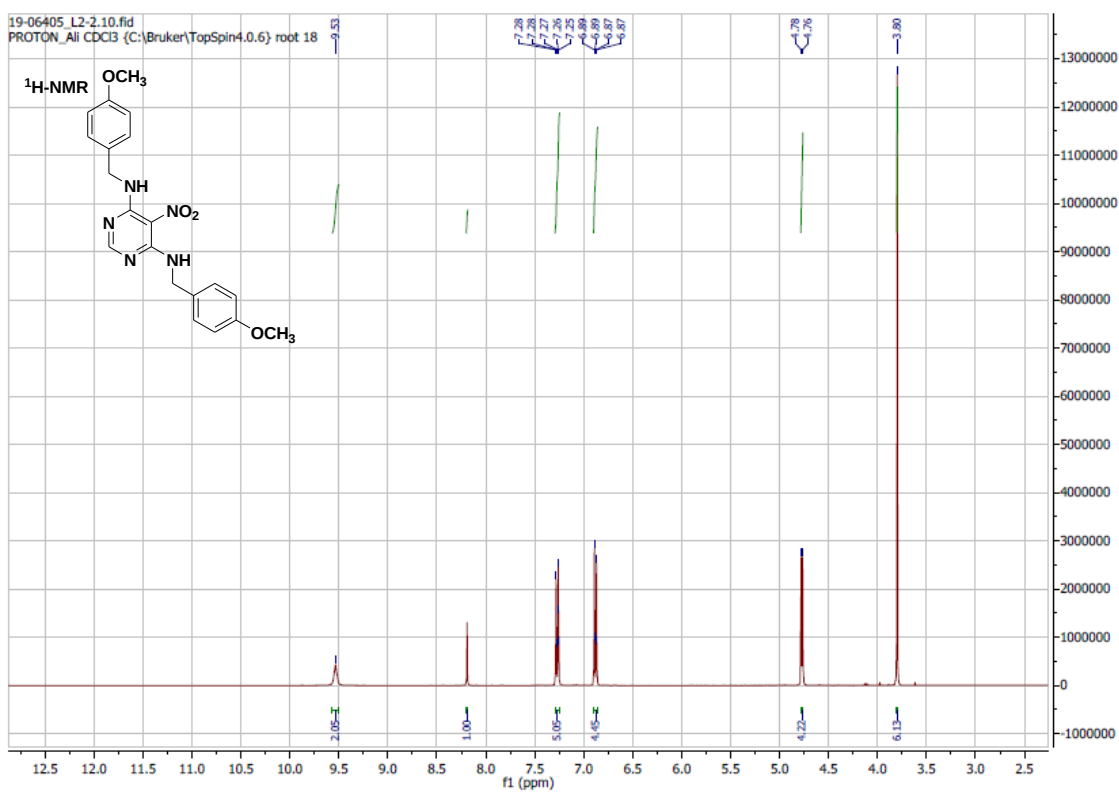
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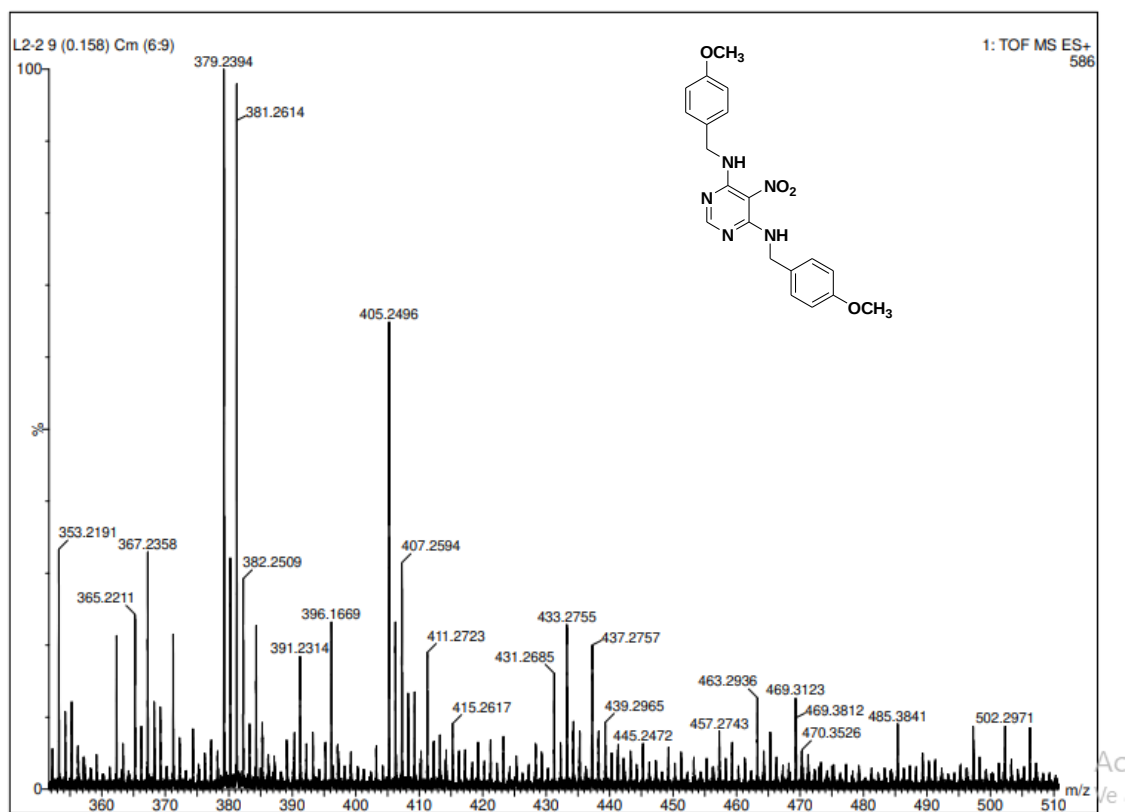
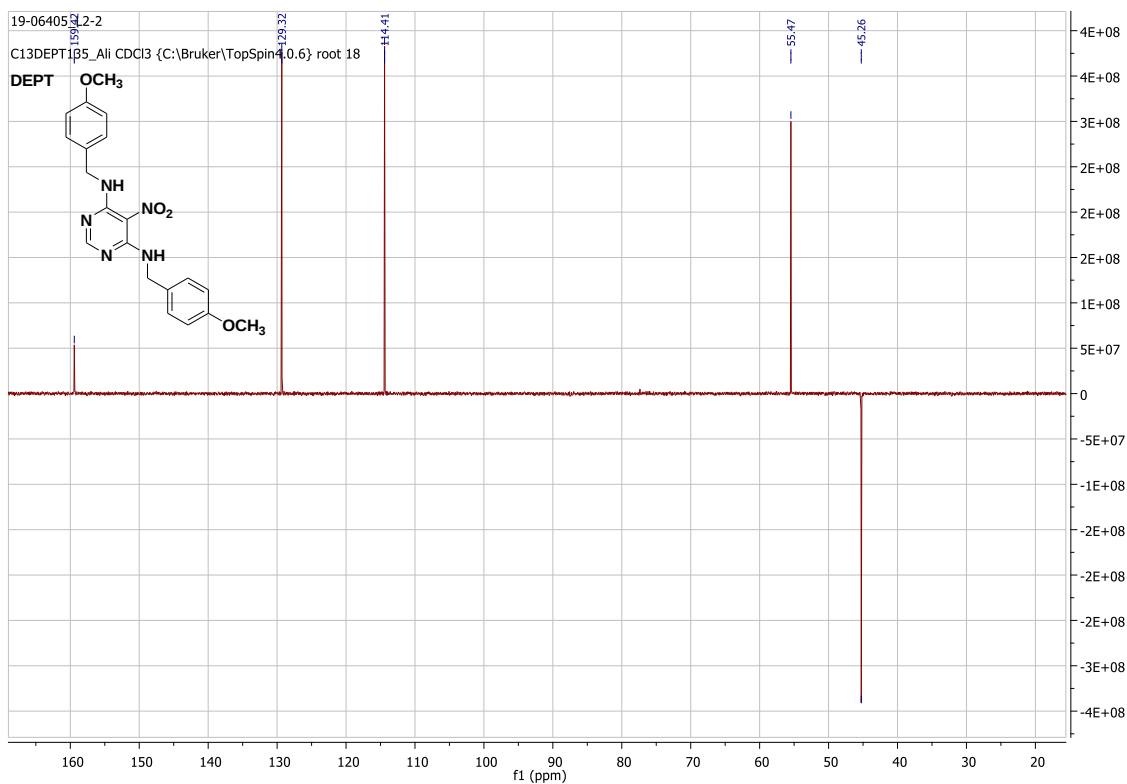
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula
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	364.1749	3.4	9.3	9.5	14.7	0.7	C18 H23 N5 O2 Na

Page 1

2.70e+001

- **N⁴,N⁶-bis (4-methoxybenzyl)-5-nitropyrimidine-4,6-diamine (5c):**





Elemental Composition Report

Page 1

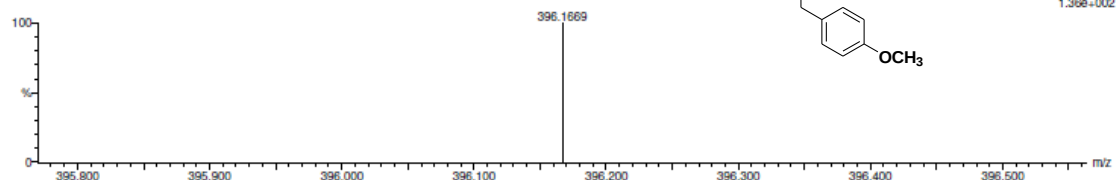
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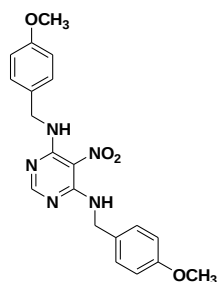
Element prediction: Off

Number of isotope peaks used for i-FIT = 3

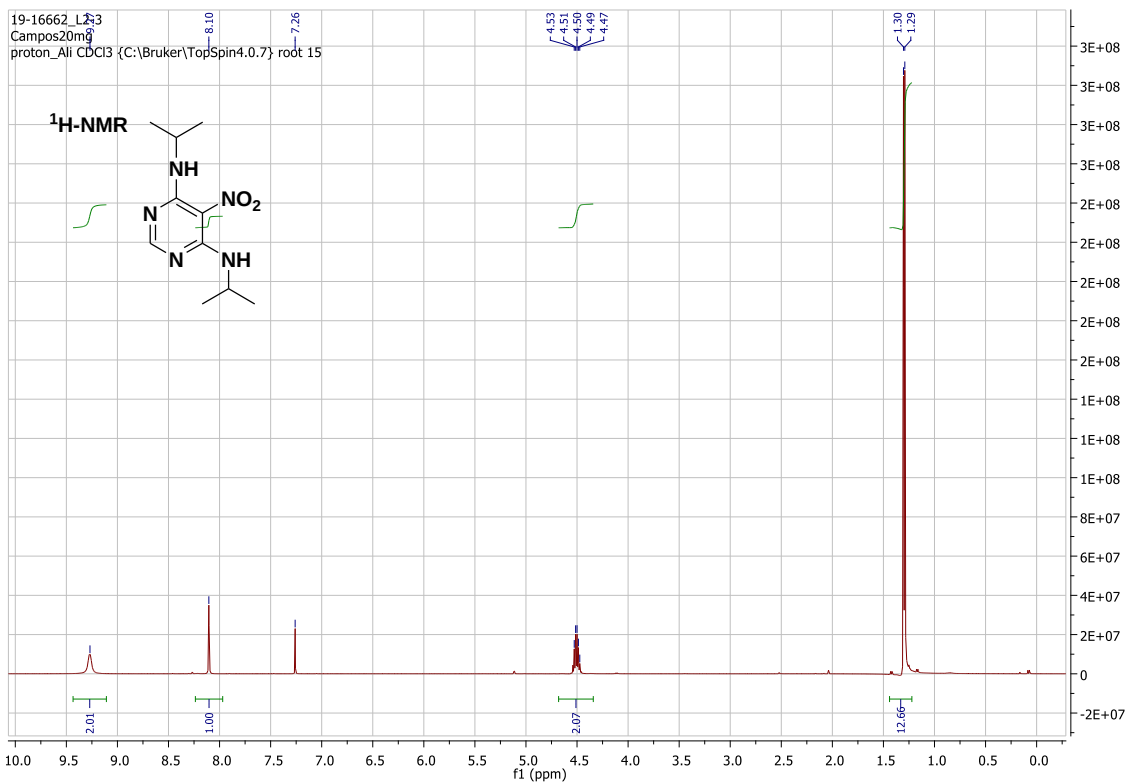
Monoisotopic Mass, Even Electron Ions
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Elements Used:
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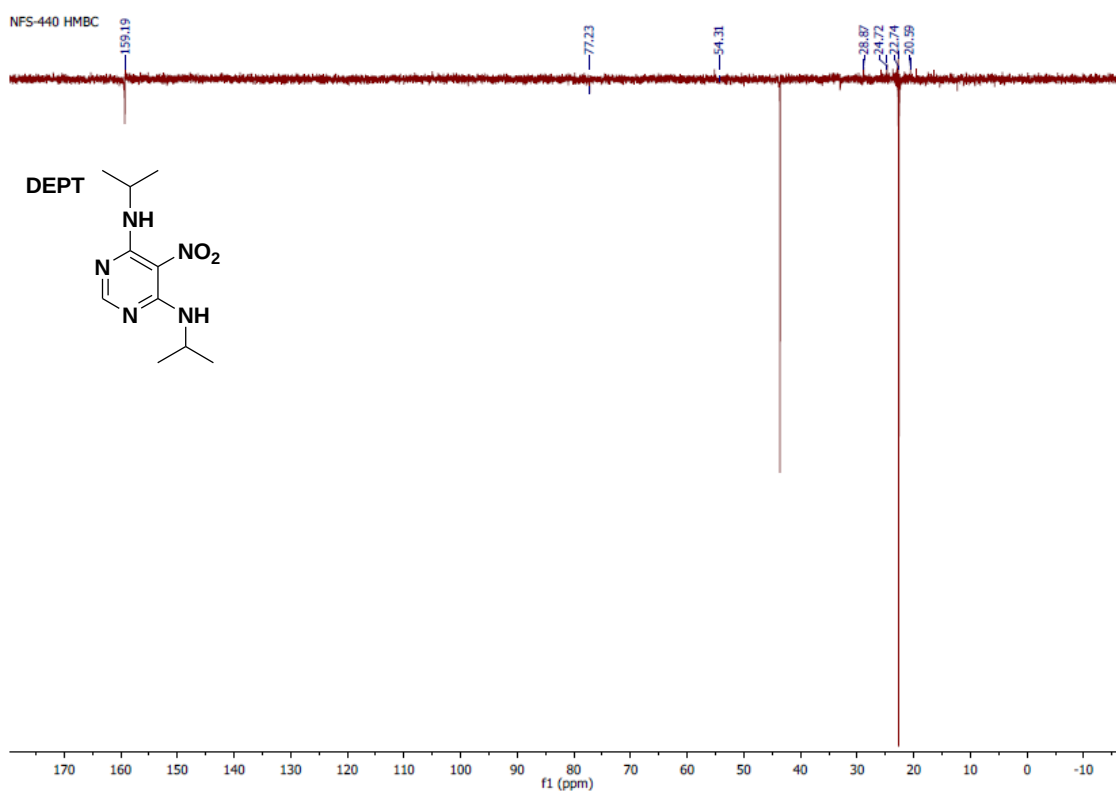
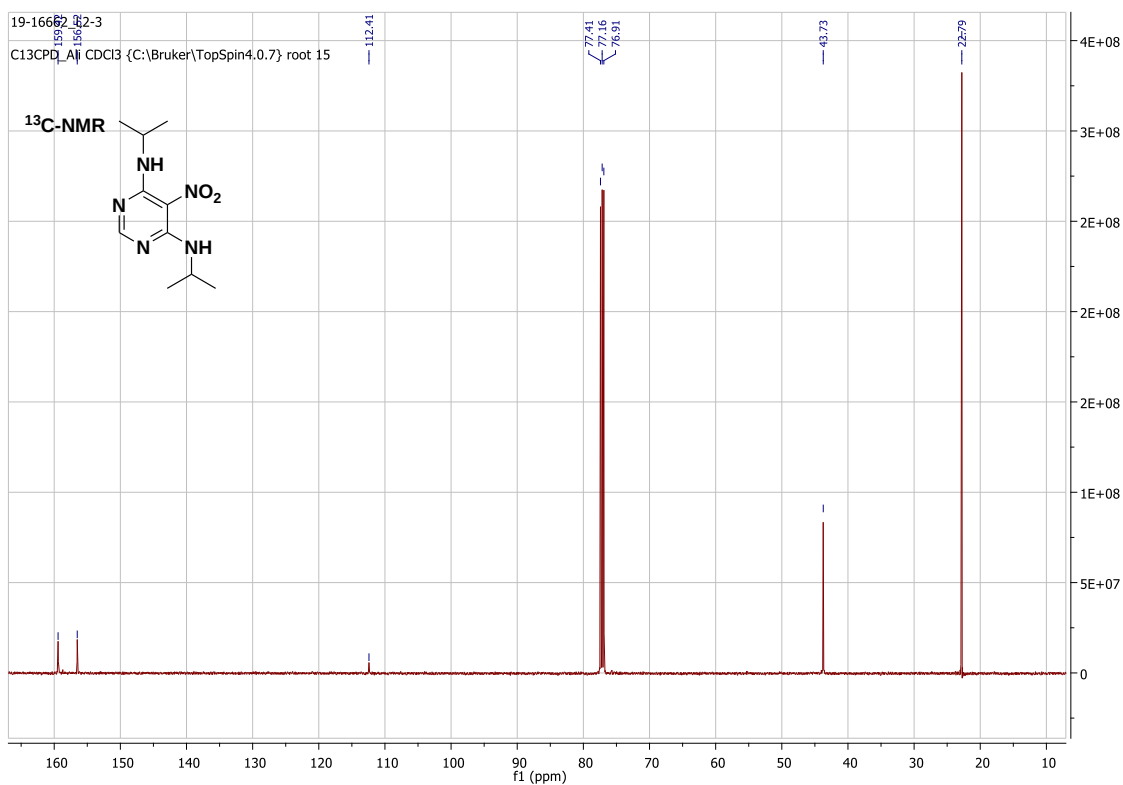


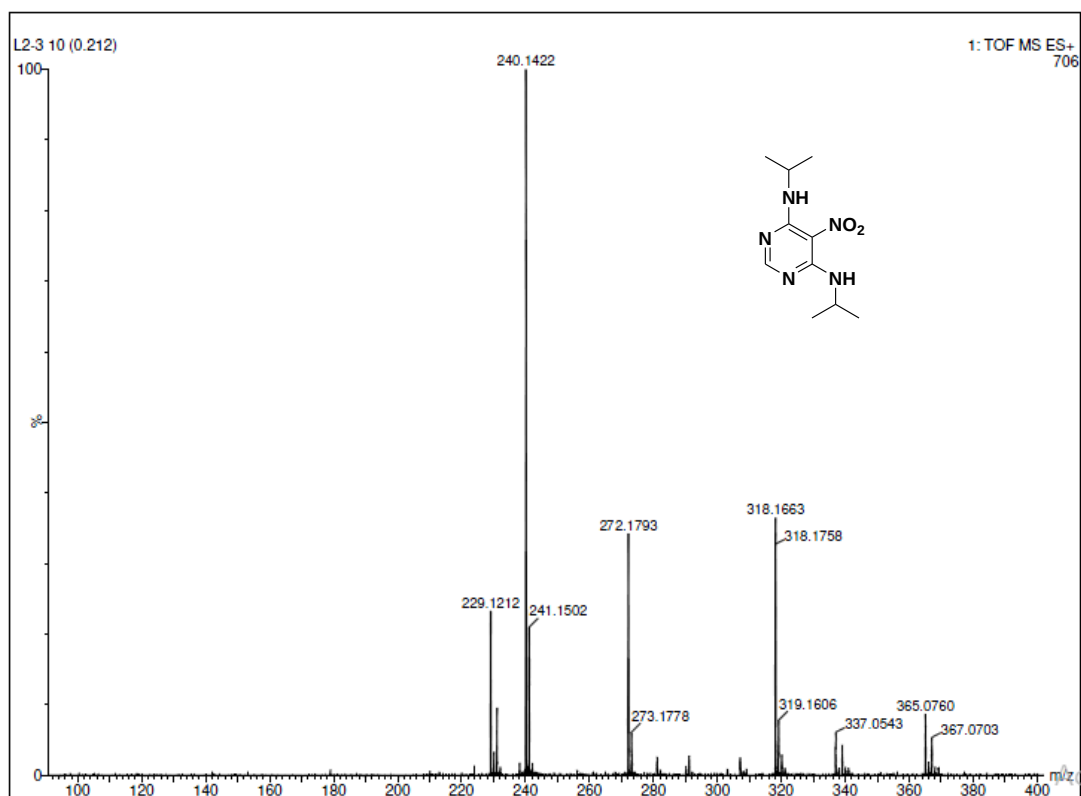
Minimum:									
Maximum:	5.0	10.0	-1.5						
			50.0						
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula		
396.1669	396.1672	-0.3	-0.8	12.5	19.2	0.6	C20	H22	N5 O4
	396.1648	2.1	5.3	9.5	19.4	0.8	C18	H23	N5 O4 Na



- **N⁴,N⁶-diisopropyl-5-nitropyrimidine-4,6-diamine (5d):**







Elemental Composition Report

Single Mass Analysis

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Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

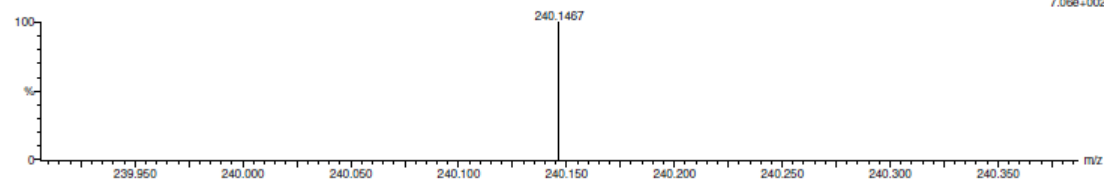
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Elements Used:

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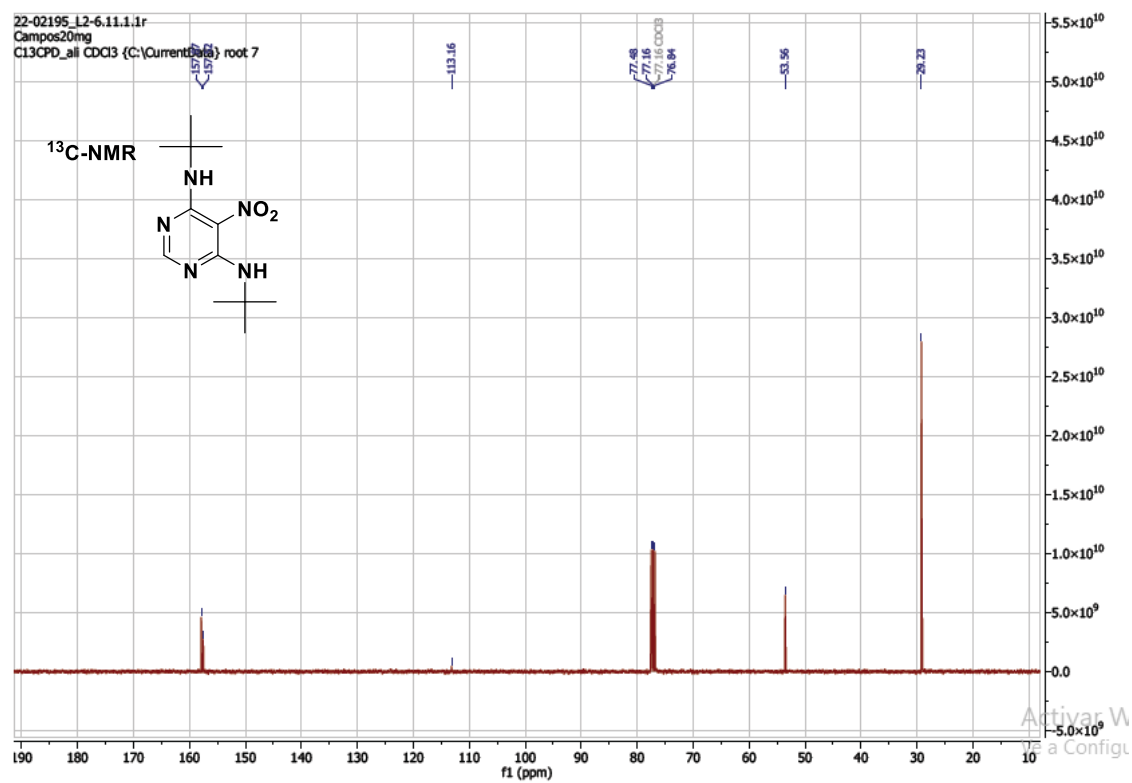
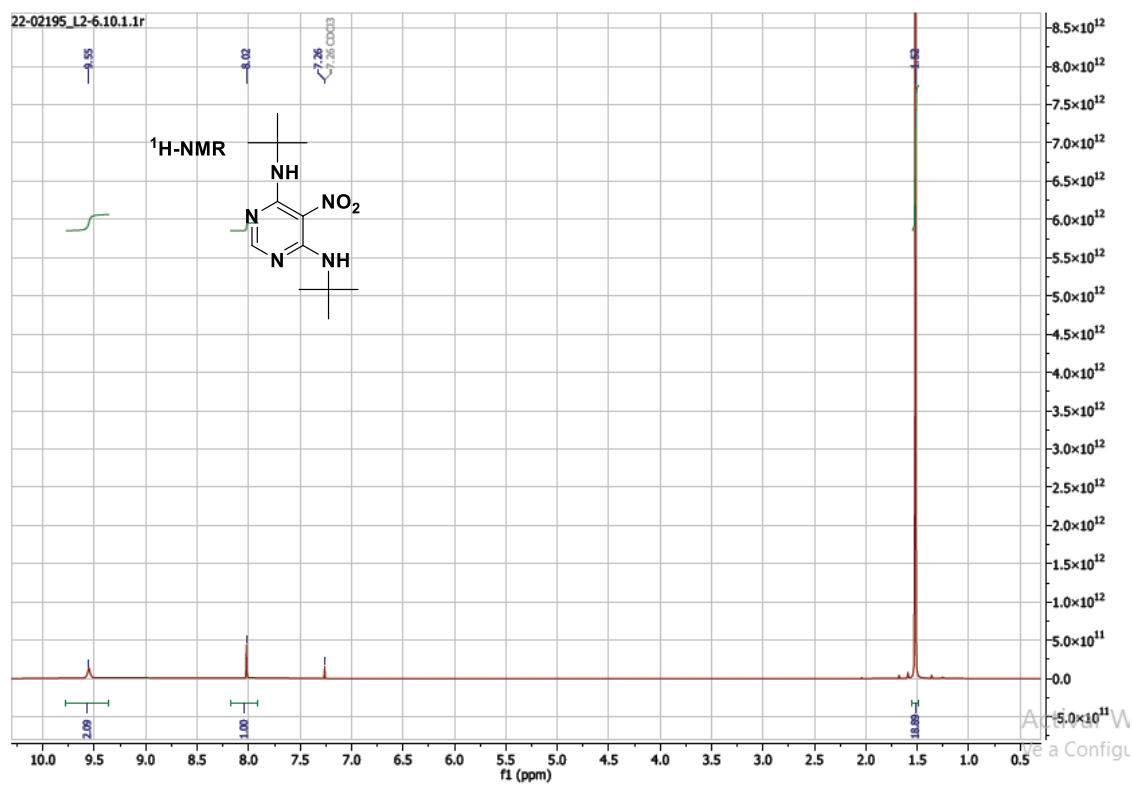
1: TOF MS ES+

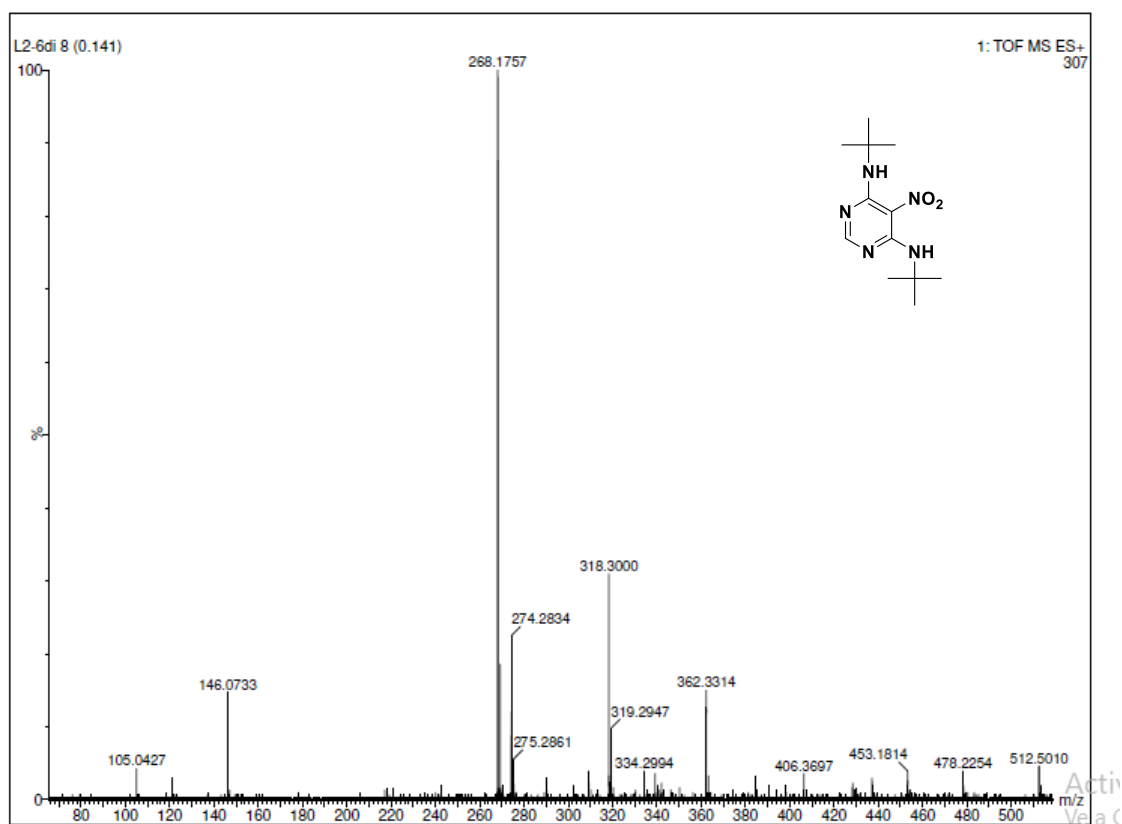
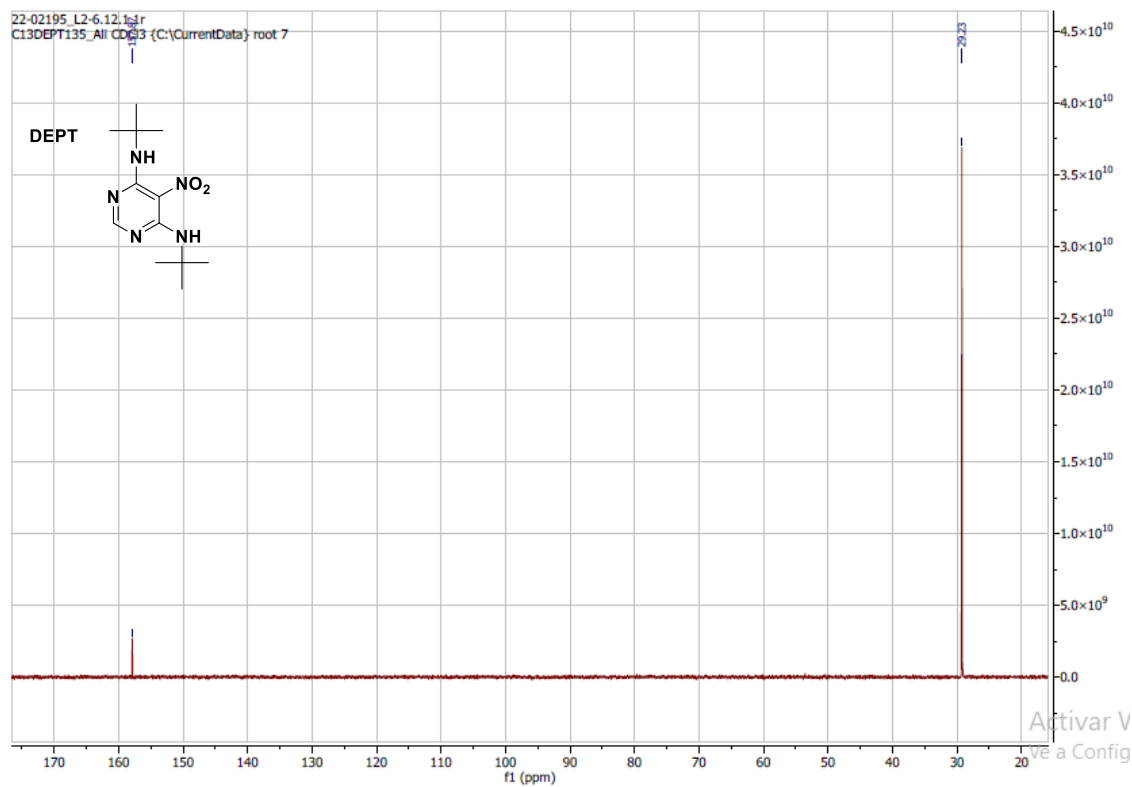


Minimum: -1.5
Maximum: 50.0

Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula
240.1467	240.1461	0.6	2.5	4.5	22.2	0.3	C10 H18 N5 O2
	240.1447	2.0	8.3	-0.5	23.2	1.2	C9 H22 N O6

- N^4,N^6 -di-tert-butyl-5-nitropyrimidine-4,6-diamine (5e):





Page 1

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

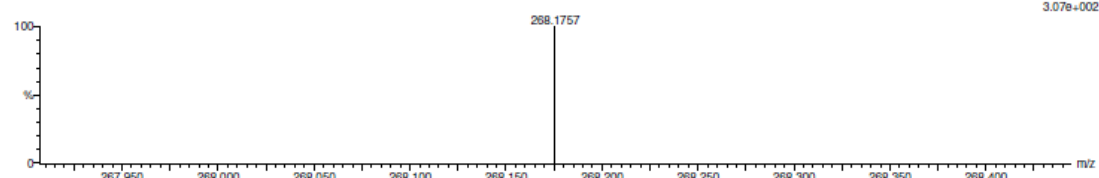
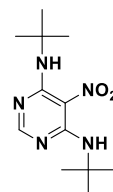
Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

76 formula(e) evaluated with 2 results within limits (up to 50 best isotopic matches for each mass)

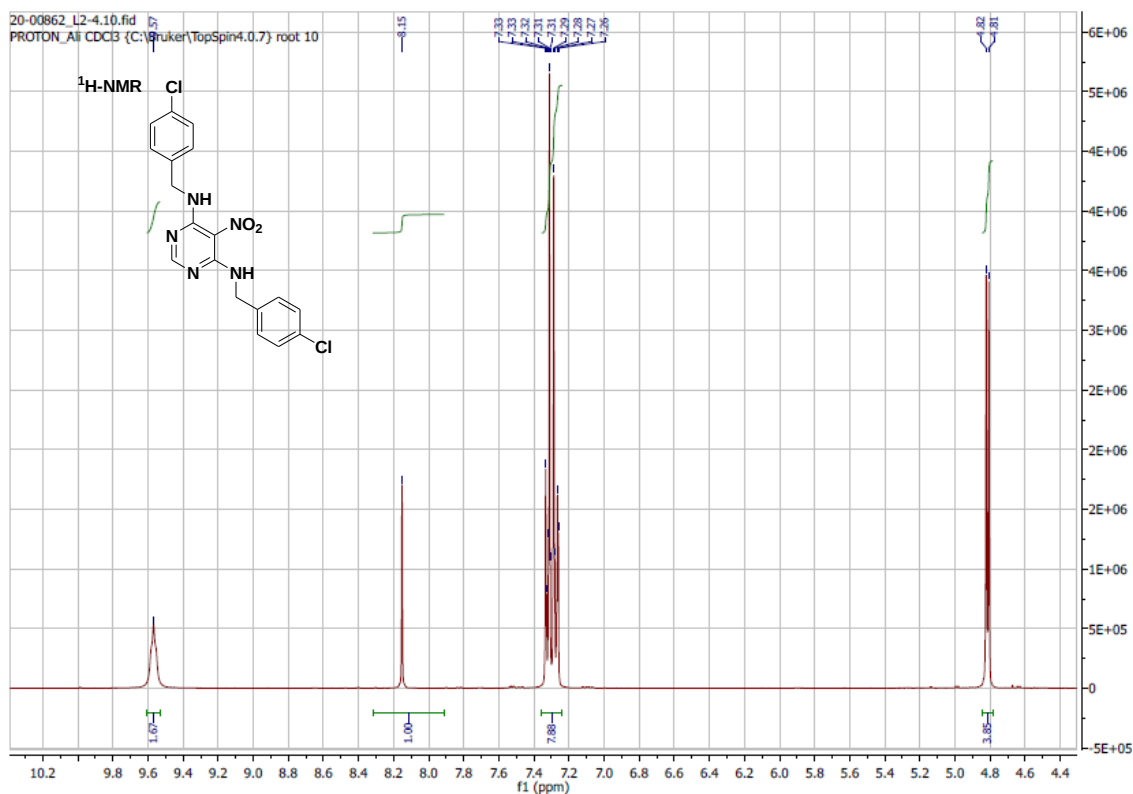
Elements Used:

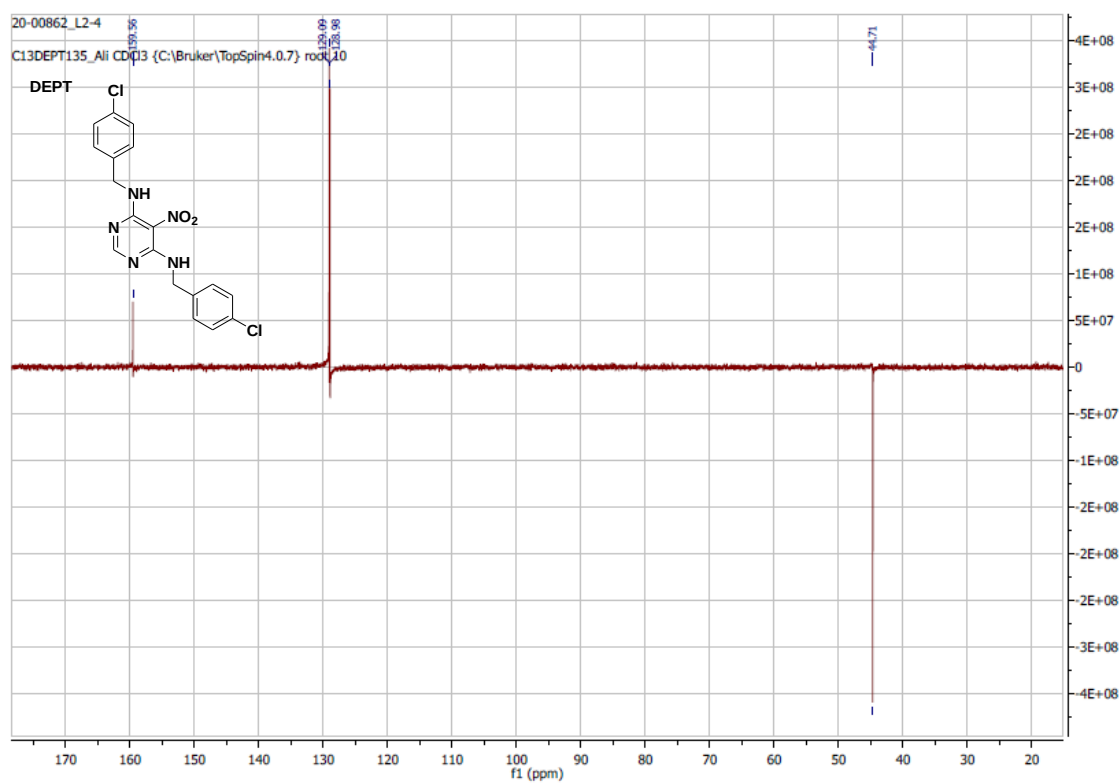
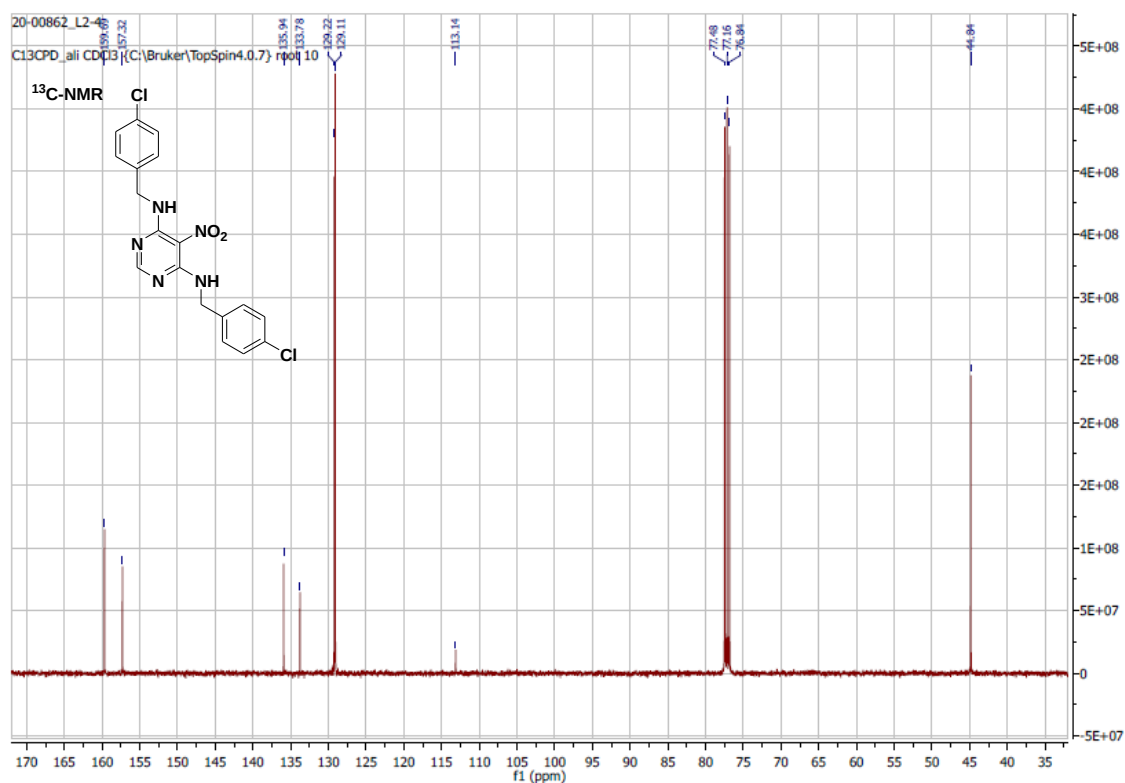
C: 0-12 H: 0-1000 N: 0-5 O: 0-7

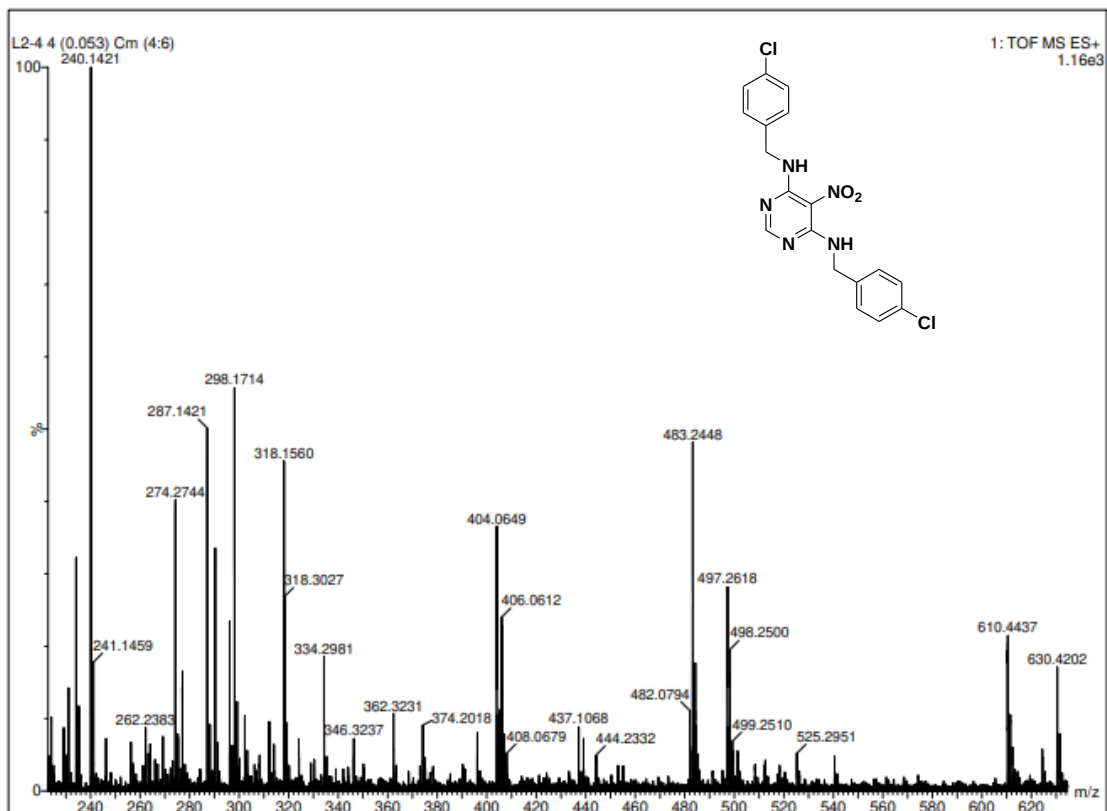
L2-6d1 8 (0.141).
1: TOF MS ES₊

Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula
268.1757	268.1774	-1.7	-6.3	4.5	20.9	1.2	C12 H22 N5 O2
	268.1760	-0.3	-1.1	-0.5	20.1	0.4	C11 H26 N O6

- **N⁴,N⁶-bis(4-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5f):**







Elemental Composition Report

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

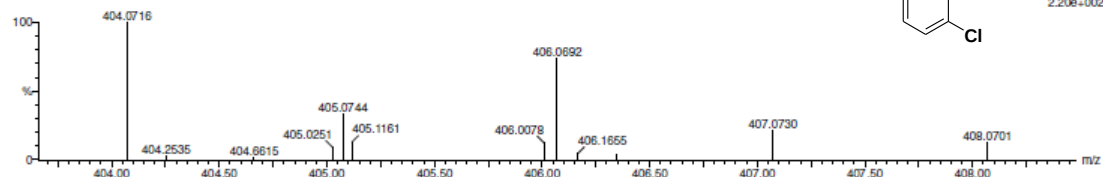
117 formula(e) evaluated with 1 results within limits (up to 50 closest results for each mass)

Elements Used:

C: 0-18 H: 0-1000 N: 0-8 O: 0-2 Cl: 0-2

L2-4 4 (0.053) AM (Cen, 6, 50.00, Ht, 5000.0, 0.00, 1.00)

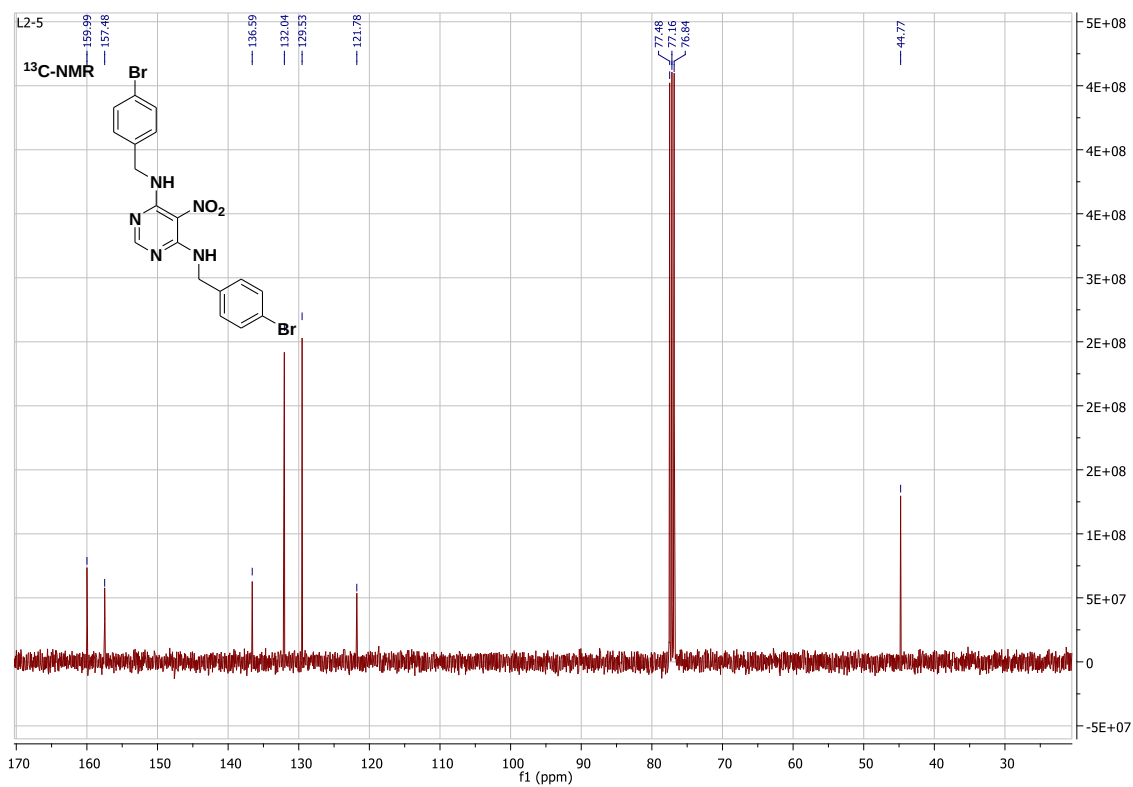
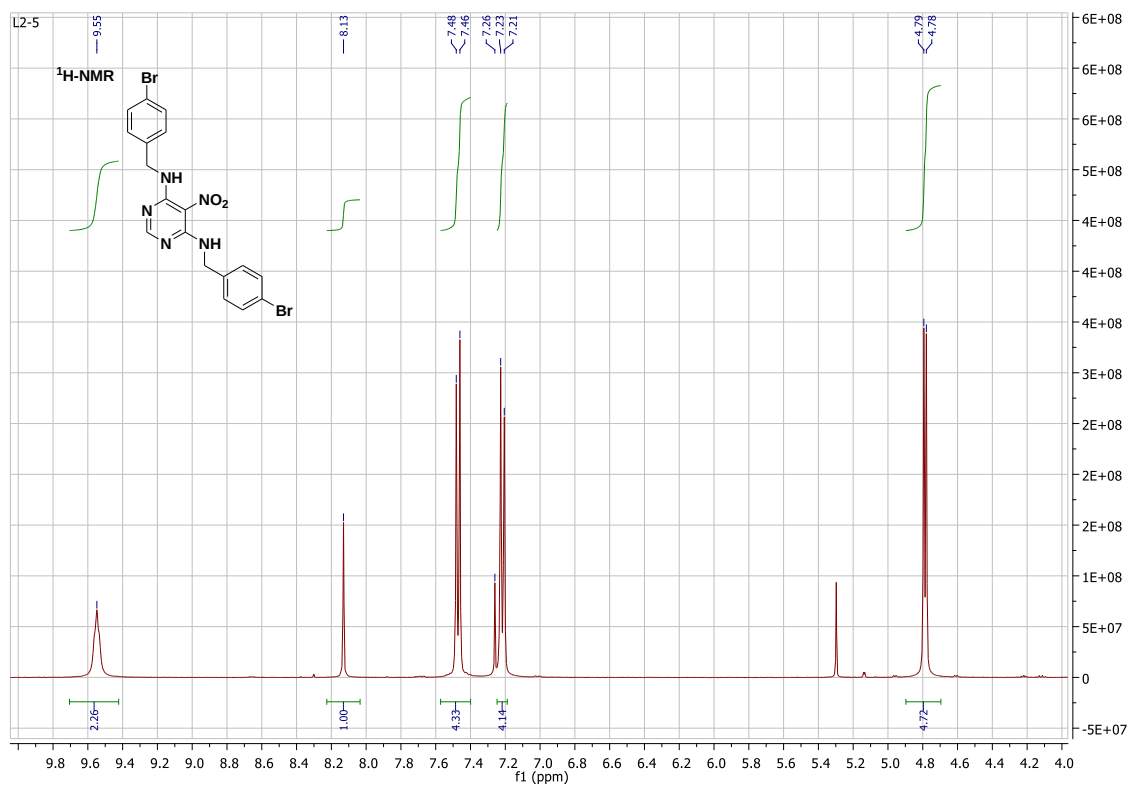
1: TOF MS ES+

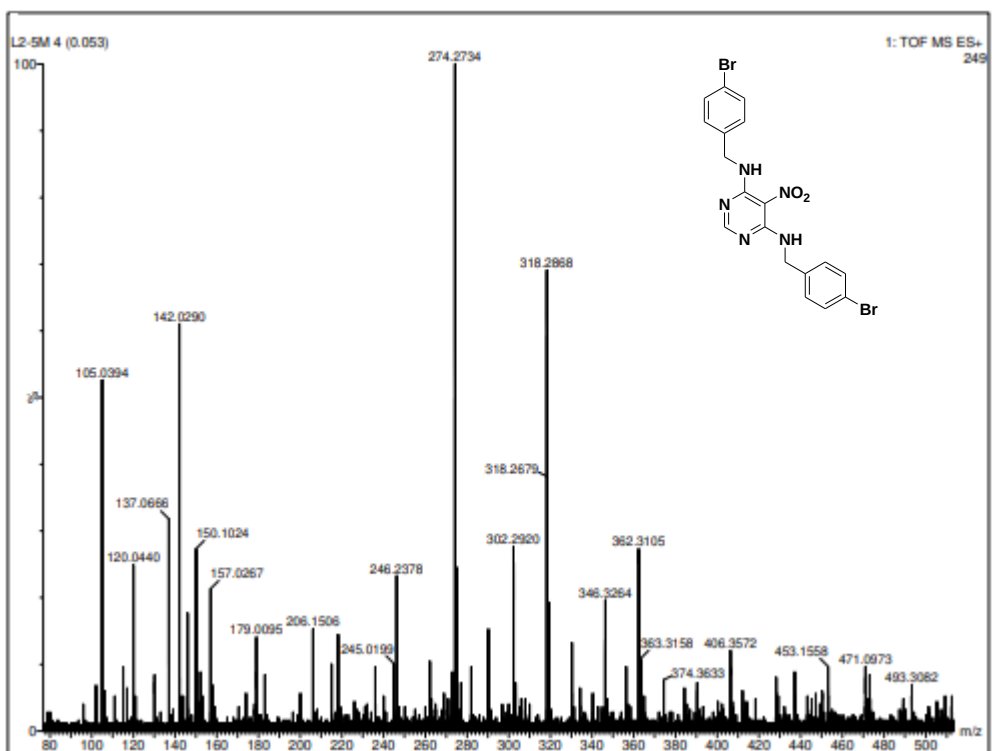
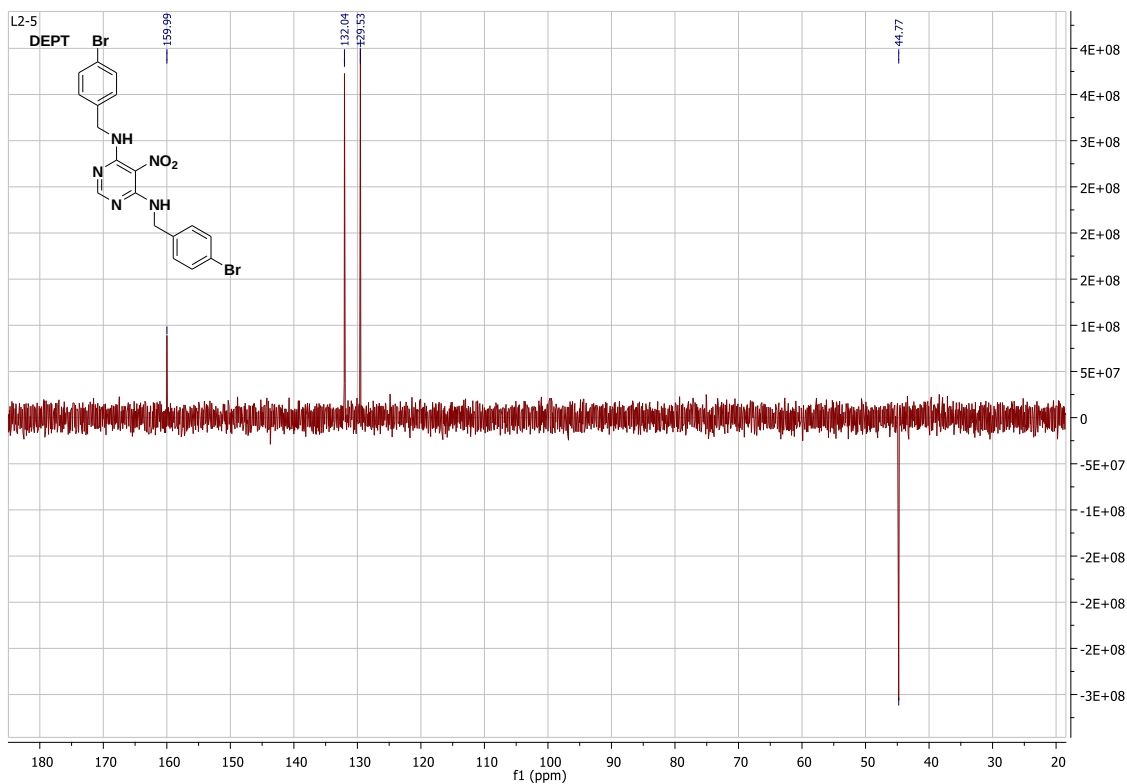


Minimum: -1.5
Maximum: 50.0

Mass	Calc. Mass	mDa	PPM	DBE	i-FIT	i-FIT (Norm)	Formula
404.0716	404.0681	3.5	8.7	12.5	44.0	0.0	C18 H16 N5 O2 Cl2

- **N^4,N^6 -bis(4-bromobenzyl)-5-nitropyrimidine-4,6-diamine (5g):**





Elemental Composition Report

Page 1

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

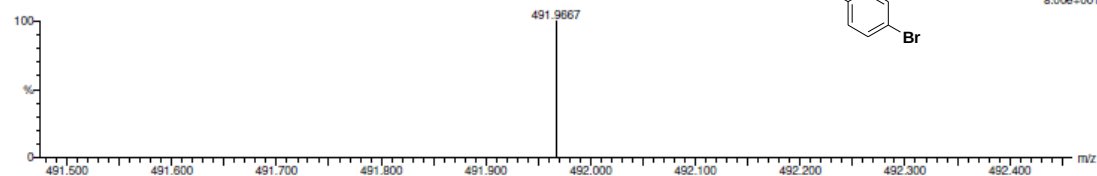
125 formula(e) evaluated with 1 results within limits (up to 50 closest results for each mass)

Elements Used:

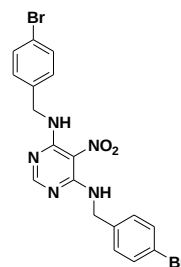
C: 0-18 H: 0-1000 N: 0-9 O: 0-2 Br: 0-2

11032020-L2-5pos 168 (3.665) AM (Top, 2, Ht, 5000.0, 0.00, 1.00)

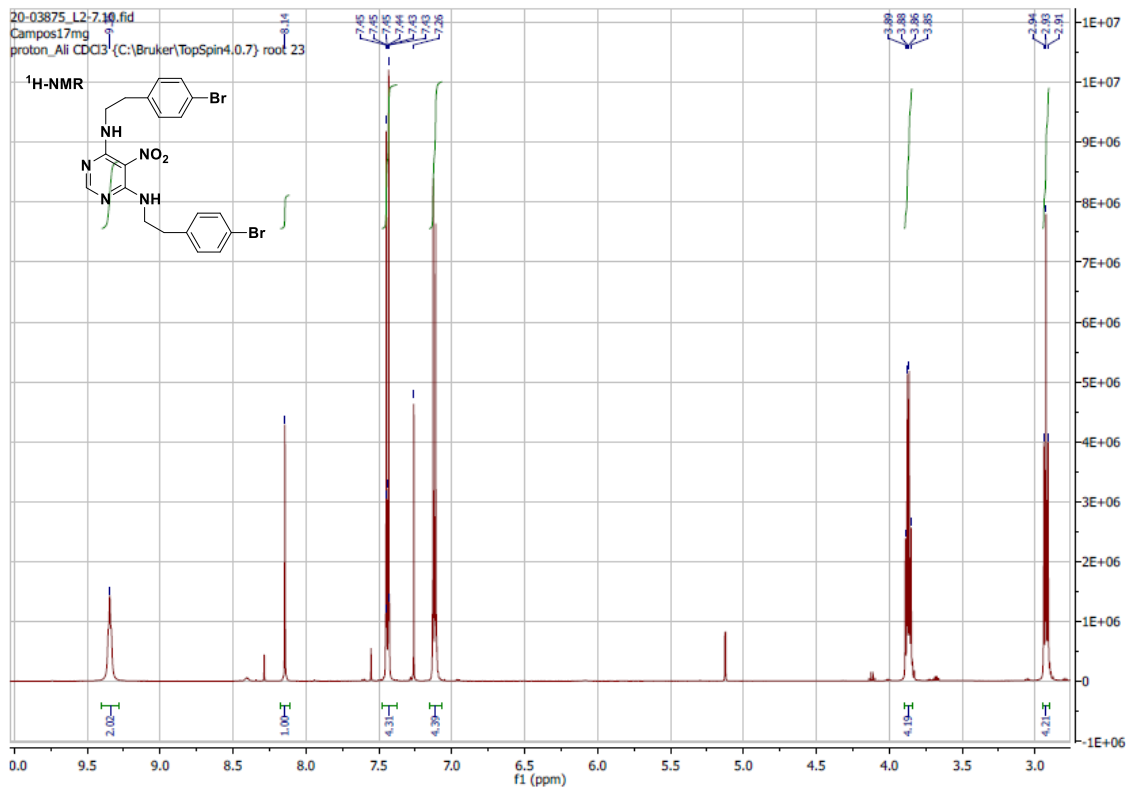
1: TOF MS ES+

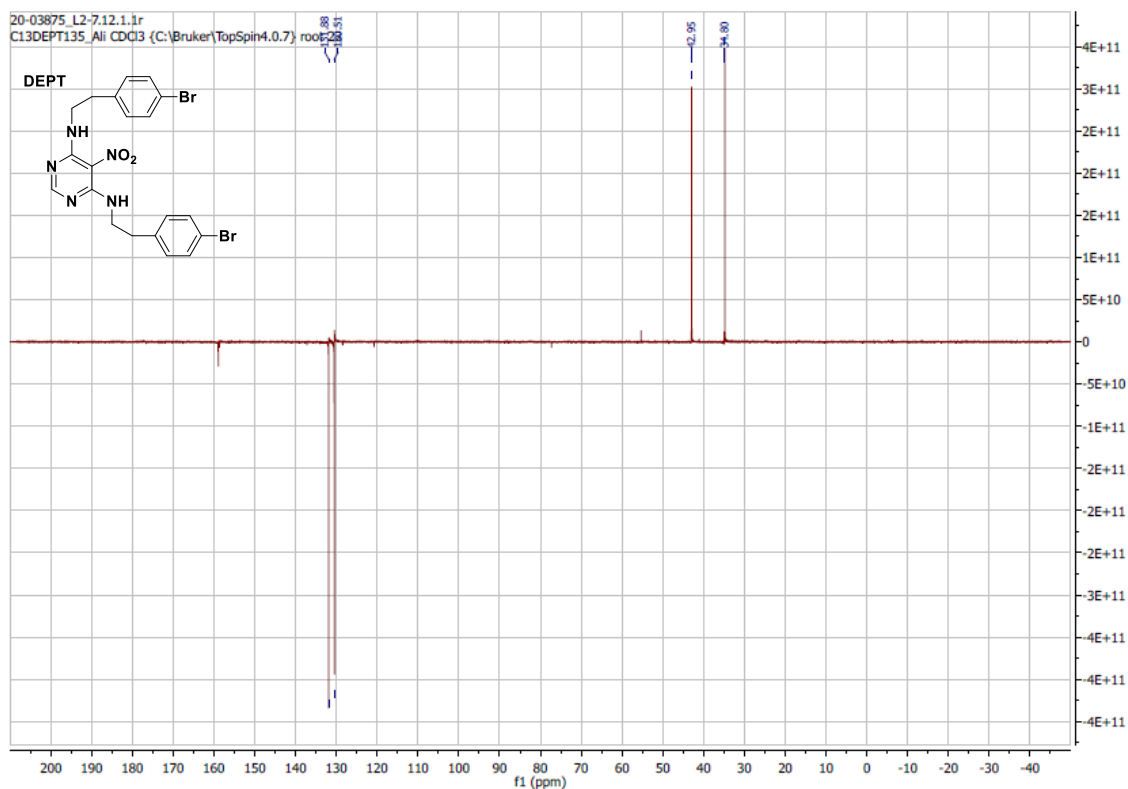
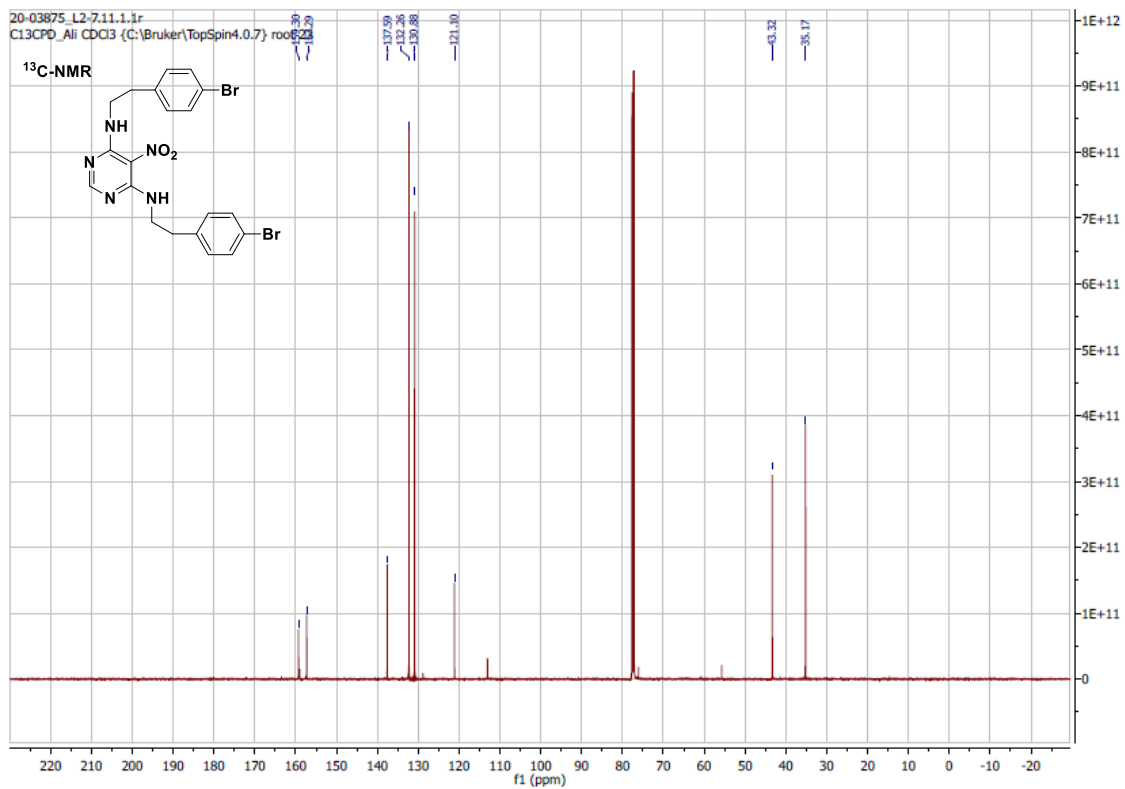


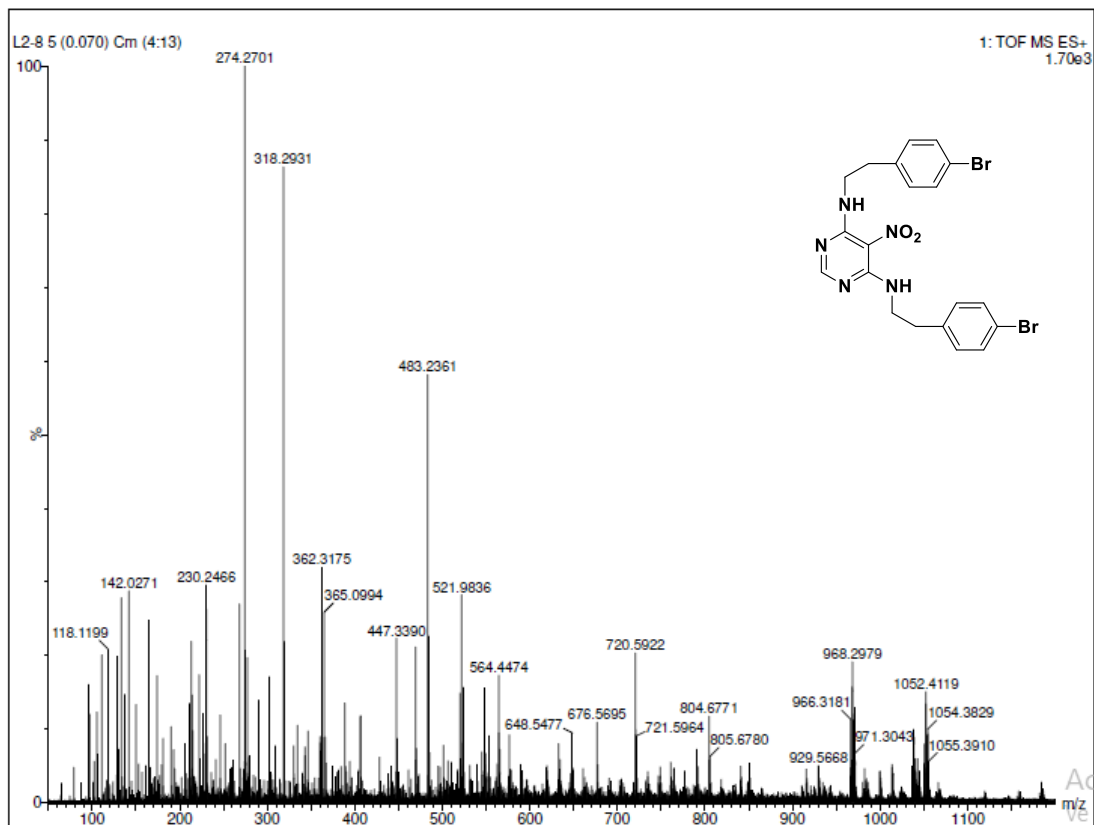
Minimum:				-1.5				
Maximum:	5.0	10.0		50.0				
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula	
491.9667	491.9671	-0.4	-0.8	12.5	19.7	0.0	C18 H16 N5 O2 Br2	



- N⁴,N⁶-bis(4-bromophenethyl)-5-nitropyrimidine-4,6-diamine (5h):







Elemental Composition Report

Page 1

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

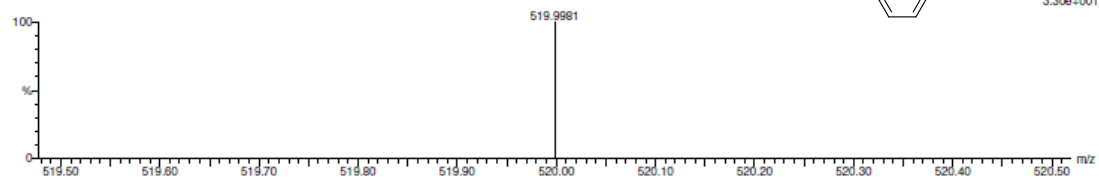
Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Odd and Even Electron Ions
58 formula(e) evaluated with 1 results within limits (up to 50 closest results for each mass)
Elements Used:

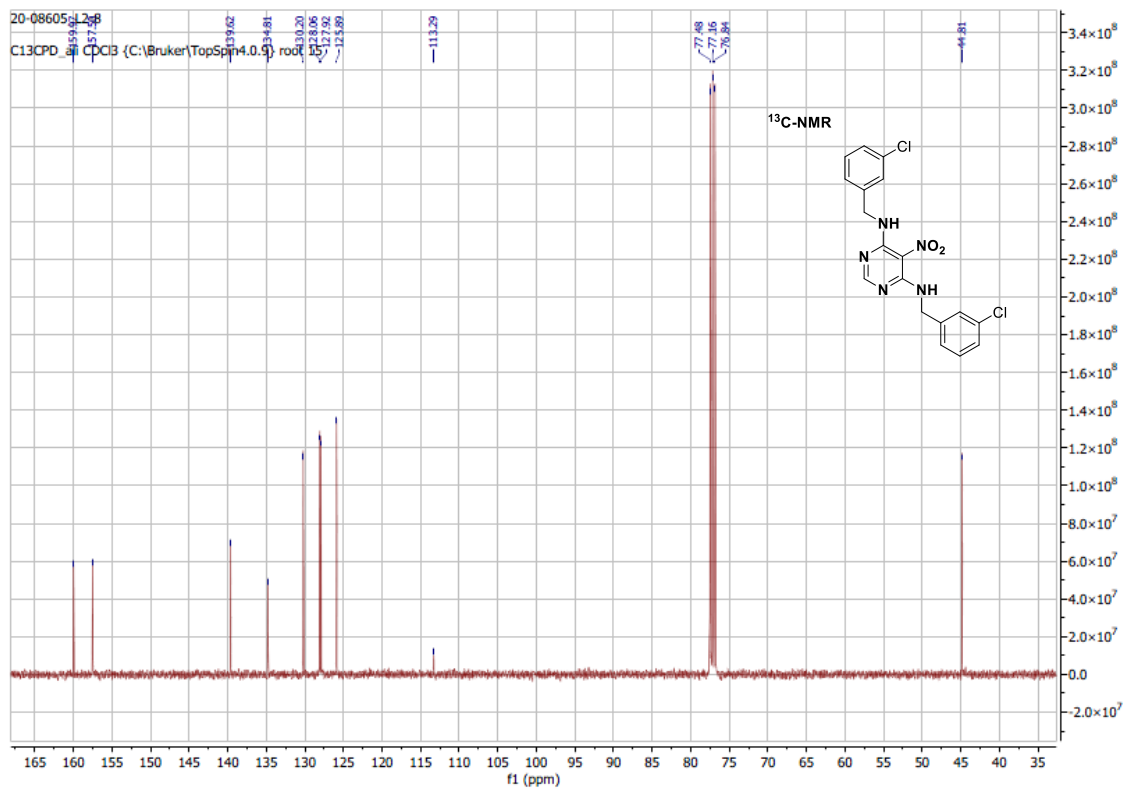
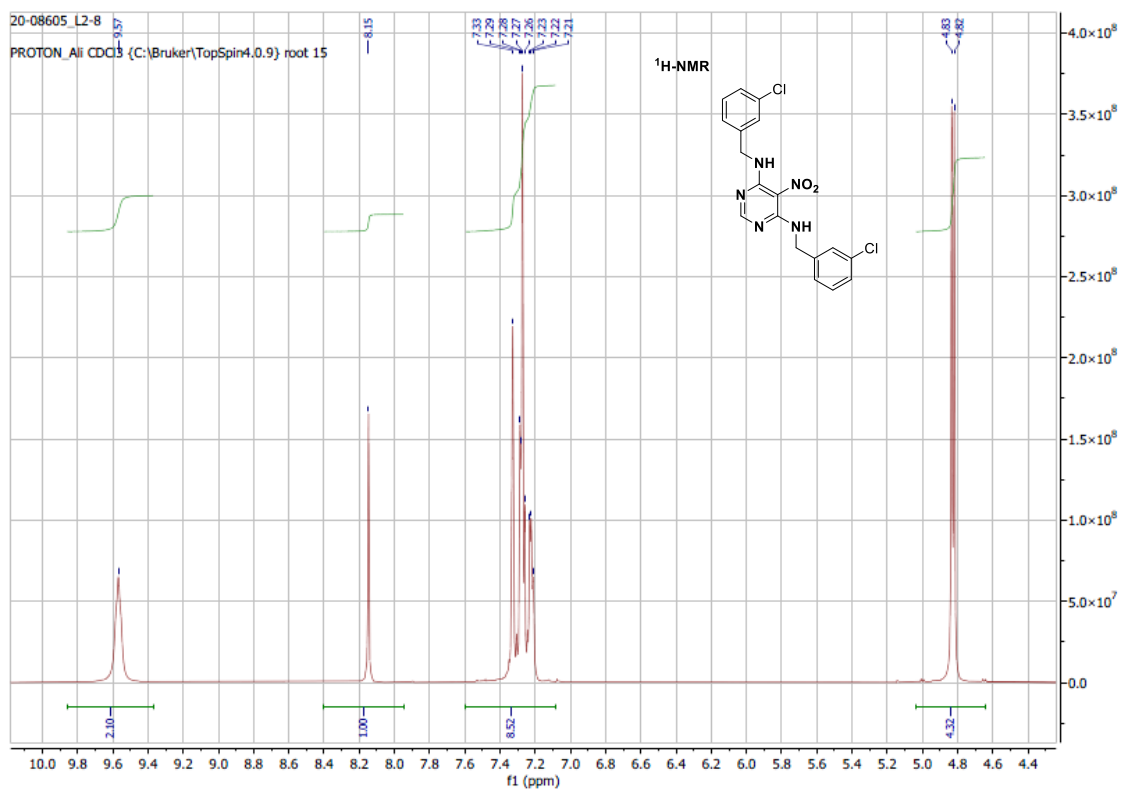
C: 0-20 H: 0-1000 N: 0-5 O: 0-2 Br: 0-2

L2-8 21 (0.422) AM (Top, 1, H: 5000.0, 0.00, 1.00)
1: TOF MS ES+



Minimum:								
Maximum:	5.0	10.0	-1.5	50.0				
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula	
519.9981	519.9984	-0.3	-0.6	12.5	17.9	0.0	C20 H20 N5 O2 Br2	

- **N⁴,N⁶-bis(3-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5i):**



Elemental Composition Report

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

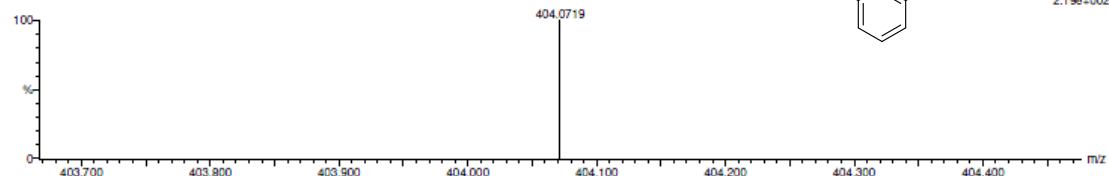
58 formula(e) evaluated with 1 results within limits (up to 50 closest results for each mass)

Elements Used:

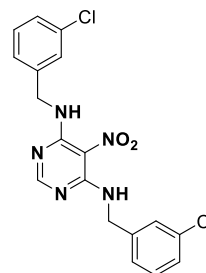
C: 0-18 H: 0-1000 N: 0-5 O: 0-2 Cl: 0-2

L2-6 27 (0.598) AM (Cen, 6, 80.00, Ht, 5000.0, 0.00, 1.00); Cm (27.31)

1: TOF MS ES+



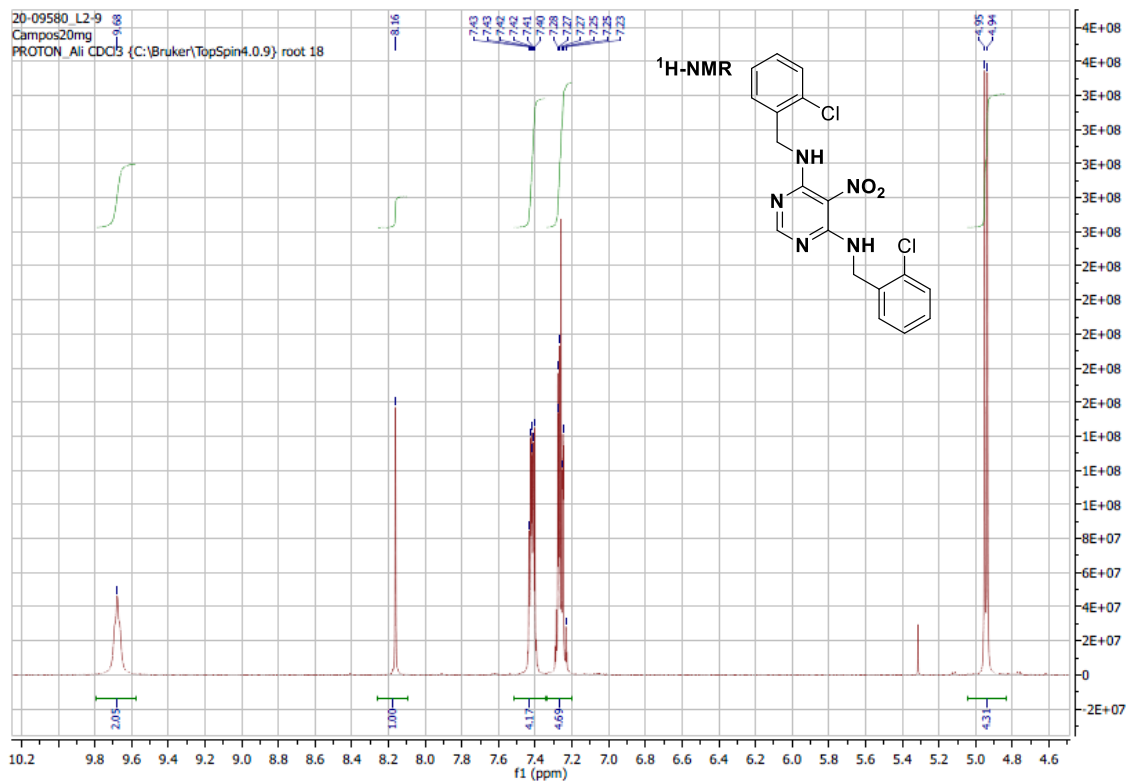
Minimum:				-1.5			
Maximum:	5.0	10.0		50.0			
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula
404.0719	404.0681	3.8	9.4	12.5	22.3	0.0	C18 H16 N5 O2 Cl2

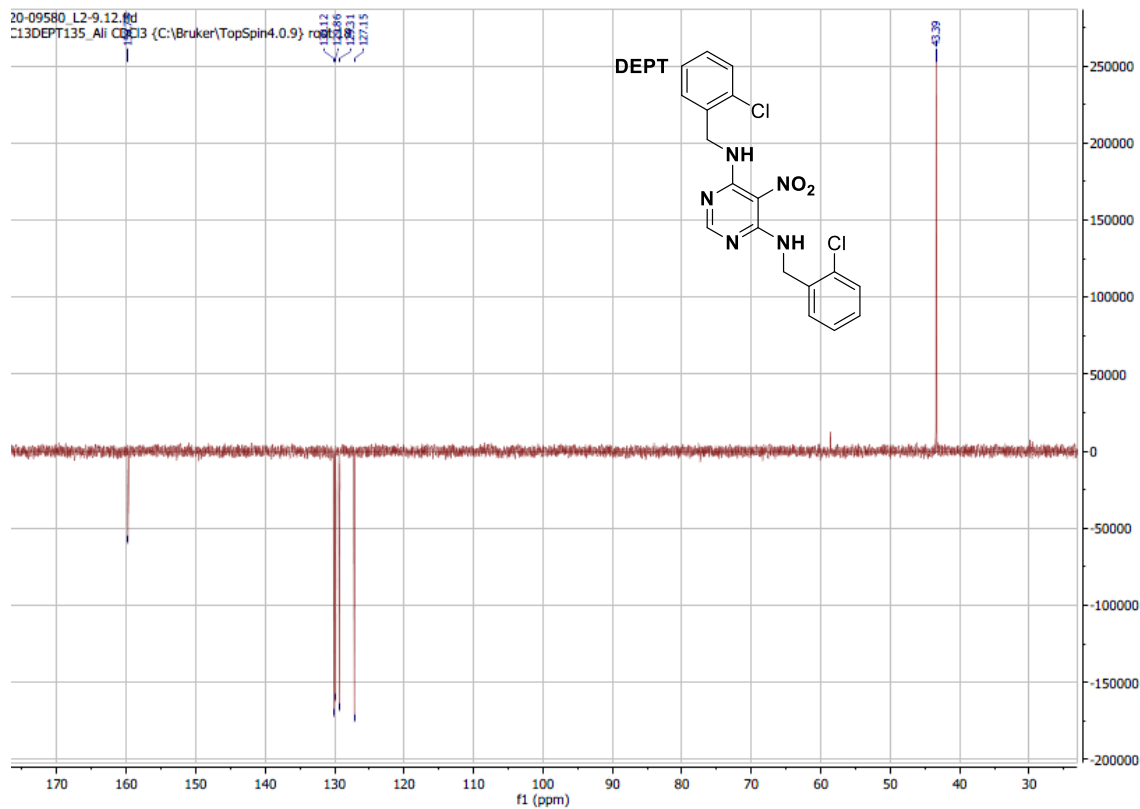
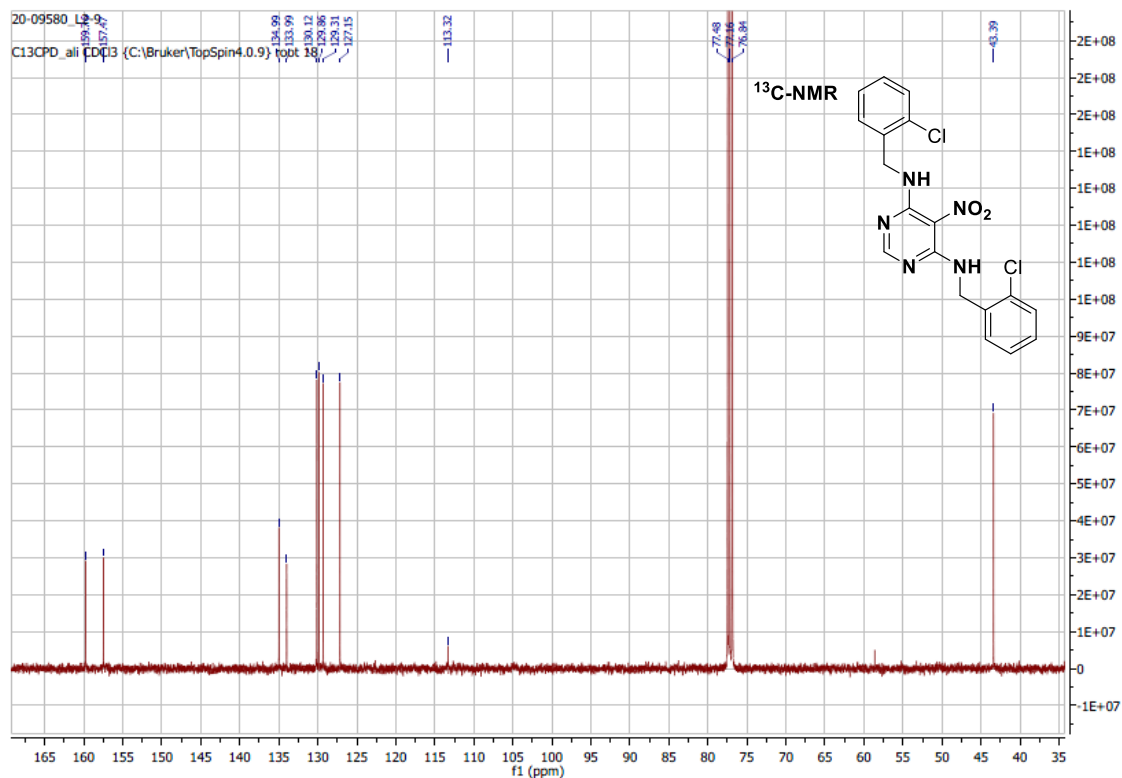


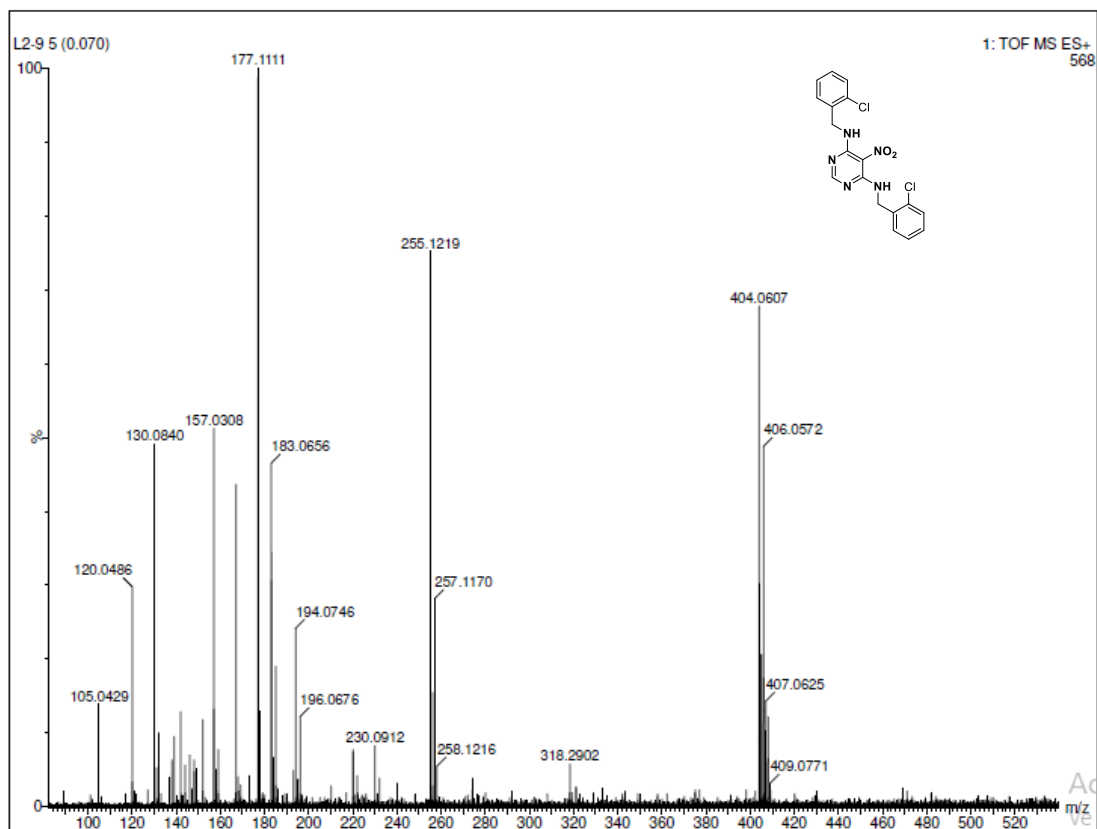
Page 1

2.19e+002

- **N⁴,N⁶-bis(2-chlorobenzyl)-5-nitropyrimidine-4,6-diamine (5j):**







Elemental Composition Report

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

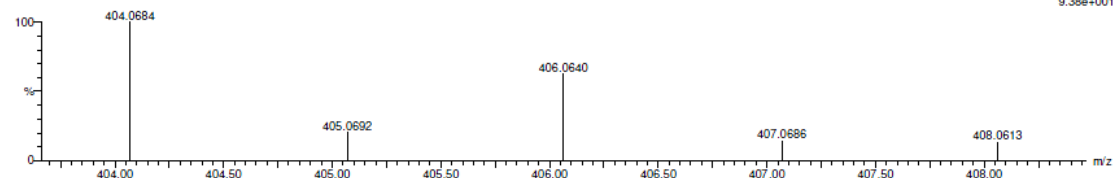
145 formula(e) evaluated with 1 results within limits (up to 50 closest results for each mass)

Elements Used:

C: 0-18 H: 0-1000 N: 0-5 O: 0-5 Cl: 0-2

L2-9 21 (0.422) AM (Gen. 6, 100.00, Ht. 5000.0, 0.00, 1.00)

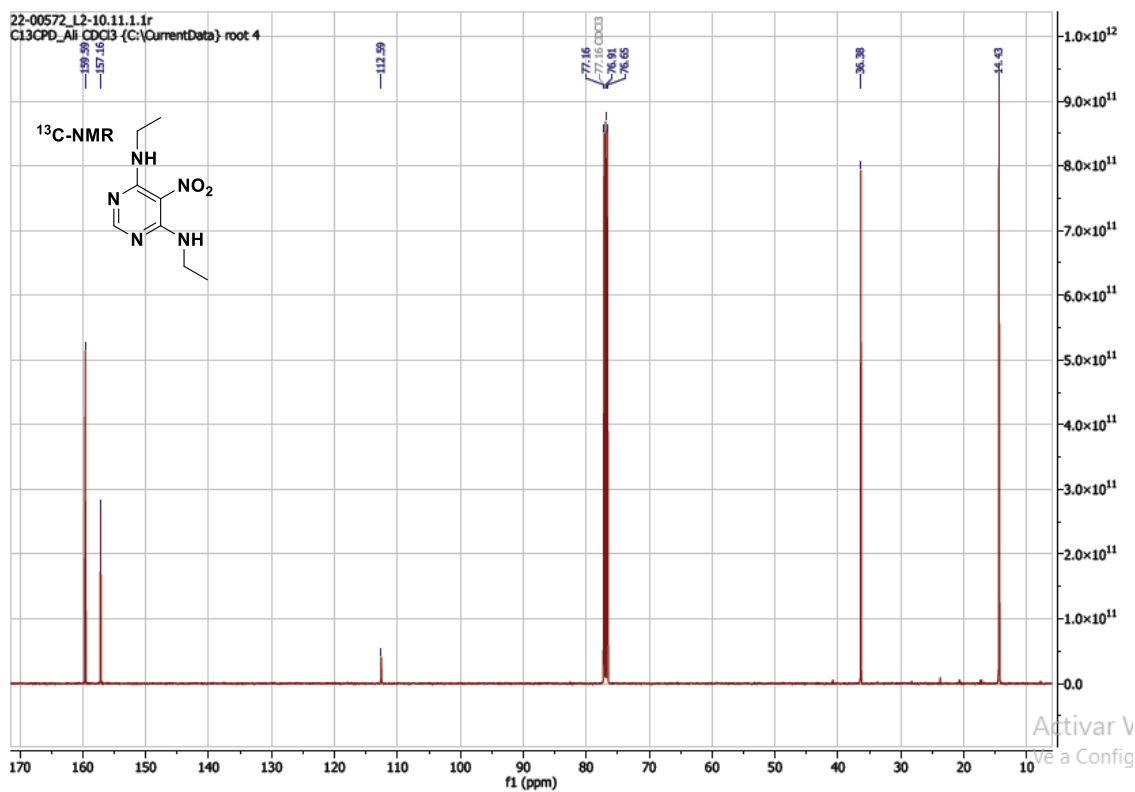
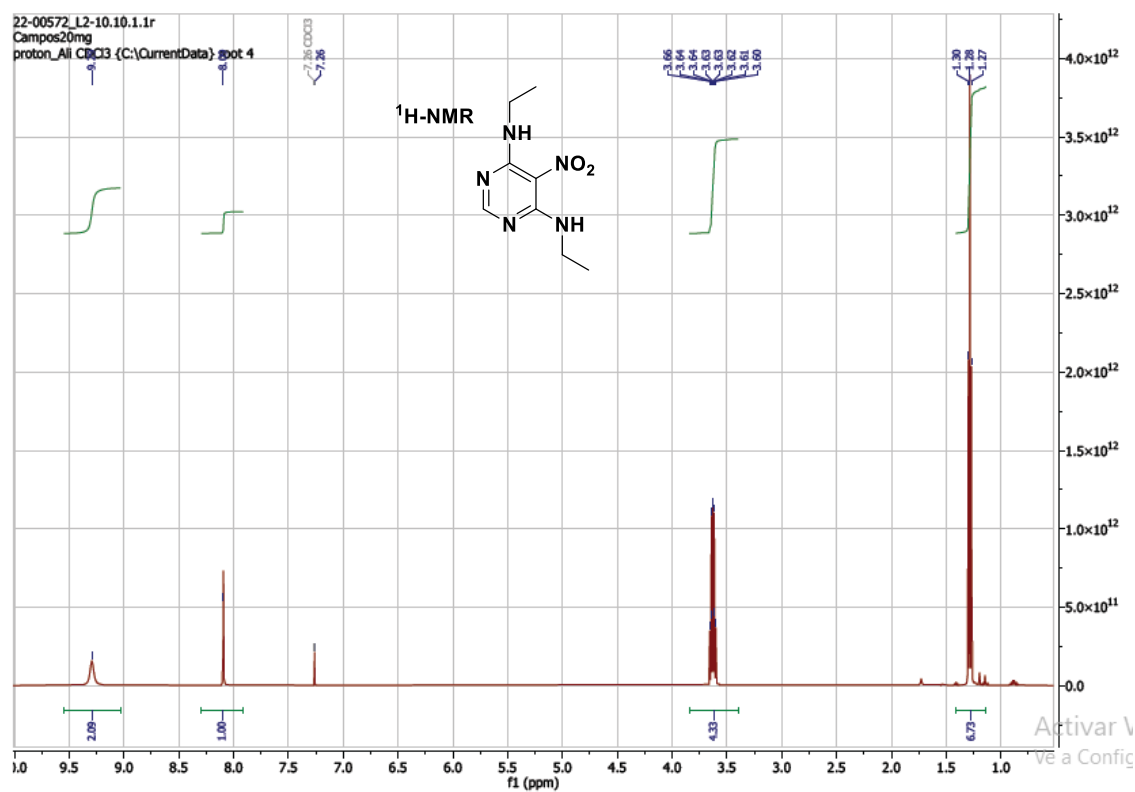
1: TOF MS ES+

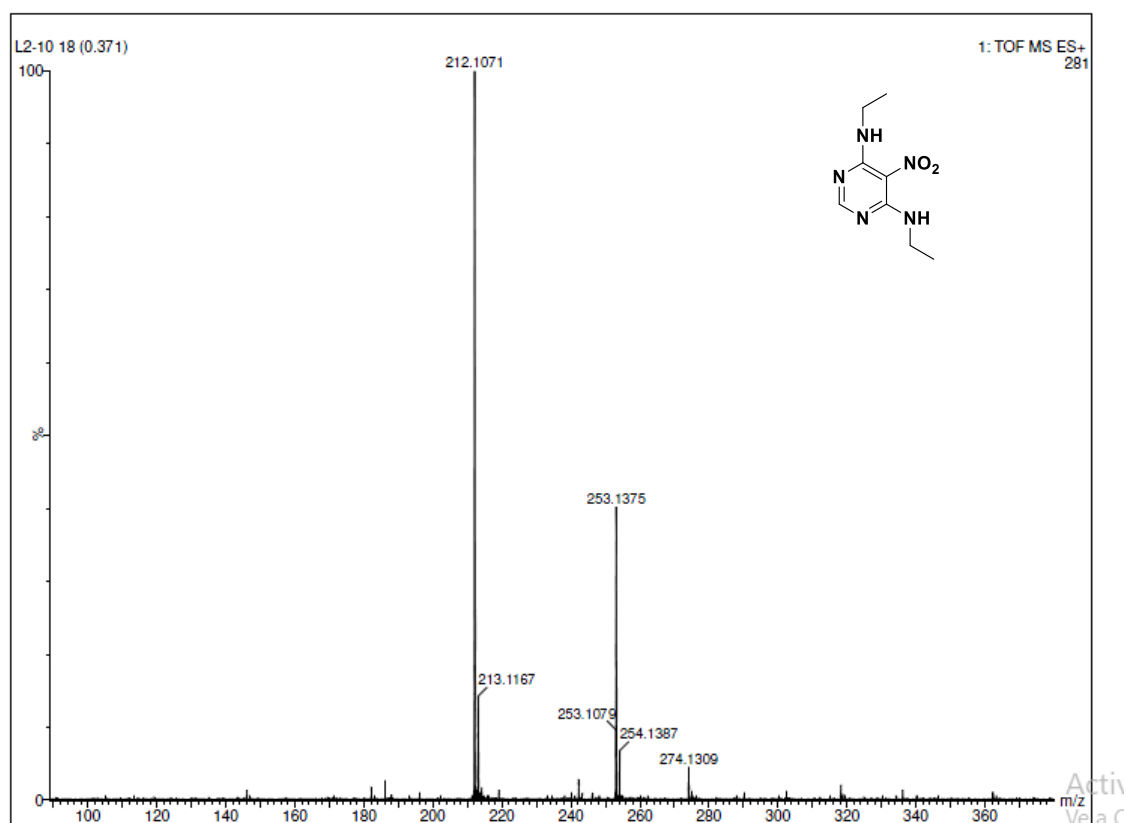
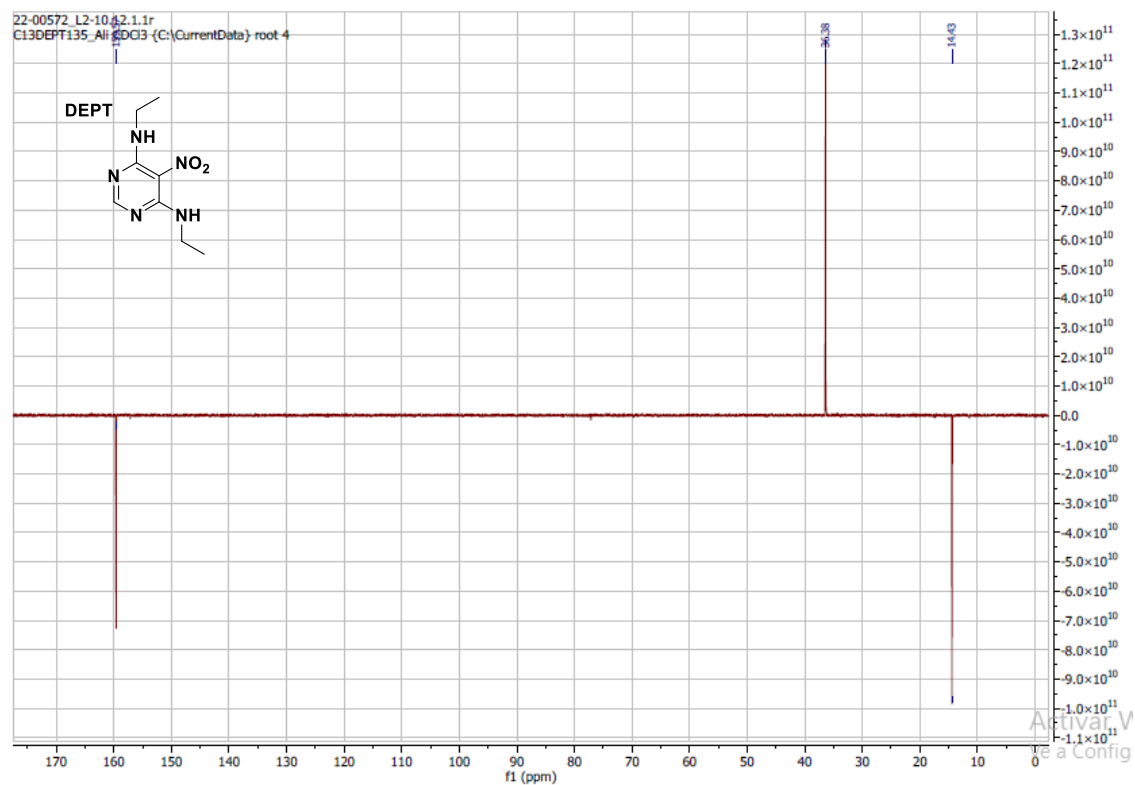


Minimum: -1.5
Maximum: 50.0

Mass	Calc. Mass	mDa	PPM	DBE	i-FIT	i-FIT (Norm)	Formula
404.0684	404.0681	0.3	0.7	12.5	5.6	0.0	C18 H16 N5 O2 Cl2

- N^4,N^6 -diethyl-5-nitropyrimidine-4,6-diamine (5k):





Elemental Composition Report

Page 1

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

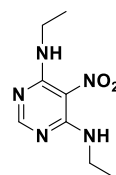
144 formula(e) evaluated with 2 results within limits (up to 50 best isotopic matches for each mass)

Elements Used:

C: 0-8 H: 0-1000 N: 0-8 O: 0-18

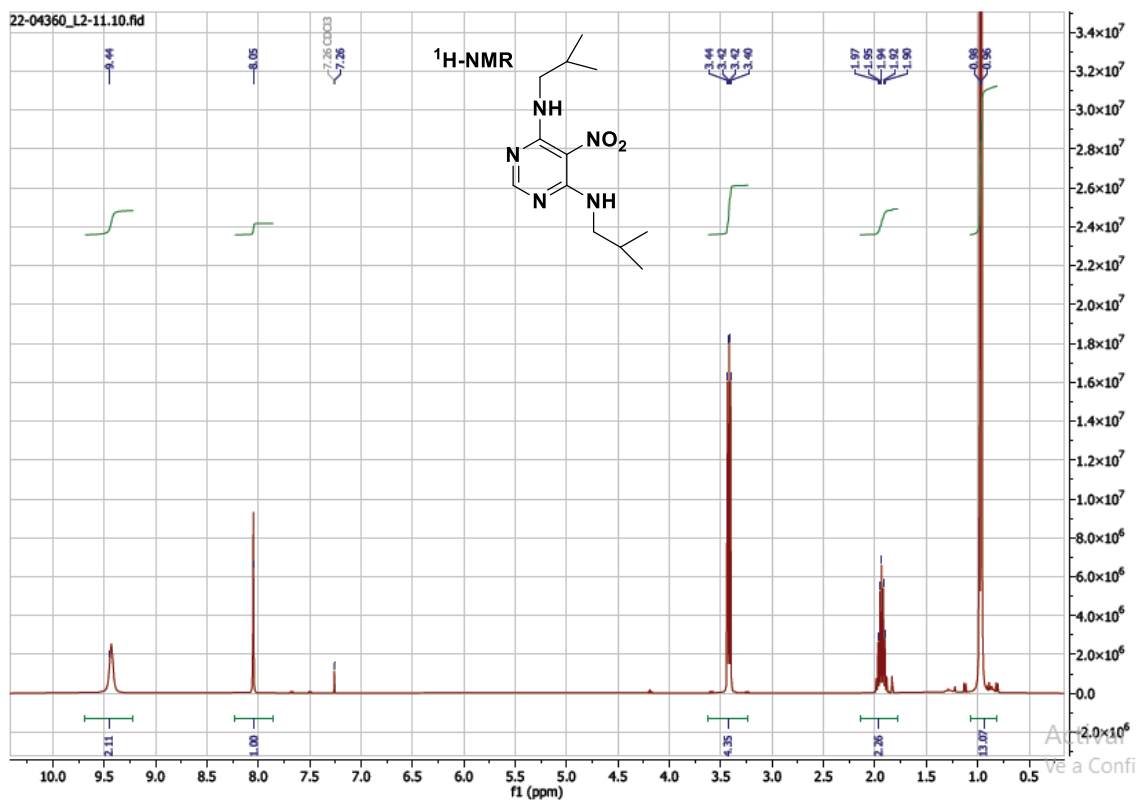
L2:10.6 (0.145) AM (Top,6, Ht,5000.0,0.00,1.00)

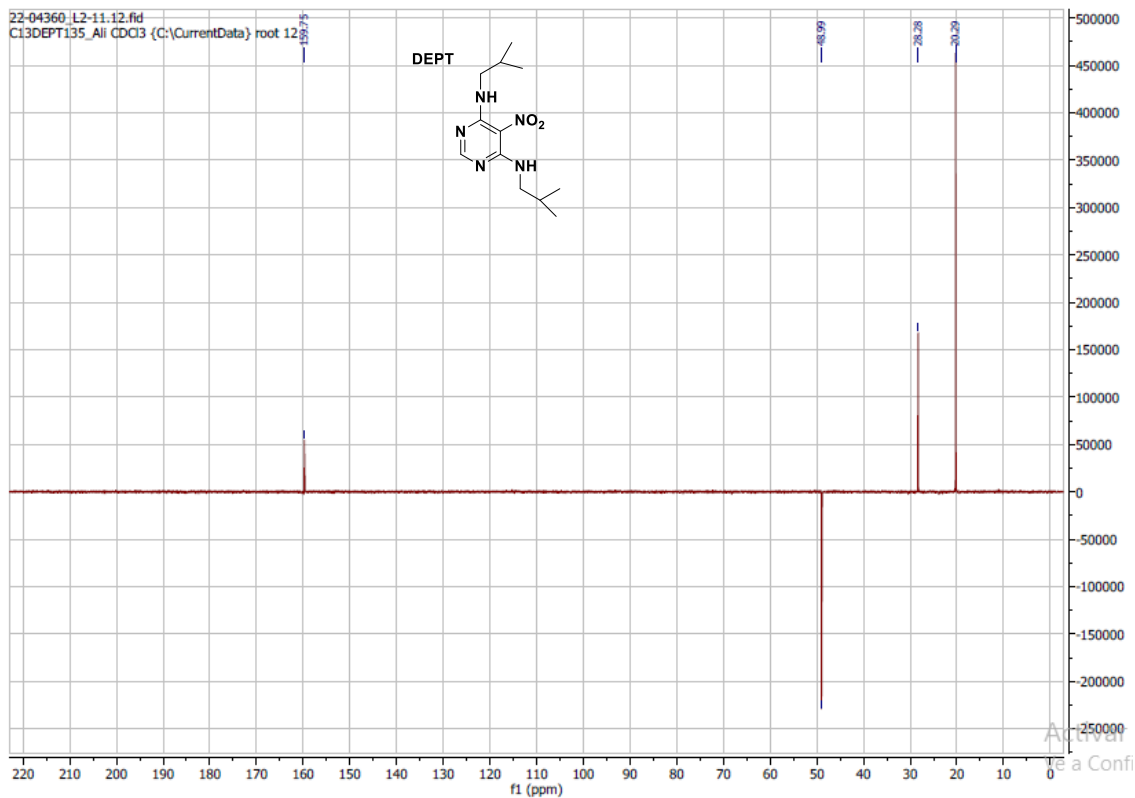
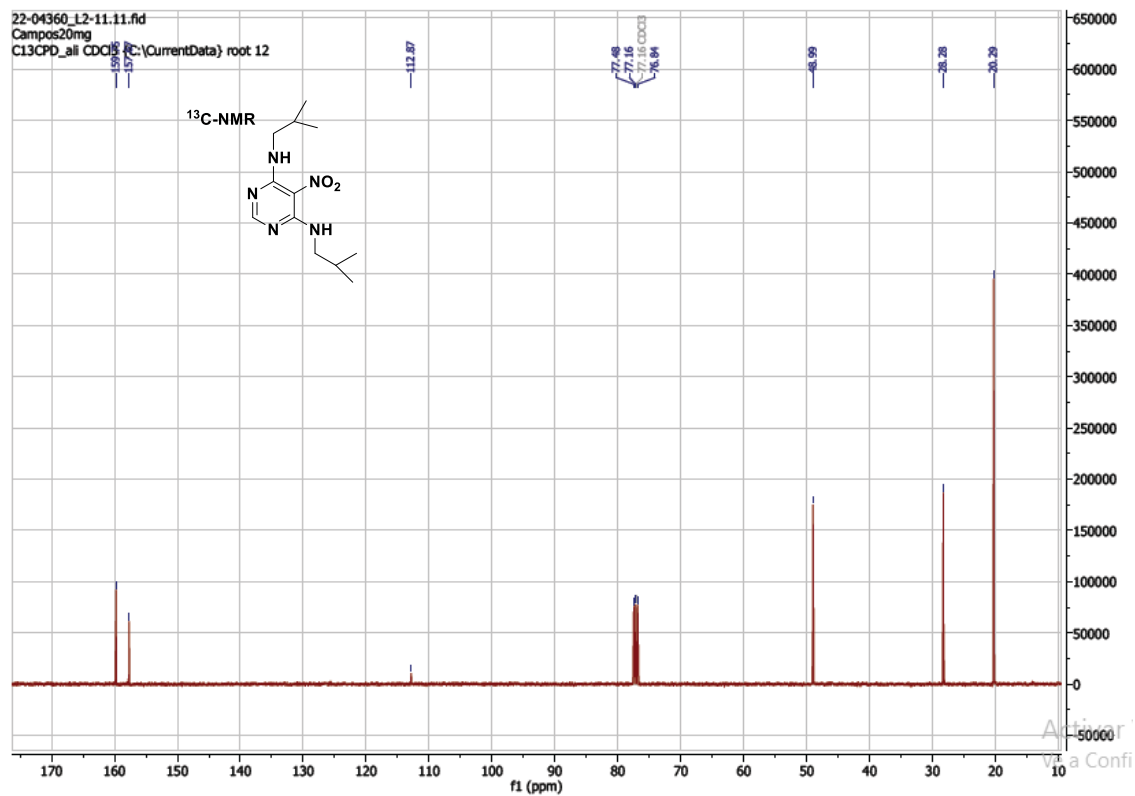
1: TOF MS ES+

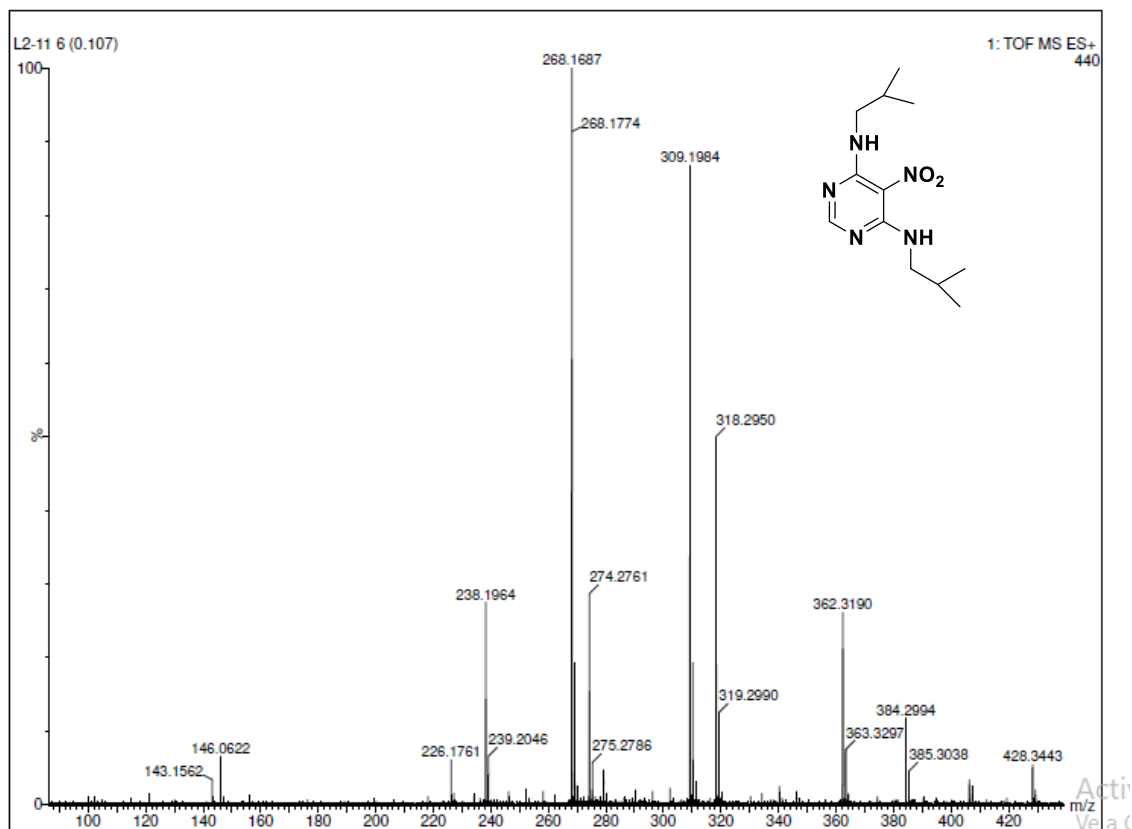


Minimum:								
Maximum:								
	5.0	10.0	-1.5					
			50.0					
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula	
212.1143	212.1147	-0.4	-1.9	4.5	23.4	0.7	C8 H14 N5 O2	
	212.1134	0.9	4.2	-0.5	23.5	0.7	C7 H18 N O6	

- **N⁴,N⁶-diisobutyl-5-nitropyrimidine-4,6-diamine (5I):**







Elemental Composition Report

Page 1

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

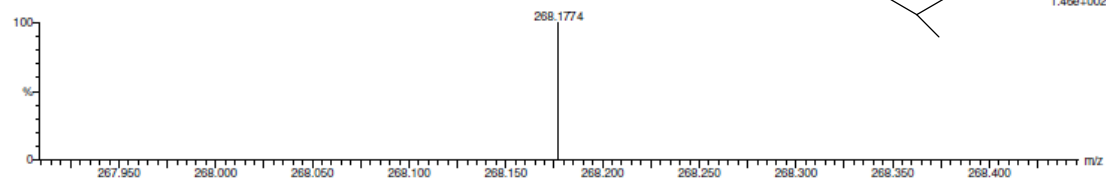
190 formula(e) evaluated with 1 results within limits (up to 50 best isotopic matches for each mass)

Elements Used:

C: 0-20 H: 0-1000 N: 0-5 O: 0-5 I: 0-1

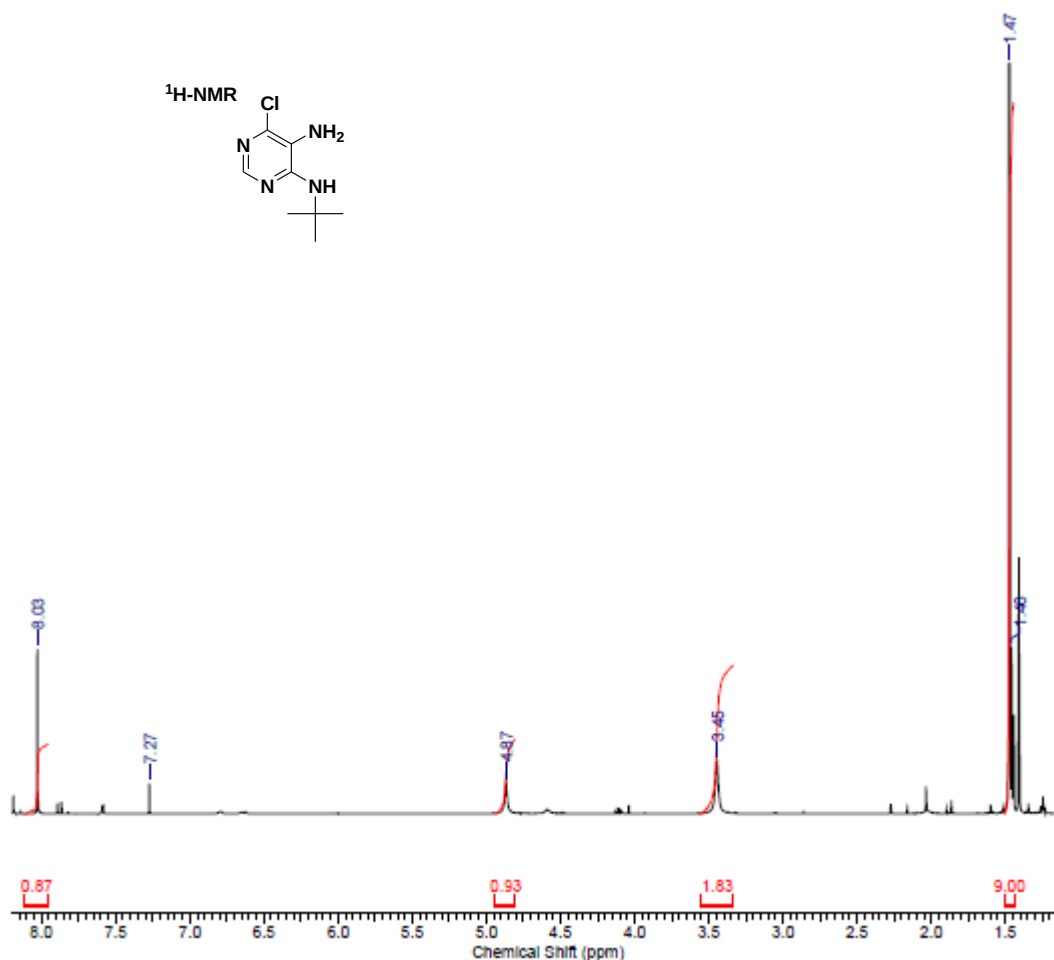
L2-11 9 (0.158) AM (Top 6, H₂ 5000.0, 0.00, 1.00)

1: TOF MS ES+



Minimum:				-1.5				
Maximum:		5.0	10.0	50.0				
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula	
268.1774	268.1774	0.0	0.0	4.5	18.3	0.0	C12 H22 N5 O2	

- 5-Amino-4-tert-butylamino-6-chloropyrimidine (6b):



Elemental Composition Report

Single Mass Analysis

Tolerance = 5.0 PPM / DBE: min = -1.5, max = 50.0
 Element prediction: Off
 Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

960 formula(e) evaluated with 3 results within limits (up to 50 closest results for each mass)

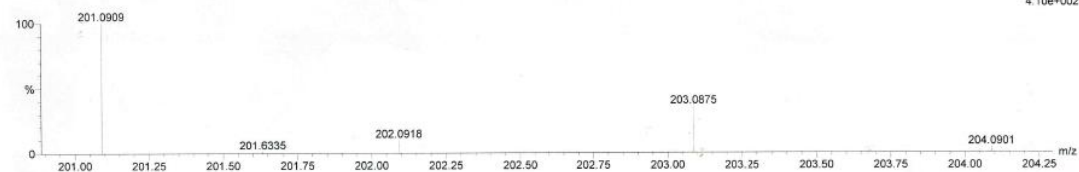
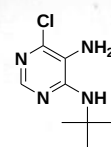
Elements Used:

C: 0-17 H: 0-1000 N: 0-10 O: 0-10 Na: 0-1 S: 0-1 Cl: 0-1

PRE-TB-F dilu 9 (0.217)

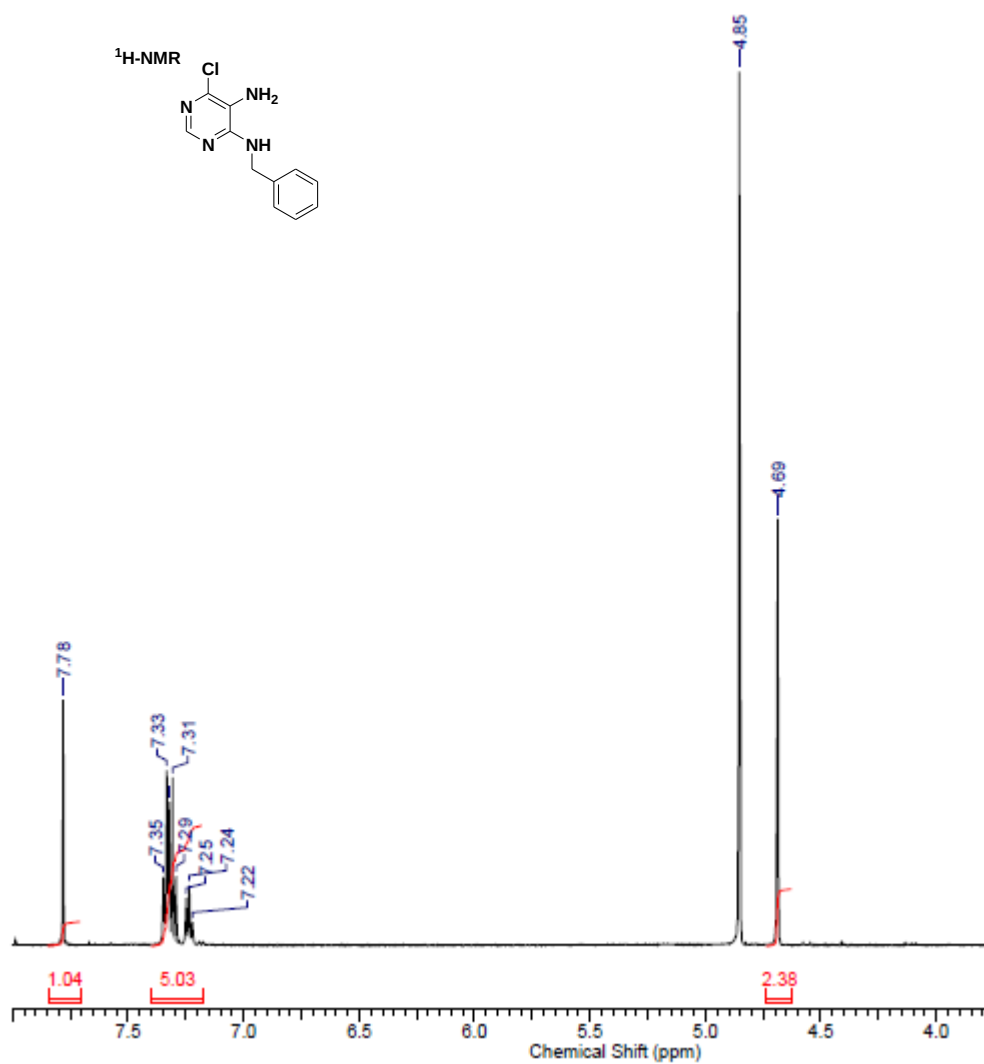
1: TOF MS ES+

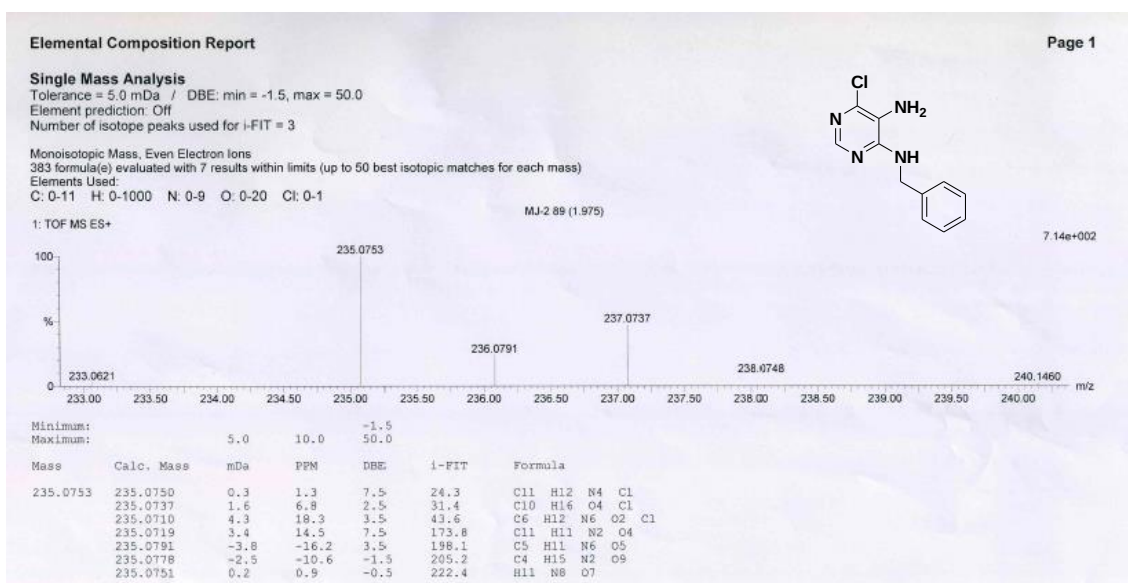
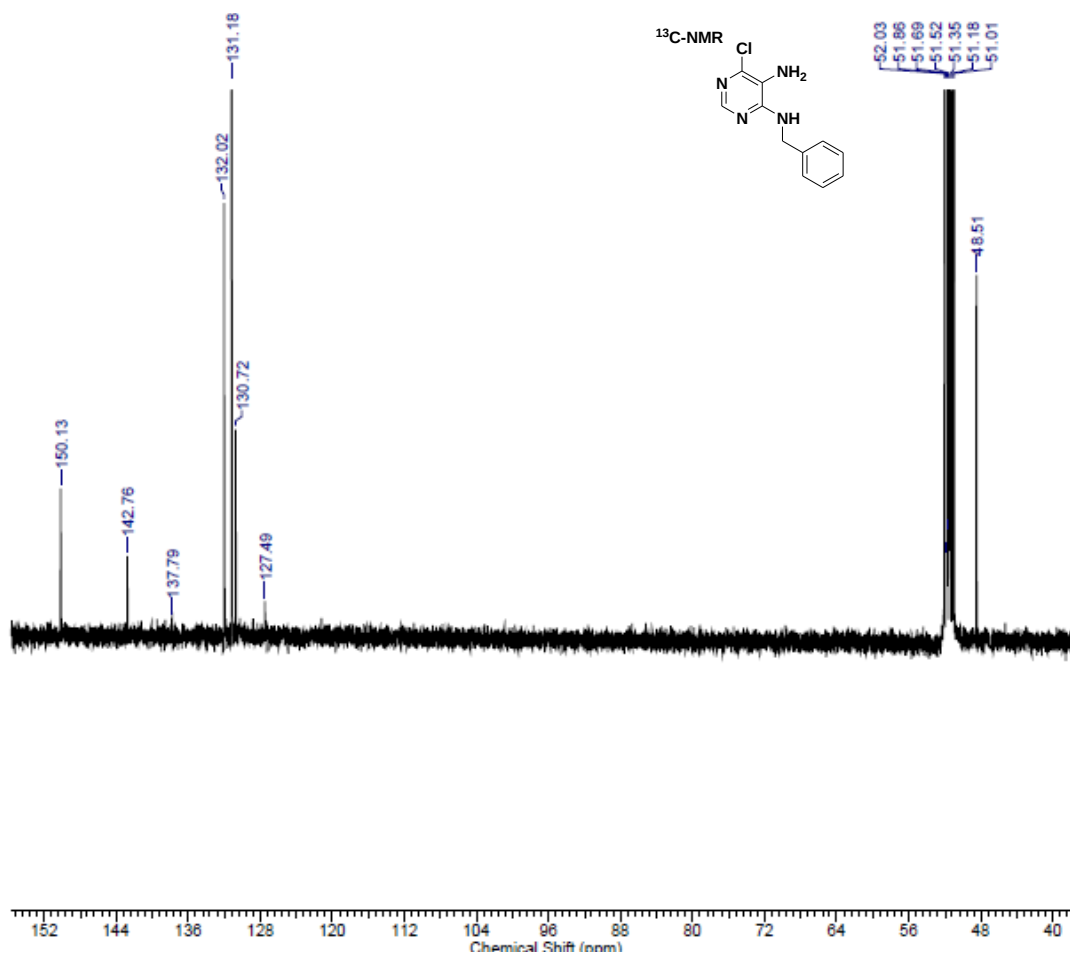
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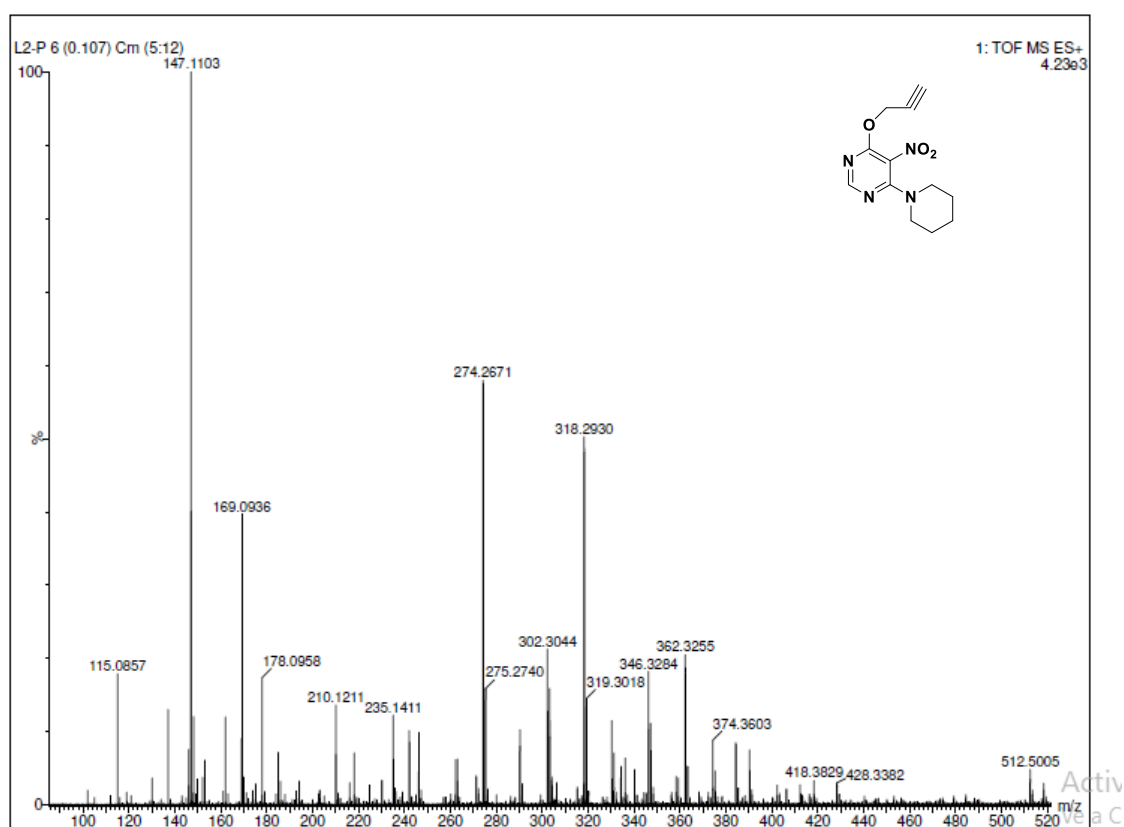
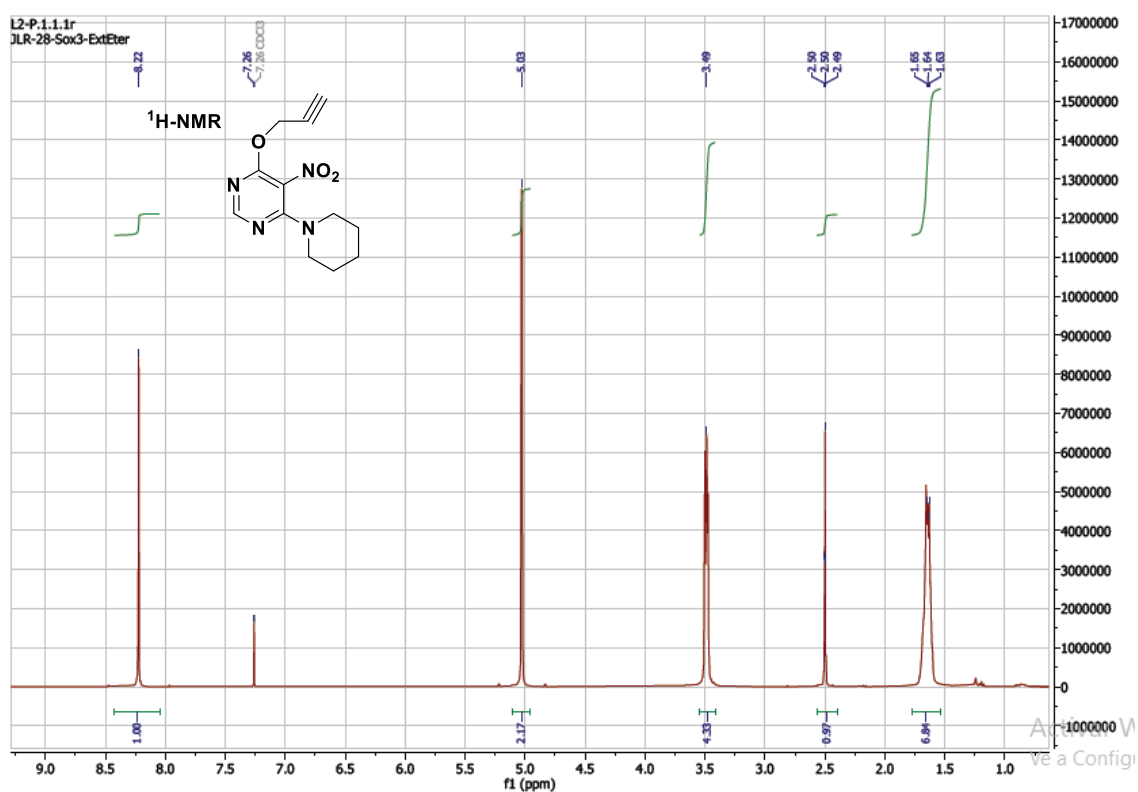
Minimum:				-1.5			
Maximum:		20.0	5.0	50.0			
Mass	Calc. Mass	mDa	PPM	DBE	i-FIT	Formula	
201.0909	201.0907	0.2	1.0	3.5	0.6	C9	H14 N4 Cl
	201.0909	0.0	0.0	-1.5	92.4	C5	H17 N2 O4 S
	201.0916	-0.7	-3.5	7.5	72.7	C13	H13 O2

- 5-Amino-4-benzylamino-6-chloropyrimidine (6c):





- 5-Nitro-4-(piperidin-1-yl)-6-(prop-2-yn-1-yloxy)pyrimidine (7):



Elemental Composition Report

Single Mass Analysis

Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0

Element prediction: Off

Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions

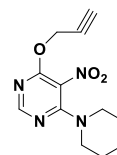
451 formula(e) evaluated with 3 results within limits (up to 50 best isotopic matches for each mass)

Elements Used:

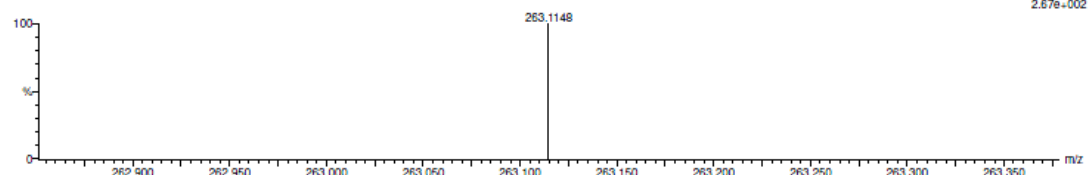
C: 0-25 H: 0-1000 N: 0-5 O: 0-8 S: 0-2

L2-P 6 (0.107) AM (Top, 6, Ht, 5000.0, 0.00, 1.00) Cm (5:12)

1: TOF MS ES+

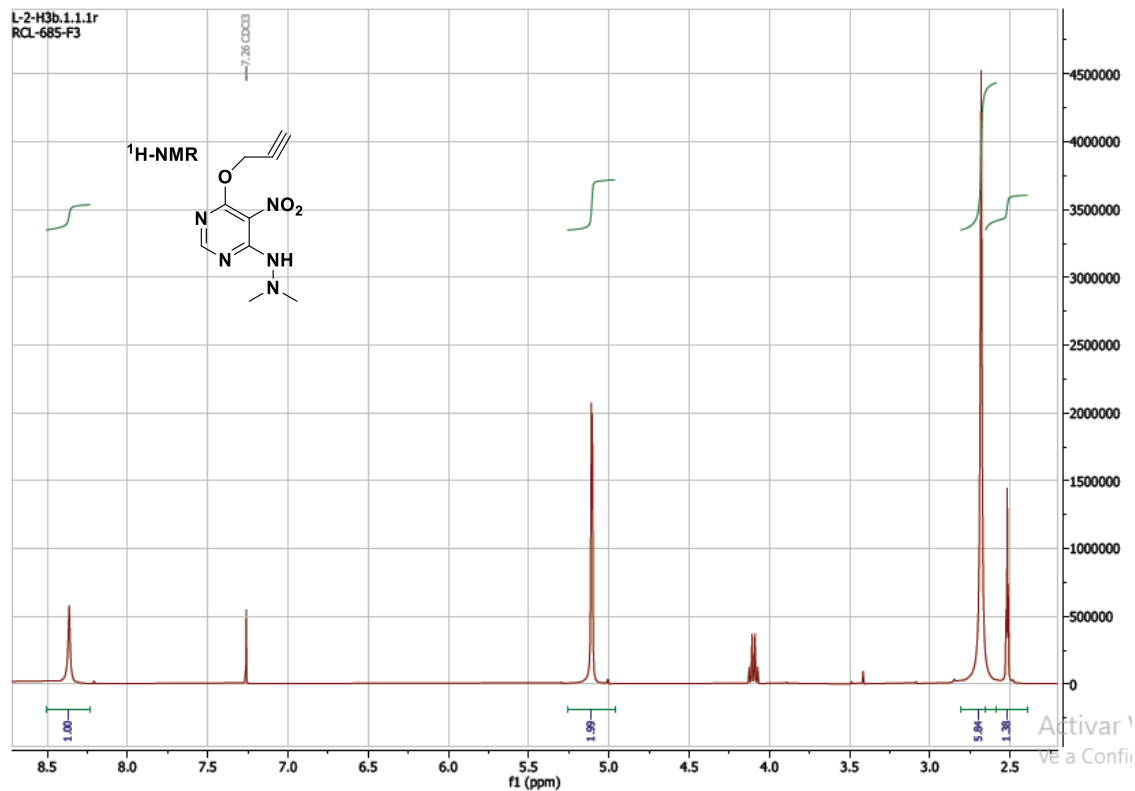


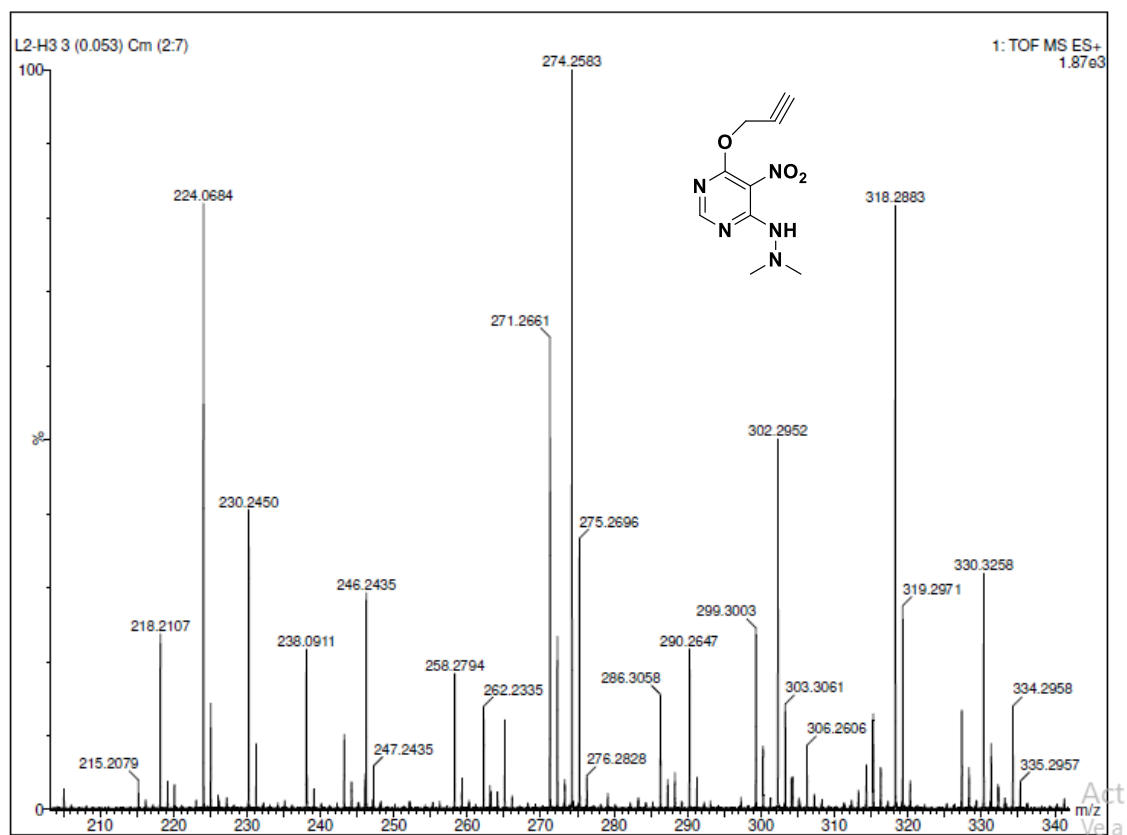
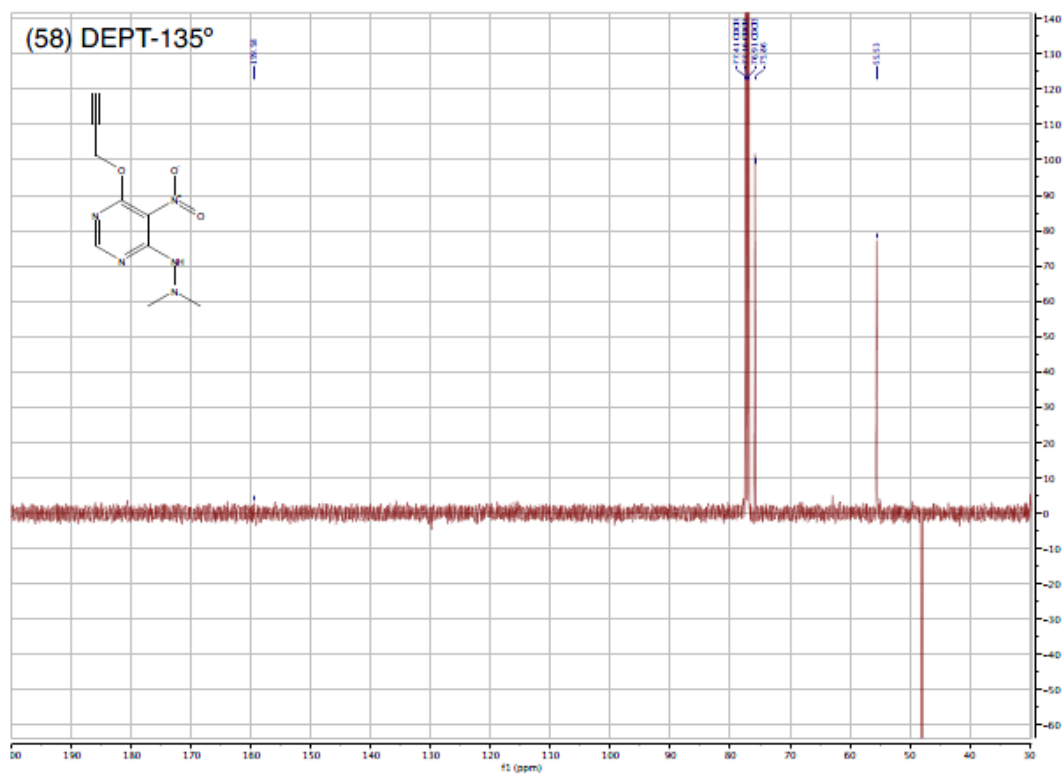
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Minimum:				-1.5				
Maximum:		5.0	10.0	50.0				
Mass	Calc. Mass	mDa	PPM	DBE	1-FIT	1-FIT (Norm)	Formula	
263.1148	263.1144	0.4	1.5	7.5	19.9	0.9	C12 H15 N4 O3	
	263.1139	0.9	3.4	1.5	20.2	1.2	C12 H23 O2 S2	
	263.1131	1.7	6.5	2.5	20.3	1.3	C11 H19 O7	

- 4-(2,2-dimethylhydrazineyl)-5-nitro-6-(prop-2-yn-1-yloxy)pyrimidine (8):

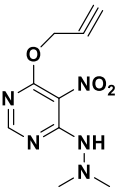




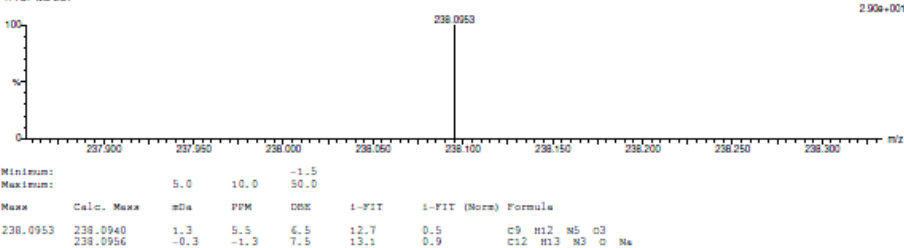
Elemental Composition Report

Single Mass Analysis
Tolerance = 10.0 PPM / DBE: min = -1.5, max = 50.0
Element prediction: Off
Number of isotope peaks used for i-FIT = 3

Monoisotopic Mass, Even Electron Ions
212 formula(e) evaluated with 2 results within limits (up to 50 closest results for each mass)
Elements Used:
C: 0-19 H: 0-1000 N: 0-6 O: 0-4 Na: 0-1
ALM:15 57 (1.252) AM (Top:1, H:5000.0,0.00,1.00)
1: TOF MS ES+



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Ac